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Class 12 Physics Chapterwise PYQs

2026 – 2004 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 8: Electromagnetic Waves

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 79 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. An electromagnetic wave passes from vacuum into a dielectric medium with relative electrical permittivity (3/2) and relative magnetic permeability (8/3). Then, its:

(A) wavelength is doubled and frequency remains unchanged.

(B) wavelength is doubled and frequency is halved.

(C) wavelength is halved and frequency remains unchanged.

(D) wavelength and frequency both will remain unchanged.

Check Solution → **[2026 • Set 55-1-1]**

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2. Electromagnetic waves used in a diagnostic tool in medicine have a wavelength range:

(A) 1 nm to 10^{-3} nm

(B) 400 nm to 1 nm

(C) 1 mm to 700 nm

(D) 0.1 m to 1 mm

Check Solution → **[2026 • Set 55-1-1]**

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3. Electromagnetic waves used for purification of water are:

(A) X-rays

(B) Ultraviolet rays

(C) Infrared rays

(D) Ultrasonic rays

Check Solution → **[2026 • Set 55-1-2]**

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4. Welders wear special glass goggles or face masks with glass windows to protect their eyes from:

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Microwaves

Check Solution → [2026 • Set 55-1-3]

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5. An electromagnetic wave is propagating along x-axis. At any instant, the phase difference (in radian) between the electric field ($\vec{E}$) and the magnetic field ($\vec{B}$) associated with the wave is:

(A) zero

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) π

Check Solution → [2026 • Set 55-2-1]

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6. The expressions of magnetic fields associated with four electromagnetic waves are given below:

I. $B_1 = (4\times 10^{-7}\ T)\sin[0{\cdot}7\times 10^{3}\ x + 1{\cdot}4\times 10^{11}\ t]$

II. $B_2 = (2\times 10^{-7}\ T)\sin[0{\cdot}6\times 10^{3}\ x + 1{\cdot}5\times 10^{11}\ t]$

III. $B_3 = (3\times 10^{-6}\ T)\sin[0{\cdot}5\times 10^{3}\ x + 1{\cdot}5\times 10^{11}\ t]$

IV. $B_4 = (5\times 10^{-7}\ T)\sin[0{\cdot}2\times 10^{4}\ x + 4{\cdot}8\times 10^{11}\ t]$

Which wave is travelling in free space?

(A) I

(B) II

(C) III

(D) IV

Check Solution → [2026 • Set 55-2-2]

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7. The electric field and the magnetic field components of an electromagnetic wave travelling in vacuum are represented as $E = m \sin(\alpha z - \beta t)$ and $B = n \sin(\alpha z - \beta t)$, where m, n, α and β are constants characterising the wave. Then:

(A) $\frac{m}{n} = \frac{\alpha}{\beta}$

(B) $\frac{m}{n} = \frac{\beta}{\alpha}$

(C) $\frac{n}{m} = \frac{\beta}{\alpha}$

(D) $\frac{n}{m} = \alpha\beta$

Check Solution → **[2026 • Set 55-2-3]**

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8. A plane electromagnetic wave travels through a medium and the magnetic field associated with it is given by $B = 5\times 10^{-8} \sin(3\times 10^{10} t - 150 x)$ T, where x is in metres and t is in seconds. The velocity of the wave is:

(A) $2{\cdot}0 \times 10^{8}\ m\ s^{-1}$

(B) $4{\cdot}5 \times 10^{7}\ m\ s^{-1}$

(C) $3{\cdot}5 \times 10^{7}\ m\ s^{-1}$

(D) $2{\cdot}5 \times 10^{8}\ m\ s^{-1}$

Check Solution → **[2026 • Set 55-3-1]**

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9. The ratio of amplitude of electric field to the amplitude of the magnetic field associated with an electromagnetic wave propagating in glass ($n = 1{\cdot}5$) is:

(A) $3 \times 10^{8}\ m\ s^{-1}$

(B) $2 \times 10^{8}\ m\ s^{-1}$

(C) $3{\cdot}3 \times 10^{-9}\ m\ s^{-1}$

(D) $5 \times 10^{-9}\ m\ s^{-1}$

Check Solution → **[2026 • Set 55-4-1]**

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10. The magnetic field in a plane electromagnetic wave travelling in glass ($n = 1.5$) is given by

$B_y = (2 \times 10^{-7}\ T) \sin(\alpha\, x + 1.5 \times 10^{11}\ t)$
where x is in metres and t is in seconds. The value of α is:

(A) $0.5 \times 10^3\ m^{-1}$

(B) $6.0 \times 10^2\ m^{-1}$

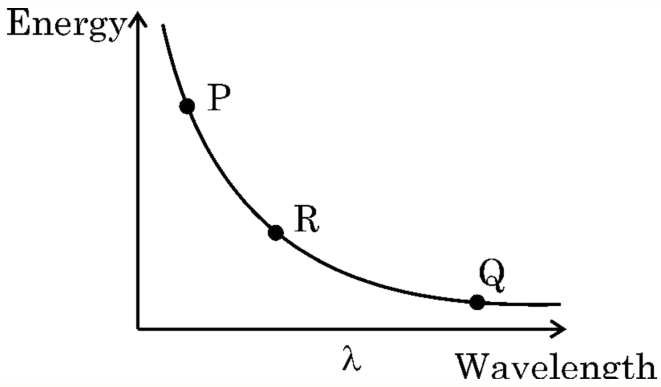
(C) $7.5 \times 10^2\ m^{-1}$

(D) $1.5 \times 10^3\ m^{-1}$

Check Solution → [2026 • Set 55-5-1]

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11. The given diagram exhibits the relationship between the wavelength of the electromagnetic waves and the energy of photon associated with them. The three points P, Q and R marked on the diagram may correspond respectively to:



(A) X-rays, microwaves, UV radiation

(B) X-rays, UV radiation, microwaves

(C) UV radiation, microwaves, X-rays

(D) Microwaves, UV radiation, X-rays

Check Solution → [2025 • Set 55-1-1]

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12. Which one of the following correctly represents the change in wave characteristics (all in vacuum) from microwaves to X-rays in electromagnetic spectrum?

(A) Speed: Remains same; Wavelength: Decreases; Frequency: Remains same

(B) Speed: Remains same; Wavelength: Decreases; Frequency: Increases

(C) Speed: Increases; Wavelength: Increases; Frequency: Decreases

(D) Speed: Remains same; Wavelength: Increases; Frequency: Remains same

Check Solution → [2025 • Set 55-1-2]

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13. X-rays are more harmful to human beings than ultraviolet radiations because X-rays:

(A) have frequency lower than that of ultraviolet radiations.

(B) have wavelength smaller than that of ultraviolet radiations.

(C) move faster than ultraviolet radiations in air.

(D) are mechanical waves but ultraviolet radiations are electromagnetic waves.

Check Solution → [2025 • Set 55-1-3]

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14. **Assertion (A):** Out of Infrared and radio waves, the radio waves show more diffraction effect.

Reason (R): Radio waves have greater frequency than infrared waves.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-2-1]

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15. The amplitude of electric field in an electromagnetic wave in free space is 1000 V m^{-1}. The amplitude of the magnetic field in this electromagnetic wave is :

(A) $3.0 \times 10^{-6}\ T$

(B) $3.33 \times 10^{-6}\ T$

(C) $3.0 \times 10^{-11}\ T$

(D) $3.33 \times 10^{-11}\ T$

Check Solution → [2025 • Set 55-4-1]

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16. The electric field in space between the plates of a parallel plate capacitor (each of area $2.5 \times 10^{-3}\ m^2$) is changing at the rate of $4 \times 10^6\ V\,m^{-1}s^{-1}$. The displacement current between the plates of the capacitor is :

(A) $1.8 \times 10^{-5}\ A$

(B) $3.47 \times 10^{-8}\ A$

(C) $8.85 \times 10^{-8}\ A$

(D) $6.32 \times 10^{-4}\ A$

Check Solution → [2025 • Set 55-4-2]

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17. In an electromagnetic wave travelling in free space, the amplitude of magnetic field is $6.0 \times 10^{-4}\ T$. The amplitude of its electric field is :

(A) $2 \times 10^{-4}\ V\,m^{-1}$

(B) $1.5 \times 10^{12}\ V\,m^{-1}$

(C) $1.8 \times 10^{5}\ V\,m^{-1}$

(D) $0.3 \times 10^{4}\ V\,m^{-1}$

Check Solution → [2025 • Set 55-4-3]

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18. Which of the following rays coming from the Sun plays an important role in maintaining the Earth's warmth?

(A) Infrared rays

(B) γ rays

(C) UV rays

(D) Visible light rays

Check Solution → [2025 • Set 55-5-1]

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19. The dimensions of (με), where ε is permittivity and μ is permeability of a medium, are:

(A) $[M^0 L^{-1} T]$

(B) $[M^0 L^{-2} T^2]$

(C) $[M^{-1} L^{-2} T^{-1}]$

(D) $[M^1 L^1 T^{-1}]$

Check Solution → [2025 • Set 55-5-1]

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20. **Assertion (A):** EM waves do not require a medium for their propagation.
Reason (R): EM waves are transverse waves.
Select the correct answer:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-7-1]

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21. In the four regions, I, II, III and IV, the electric fields are described as:
Region I: $E_y = E_0 \sin(kz - \omega t)$
Region II: $E_y = E_0$
Region III: $E_y = E_0 \sin kz$
Region IV: $E_y = E_0 \cos kz$
The displacement current will exist in the region:

(A) I

(B) IV

(C) II

(D) III

Check Solution → [2024 • Set 55-1-1]

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22. The electric and magnetic fields of electromagnetic waves are:

(A) In the same phase and perpendicular to each other.

(B) In the same phase and not perpendicular to each other.

(C) Not in the same phase but are perpendicular to each other.

(D) Neither in the same phase nor perpendicular to each other.

Check Solution → [2024 • Set 55-1-2]

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23. The first scientist who produced and observed electromagnetic waves of wavelengths in the range 25 mm – 5 mm was:

(A) J.C. Maxwell

(B) H.R. Hertz

(C) J.C. Bose

(D) G. Marconi

Check Solution → [2024 • Set 55-1-3]

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24. Electromagnetic waves with wavelength 10 nm are called:

(A) Infrared waves

(B) Ultraviolet rays

(C) Gamma rays

(D) X-rays

Check Solution → [2024 • Set 55-2-1]

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25. Welders wear special glass goggles or face masks with glass windows to protect their eyes from radiations produced by welding arcs. These radiations are:

(A) X-rays

(B) Ultraviolet rays

(C) Infrared waves

(D) Gamma rays

Check Solution → [2024 • Set 55-2-2]

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26. In four regions I, II, III and IV, the magnetic field is given by:

I. $B_y = B_0 \sin kz$

II. $B_y = B_0 \cos kz$

III. $B_y = B_0 \sin(kz - \omega t)$

IV. $B_y = B_0 \sin kz + B_0 \cos kz$

The electromagnetic wave will exist in the region:

(A) IV

(B) I

(C) III

(D) II

Check Solution → **[2024 • Set 55-2-3]**

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27. Which one of the following has the highest frequency?

(A) Infrared rays

(B) Gamma rays

(C) Radio waves

(D) Microwaves

Check Solution → **[2024 • Set 55-3-1]**

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28. Electromagnetic waves with frequency 1.0×10^{18} Hz are known as:

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Gamma rays

Check Solution → **[2024 • Set 55-3-2]**

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29. A plane electromagnetic wave is travelling in air in $+x$ direction. At a particular moment, its electric field $\vec{E}$ is along $+y$ direction. At that moment, the magnetic field $\vec{B}$ is along:

(A) $+z$ direction and in phase with $\vec{E}$

(B) $-z$ direction and in phase with $\vec{E}$

(C) $+z$ direction and out of phase with $\vec{E}$

(D) $-z$ direction and out of phase with $\vec{E}$

Check Solution → [2024 • Set 55-3-3]

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30. The phase difference between electric field $\vec{E}$ and magnetic field $\vec{B}$ in an electromagnetic wave propagating along z-axis is

(A) zero

(B) π

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4}$

Check Solution → [2024 • Set 55-4-1]

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31. The electromagnetic waves used to purify water are:

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Gamma rays

Check Solution → [2024 • Set 55-5-1]

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32. In the process of charging of a capacitor, the current produced between the plates of the capacitor is (where symbols have their usual meanings):

(A) $\mu_0 \frac{d\phi_E}{dt}$

(B) $\frac{1}{\mu_0} \frac{d\phi_E}{dt}$

(C) $\varepsilon_0 \frac{d\phi_E}{dt}$

(D) $\frac{1}{\varepsilon_0} \frac{d\phi_E}{dt}$

Check Solution → [2023 • Set 55-1-1]

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33. $\vec{E}$ and $\vec{B}$ represent the electric and the magnetic field of an electromagnetic wave respectively. The direction of propagation of the wave is along

(A) $\vec{B}$

(B) $\vec{E}$

(C) $\vec{E} \times \vec{B}$

(D) $\vec{B} \times \vec{E}$

Check Solution → [2023 • Set 55-2-1]

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34. The ratio of the magnitudes of the electric field and magnetic field of a plane electromagnetic wave is

(A) 1

(B) $\frac{1}{c}$

(C) c

(D) c^2

Check Solution → [2023 • Set 55-2-2]

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35. An electromagnetic wave is produced by a charge

(A) moving with a constant velocity

(B) moving with a constant speed parallel to a magnetic field

(C) moving with an acceleration

(D) at rest

Check Solution → [2023 • Set 55-2-3]

36. The electromagnetic radiations used to kill germs in water purifiers are called:

(A) Infrared waves

(B) X-rays

(C) Gamma rays

(D) Ultraviolet rays

Check Solution → [2023 • Set 55-3-1]

37. Electromagnetic waves of wavelength of the order of a few meters were first produced and detected in the laboratory by:

(A) J.C. Maxwell

(B) J.C. Bose

(C) H.R. Hertz

(D) G. Marconi

Check Solution → [2023 • Set 55-3-2]

38. A capacitor is charged by a battery. Between the plates, during process of charging:

(A) only displacement current exists.

(B) only conduction current exists.

(C) both displacement current and conduction current exist.

(D) no current exists.

Check Solution → [2023 • Set 55-3-3]

39. The electromagnetic waves used in radar systems are:

(A) Infrared waves

(B) Ultraviolet rays

(C) Microwaves

(D) X-rays

Check Solution → [2023 • Set 55-4-1]

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40. Name the electromagnetic waves also known as 'heat waves'.

(A) Radio waves

(B) Microwaves

(C) X-rays

(D) Infrared waves

Check Solution → [2023 • Set 55-4-3]

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41. Which one of the following electromagnetic radiation has the least wavelength?

(A) Gamma rays

(B) Microwaves

(C) Visible light

(D) X-rays

Check Solution → [2023 • Set 55-5-1]

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42. Choose the correct option related to wavelengths (λ) of different parts of electromagnetic spectrum.

(A) $\lambda_{X\text{–rays}} < \lambda_{\text{micro waves}} < \lambda_{\text{visible}}$

(B) $\lambda_{\text{visible}} > \lambda_{X\text{–rays}} > \lambda_{\text{radio waves}}$

(C) $\lambda_{\text{radio waves}} > \lambda_{\text{visible}} > \lambda_{X\text{–rays}}$

(D) $\lambda_{\text{visible}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{X\text{–rays}}$

Check Solution → [2023 • Set 55-5-2]

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43. What oscillates in an electromagnetic wave of frequency 10 MHz?

Check Solution → [2021]

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44. Name the electromagnetic radiation used for killing germs in water purifiers.

Check Solution → [2021]

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45. Write the mathematical form of Ampere-Maxwell circuital law.

Check Solution → [2020 • Set 55-1-1]

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46. How is displacement current produced between the plates of a parallel plate capacitor during charging?

Check Solution → [2020 • Set 55-1-2]

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47. Differentiate between conduction current and displacement current.

Check Solution → [2020 • Set 55-1-3]

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48. Displacement current exists only when

(A) electric field is changing.

(B) magnetic field is changing.

(C) electric field is not changing.

(D) magnetic field is not changing.

Check Solution → [2020 • Set 55-2-1]

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49. Electromagnetic waves used as a diagnostic tool in medicine are

(A) X-rays.

(B) ultraviolet rays.

(C) infrared radiation.

(D) ultrasonic waves.

Check Solution → [2020 • Set 55-2-1]

50. A welder wears special glasses to protect his eyes mostly from the harmful effect of

(A) very intense visible light.

(B) infrared radiation.

(C) ultraviolet rays.

(D) microwaves.

Check Solution → [2020 • Set 55-2-2]

51. In a travelling plane electromagnetic wave, which of the following have zero average value?

(A) Magnetic energy and electric energy

(B) Magnetic field and electric field

(C) Magnetic energy and magnetic field

(D) Electric energy and electric field

Check Solution → [2020 • Set 55-2-3]

52. Mention the contribution of Indian physicist J.C. Bose in the production of electromagnetic waves.

Check Solution → [2020 • Set 55-3-1]

53. Write one use of the electromagnetic waves of frequency range from 10^{16} Hz to 10^{20} Hz.

Check Solution → [2020 • Set 55-3-1]

54. Which physical quantity associated with an electromagnetic wave does not change when it propagates from one medium into another medium?

Check Solution → [2020 • Set 55-3-2]

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55. Name the electromagnetic radiations which may cause sunburn. Write their wavelength range.

Check Solution → [2020 • Set 55-3-3]

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56. An A.C. source with variable frequency is connected to a parallel plate capacitor. How will the displacement current be affected with the decrease in frequency of the source?

Check Solution → [2020 • Set 55-4-1]

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57. Depict the fields diagram of an electromagnetic wave propagating along positive *X*-axis with its electric field along *Y*-axis.

Check Solution → [2020 • Set 55-5-1]

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58. Which part of the electromagnetic spectrum is used in RADAR? Give its frequency range.

OR

How are electromagnetic waves produced by accelerating charges?

Check Solution → [2019 • Set 55-2-1]

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59. The small ozone layer on top of the stratosphere is crucial for human survival. Why?

OR

Illustrate by giving suitable examples, how you can show that electromagnetic waves carry both energy and momentum.

Check Solution → [2019 • Set 55-4-1]

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60. Name the electromagnetic radiations used for (a) water purification, and (b) eye surgery.

Check Solution → [2018]

---

61. How is the speed of em-waves in vacuum determined by the electric and magnetic fields?

Check Solution → [2017]

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62. Write the expression for speed of electromagnetic waves in a medium of electrical permittivity ε and magnetic permeability μ.

Check Solution → [2017]

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63. Do electromagnetic waves carry energy and momentum?

Check Solution → [2017]

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64. Write the relation for the speed of electromagnetic waves in terms of the amplitudes of electric and magnetic fields.

Check Solution → [2017]

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65. In which directions do the electric and magnetic field vectors oscillate in an electromagnetic wave propagating along the x-axis?

Check Solution → [2017]

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66. In which situation is there a displacement current but no conduction current?

Check Solution → [2016]

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67. To which part of the electromagnetic spectrum does a wave of frequency 5×10^{19} Hz belong ?

Check Solution → [2014]

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68. A capacitor has been charged by a dc source. What are the magnitudes of conduction and displacement currents, when it is fully charged?

Check Solution → [2013]

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69. Name the part of electromagnetic spectrum whose wavelength lies in the range of $10^{-10}\ m$. Give its one use.

Check Solution → [2012]

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70. What are the directions of electric and magnetic field vectors relative to each other and relative to the direction of propagation of electromagnetic waves?

Check Solution → [2012]

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71. Name the physical quantity which remains same for microwaves of wavelength 1 mm and UV radiations of 1600 $\circ A$ in vacuum.

Check Solution → [2012]

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72. Name the electromagnetic waves, which (i) maintain the Earth's warmth and (ii) are used in aircraft navigation.

Check Solution → [2012]

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73. A plane electromagnetic wave travels in vacuum along z-direction. What can you say about the direction of electric and magnetic field vectors?

Check Solution → [2011 • Set 55-1-1]

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74. Which part of electromagnetic spectrum has largest penetrating power?

Check Solution → [2010]

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75. Write the following radiations in ascending order in respect of their frequencies: X-rays, microwaves, UV rays and radio waves.

Check Solution → [2009]

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76. Name the EM waves used for studying crystal structure of solids. What is its frequency range?

Check Solution → [2009]

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77. Name the part of the electromagnetic spectrum of wavelength 10^{-2} m and mention its one application.

Check Solution → [2008]

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78. Identify the part of the electromagnetic spectrum to which the following wavelengths belong: (i) 1 mm (ii) 10^{-11} m.

Check Solution → [2008]

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79. What is the name given to that part of electromagnetic spectrum which is used for taking photographs of earth under foggy conditions from great heights?

Check Solution → [2004]

SECTION B

2-Mark Questions

VSA · 39 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

80. Consider an induced magnetic field due to changing electric field and an induced electric field due to changing magnetic field. Which one is more easily observed? Justify your answer.

Check Solution → **[2023 • Set 55-1-1]**

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81. A plane electromagnetic wave propagates in vacuum along x-axis.
(a) Give the direction of electric field and magnetic field vectors.
(b) E_0 and B_0 are the magnitudes of electric and magnetic fields respectively in a plane electromagnetic wave. Find the ratio of energy densities of electric field and magnetic field.

Check Solution → **[2023 • Set 55-1-2]**

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82. Identify the part of electromagnetic spectrum which is:
(a) next to the lowest frequency end of the visible part of electromagnetic spectrum, and
(b) produced by bombarding a metal target by high energy electrons.
Give one use of each of them.

Check Solution → **[2023 • Set 55-1-3]**

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83. How are infrared waves produced? Why are these waves referred to as heat waves? Give any two uses of infrared waves.

Check Solution → **[2023 • Set 55-2-1]**

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84. How are X-rays produced? Give any two uses of these.

Check Solution → [2023 • Set 55-2-1]

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85. What is meant by the term 'displacement current'? Briefly explain how this current is different from a conduction current.

Check Solution → [2023 • Set 55-3-1]

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86. (a) What is a displacement current? How is it different from a conduction current?

OR

(b) Write any two characteristics of an electromagnetic wave. Why are microwaves used in radar systems?

Check Solution → [2023 • Set 55-4-1]

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87. Identify the electromagnetic wave whose wavelengths range is from about
(a) 10^{-12} m to about 10^{-8} m.
(b) 10^{-3} m to about 10^{-1} m.
Write one use of each.

Check Solution → [2023 • Set 55-5-1]

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88. Identify the electromagnetic radiation and write its wavelength range, which is used to kill germs in water purifier. Name the two sources of these radiations.

Check Solution → [2023 • Set 55-5-2]

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89. How are electromagnetic waves produced? Write their two characteristics.

Check Solution → [2023 • Set 55-5-3]

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90. (a) The electric field of an electromagnetic wave is represented as $E_x = E_0 \sin(\omega t + kz)$.
(i) In which direction is the wave propagating?

(ii) In which direction does the magnetic field oscillate?
(b) Write two characteristics of electromagnetic waves.

Check Solution → [2021]

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91. Which of the following electromagnetic waves has (a) minimum wavelength, and (b) minimum frequency? Write one use of each of these two waves.
Infrared waves, Microwaves, γ-rays and X-rays

Check Solution → [2020 • Set 55-2-1]

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92. How does an oscillating charge radiate an electromagnetic wave? Give the relation between the frequency of radiated wave and the frequency of oscillating charge.

OR

(a) Explain briefly the fact that electromagnetic waves carry energy.
(b) Why do we not feel the pressure due to sunshine?

Check Solution → [2020 • Set 55-3-1]

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93. Gamma rays and radio waves travel with the same velocity in free space. Distinguish between them in terms of their origin and the main application.

Check Solution → [2020 • Set 55-5-1]

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94. A capacitor made of two parallel plates, each of area 'A' and separation 'd' is charged by an external dc source. Show that during charging, the displacement current inside the capacitor is the same as the current charging the capacitor.

Check Solution → [2019 • Set 55-2-1]

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95. How is the equation for Ampere's circuital law modified in the presence of displacement current? Explain.

Check Solution → [2019 • Set 55-2-2]

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96. How are electromagnetic waves produced by oscillating charges? What is the source of the energy associated with the em waves?

Check Solution → [2019 • Set 55-2-3]

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97. A parallel plate capacitor of plate area A each and separation d, is being charged by an ac source. Show that the displacement current inside the capacitor is the same as the current charging the capacitor.

Check Solution → [2019 • Set 55-4-1]

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98. (a) Why are infra-red waves often called heat waves ? Explain.
(b) What do you understand by the statement, "Electromagnetic waves transport momentum" ?

Check Solution → [2018]

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99. How does Ampere-Maxwell law explain the flow of current through a capacitor when it is being charged by a battery? Write the expression for the displacement current in terms of the rate of change of electric flux.

Check Solution → [2017]

---

100. Identify the electromagnetic waves whose wavelengths vary as
(a) $10^{-12}\ m < \lambda < 10^{-8}$ m
(b) $10^{-3}\ m < \lambda < 10^{-1}$ m
Write one use for each.

Check Solution → [2017]

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101. (i) How are infrared waves produced? Write their one important use.
(ii) The thin Ozone layer on top of the stratosphere is crucial for human survival. Why?

Check Solution → [2016]

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102. State two properties of electromagnetic waves. How can we show that em waves carry momentum?

Check Solution → [2016]

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103. Considering the case of a parallel plate capacitor being charged, show how one is required to generalize Ampere's circuital law to include the term due to displacement current.

Check Solution → [2014]

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104. (a) An em wave is travelling in a medium with a velocity $\vec{v} = v\,\hat{i}$. Draw a sketch showing the propagation of the em wave, indicating the direction of the oscillating electric and magnetic fields.
(b) How are the magnitudes of the electric and magnetic fields related to the velocity of the em wave?

Check Solution → [2013]

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105. Draw a sketch of a plane electromagnetic wave propagating along the z-direction. Depict clearly the directions of electric and magnetic fields varying sinusoidally with z.

Check Solution → [2012]

---

106. A capacitor of capacitance 'C' is being charged by connecting it across a dc source along with an ammeter. Will the ammeter show a momentary deflection during the process of charging? If so, how would you explain this momentary deflection and the resulting continuity of current in the circuit? Write the expression for the current inside the capacitor.

Check Solution → [2012]

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107. When an ideal capacitor is charged by a dc battery, no current flows. However, when an ac source is used, the current flows continuously. How does one explain

this, based on the concept of displacement current?

Check Solution → [2012]

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108. Explain briefly how electromagnetic waves are produced by an oscillating charge. How is the frequency of the em waves produced related to that of the oscillating charge?

Check Solution → [2012]

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109. What is sky wave communication? Why is this mode of propagation restricted to the frequencies only upto few MHz?

Check Solution → [2011]

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110. A parallel plate capacitor is being charged by a time varying current. Explain briefly how Ampere's circuital law is generalized to incorporate the effect due to the displacement current.

Check Solution → [2011]

---

111. How are infrared waves produced? Why are these referred to as *heat waves*? Write their one important use.

Check Solution → [2011 • Set 55-1-1]

---

112. Show how the equation for Ampere's circuital law, viz.
$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$
is modified in the presence of displacement current.

Check Solution → [2011 • Set 55-2-1]

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113. Arrange the following electromagnetic radiations in ascending order of their frequencies:
(i) Microwave

(ii) Radio wave
(iii) X-rays
(iv) Gamma rays
Write two uses of any one of these.

Check Solution → [2010]

---

114. How does a charge q oscillating at certain frequency produce electromagnetic waves?
Sketch a schematic diagram depicting electric and magnetic fields for an electromagnetic wave propagating along the Z-direction.

Check Solution → [2009]

---

115. Answer the following questions:
(a) Optical and radio telescopes are built on the ground while X-ray astronomy is possible only from satellites orbiting the Earth. Why?
(b) The small ozone layer on top of the stratosphere is crucial for human survival. Why?

Check Solution → [2009]

---

116. Define magnetic susceptibility of a material. Name two elements, one having positive susceptibility and the other having negative susceptibility. What does negative susceptibility signify?

Check Solution → [2008]

---

117. What is meant by the transverse nature of electromagnetic waves? Draw a diagram showing the propagation of an electromagnetic wave along the x-direction, indicating clearly the directions of the oscillating electric and magnetic fields associated with it.

Check Solution → [2008]

---

118. Find the wavelength of electromagnetic waves of frequency 5×10^{19} Hz in free space. Give its two applications.

Check Solution → [2004]

SECTION C

3-Mark Questions

SA · 53 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

119. Name the electromagnetic waves which are used for:
(i) detection of fractures in bones
(ii) physiotherapy
(iii) radar systems
Also write their wavelength range.

Check Solution → **[2026 • Set 55-2-1]**

---

120. Name the electromagnetic waves which are used:
(i) as a diagnostic tool in medicine
(ii) in remote switches for TV sets
(iii) in water purifiers
Also write their wavelength range.

Check Solution → **[2026 • Set 55-2-2]**

---

121. Name the electromagnetic waves which are (i) also known as 'heat waves', (ii) used in LASIK eye surgery, (iii) used to destroy cancer cells. Also write their wavelength range.

Check Solution → **[2026 • Set 55-2-3]**

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122. (a) Depict the variation of electric field ($\vec{E}$) and magnetic field ($\vec{B}$) with respect to the direction of propagation of an electromagnetic wave. Write their two important characteristics.

(b) Show that $\frac{1}{\sqrt{\varepsilon_0 \mu_0}}$ gives the velocity of an electromagnetic wave in free space.

Check Solution → [2026 • Set 55-3-1]

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123. What are microwaves? How are they produced? Give any two uses of microwaves.

Check Solution → [2026 • Set 55-4-2]

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124. What is displacement current (i_d)? Considering the case of charging of a capacitor, show that $i_d = \varepsilon_0 \frac{d\Phi_E}{dt}$. What is the value of i_d for a conductor across which a constant voltage is applied?

Check Solution → [2026 • Set 55-5-1]

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125. Differentiate between conduction current and displacement current. A capacitor is connected across a source providing time-dependent current. Explain how the total current at an instant is the sum of conduction current and displacement current in the circuit at that instant.

Check Solution → [2026 • Set 55-5-2]

---

126. **(a)** A parallel plate capacitor is charged by an ac source. Show that the sum of conduction current (I_c) and the displacement current (I_d) has the same value at all points of the circuit.

(b) In case (a) above, is Kirchhoff's first rule (junction rule) valid at each plate of the capacitor? Explain.

Check Solution → [2025 • Set 55-1-1]

---

127. (a) The electric field $\vec{E}$ of an electromagnetic wave propagating in north direction is oscillating in up and down direction. Describe the direction of magnetic field $\vec{B}$ of the wave.
(b) Are the wave length of radio waves and microwaves longer or shorter than those detectable by human eyes?
(c) Write main use of each of the following in human life:
(i) Infrared waves (ii) Gamma rays

Check Solution → [2025 • Set 55-2-1]

---

128. (a) How is an electromagnetic wave produced?
(b) An electromagnetic wave is travelling in vertically upward direction. At an instant, its electric field vector points in west direction. In which direction does the magnetic field vector point at that instant?
(c) Estimate the ratio of shortest wave length of radio waves to the longest wave length of gamma waves.

Check Solution → [2025 • Set 55-2-2]

---

129. (a) The amplitude of a light wave becomes n times. This results in intensity of the wave becoming m times. What is the relation between n and m?
(b) White light is incident on three identical surfaces - a black surface, a yellow surface and a white surface, one by one. For which surface, the pressure exerted on the surface by the incident light will be (i) maximum (ii) minimum? Justify your answer.

Check Solution → [2025 • Set 55-2-3]

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130. **(a)** State any three characteristics of electromagnetic waves.
(b) Briefly explain how and where the displacement current exists during the charging of a capacitor.

Check Solution → [2025 • Set 55-4-1]

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131. Name the electromagnetic wave used (i) in radar, (ii) in eye surgery and (iii) as diagnostic tool in medicine. Write their wavelength range also.

Check Solution → [2025 • Set 55-6-1]

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132. **(a)** Differentiate between 'conduction current' and 'displacement current', giving one similarity and one dissimilarity between them.
(b) Explain the existence of electromagnetic waves in free space, using the concept of displacement current.

Check Solution → [2025 • Set 55-7-1]

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133. **(a)** How are electromagnetic waves produced?
(b) Write the wavelength range and one use of:
(i) Microwaves, and
(ii) Ultraviolet waves.

Check Solution → [2025 • Set 55-7-2]

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134. Name the electromagnetic radiations and write their frequency range:
(a) Used to take the photograph of the bones
(b) Produced by hot bodies
(c) Used in television communication systems

Check Solution → [2025 • Set 55-7-3]

---

135. (a) "The wavelength of the electromagnetic wave is often correlated with the characteristic size of the system that radiates." Give two examples to justify this statement.
(b) (i) Long distance radio broadcasts use short-wave bands. Why?
(ii) Optical and radio telescopes are built on the ground, but X-ray astronomy is possible only from satellites orbiting the Earth. Why?

Check Solution → [2024 • Set 55-1-1]

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136. (a) On what factors does the speed of an electromagnetic wave in a medium depend?
(b) How is an electromagnetic wave produced?
(c) Sketch a schematic diagram depicting the electric and magnetic fields for an electromagnetic wave propagating along z-axis.

Check Solution → [2024 • Set 55-2-1]

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137. The electric field in an electromagnetic wave in vacuum is given by:
$\vec{E} = (6.3\ N/C)\ [\cos((1.5\ \text{rad}/m)\ y + (4.5 \times 10^8\ \text{rad}/s)\ t)]\ \hat{i}$
(a) Find the wavelength and frequency of the wave.
(b) What is the amplitude of the magnetic field of the wave?
(c) Write an expression for the magnetic field of this wave.

Check Solution → [2024 • Set 55-3-1]

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138. Name the electromagnetic waves with their wavelength range which are used for
(i) FM radio broadcast
(ii) detection of fracture in bones
(iii) treatment of muscular strain

Check Solution → [2024 • Set 55-4-1]

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139. What is meant by displacement current? How is this current different from the conduction current? A capacitor is being charged by a source of emf. Justify the continuity of current in the circuit.

Check Solution → [2024 • Set 55-4-2]

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140. Explain the following giving reasons:
(i) 'Electromagnetic waves differ considerably in their mode of interaction with matter'.
(ii) 'Food items to be heated in microwave oven must contain water'.
(iii) 'Welders wear face mask with glasses during welding'.

Check Solution → [2024 • Set 55-4-3]

141. (a) Name the parts of the electromagnetic spectrum which are (i) also known as 'heat waves' and (ii) absorbed by ozone layer in the atmosphere.
(b) Write briefly one method each, of the production and detection of these radiations.

Check Solution → [2024 • Set 55-5-1]

142. Name the part of the electromagnetic spectrum which are:
(i) stopped by face mask worn by welders.
(ii) used in detectors in Earth satellites.
(iii) used in 'short-wave band' in communication.
Also write the order of wavelengths, in each case.

Check Solution → [2024 • Set 55-5-2]

143. (i) Depict a plane electromagnetic wave propagating along the x-axis. Write the expressions for its oscillating electric and magnetic fields.
(ii) Write three characteristics of electromagnetic waves.

Check Solution → [2022 • Set 55-2-1]

144. Name the electromagnetic waves which are produced by the following:
(i) Radioactive decays of nucleus
(ii) Welding arcs
(iii) Hot bodies
Write one use each of these waves.

Check Solution → [2022 • Set 55-2-1]

145. Electromagnetic waves of wavelengths λ_1, λ_2 and λ_3 are used in radar systems, in water purifiers and in remote switches of TV, respectively.
(i) Identify the electromagnetic waves, and
(ii) Write one source of each of them.

Check Solution → [2022 • Set 55-3-1]

146. Identify electromagnetic waves which:
(i) are used in radar system.
(ii) affect a photographic plate.
(iii) are used in surgery.
Write their frequency range.

Check Solution → [2022 • Set 55-4-1]

147. (i) Arrange the following electromagnetic radiation in the ascending order of their frequencies:
X-rays, microwaves, gamma rays, radio waves
(ii) Write two uses of any two of these radiation.

Check Solution → [2022 • Set 55-5-1]

148. Write the expression for the speed of light in a material medium of relative permittivity ε_r and relative magnetic permeability μ_r.

Check Solution → [2020 • Set 55-1-1]

149. Write the wavelength range and name of the electromagnetic waves which are used in (i) radar systems for aircraft navigation, and (ii) Earth satellites to observe the growth of the crops.

Check Solution → [2020 • Set 55-1-1]

150. Name the electro-magnetic waves with their frequency range, produced in
(a) some radioactive decay
(b) sparks during electric welding
(c) TV remote

Check Solution → [2020 • Set 55-4-1]

151. (a) We feel the warmth of the sun light but not the pressure on our hands. Explain.
(b) Which out of wavelength, frequency and speed of an electro-magnetic wave does not change on passing from one medium to another?
(c) A thin ozone layer in the upper atmosphere is crucial for human survival on earth, why?

Check Solution → **[2020 • Set 55-4-2]**

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152. (a) How are electro-magnetic waves produced? Depict an electro-magnetic wave propagating in Z-direction with its magnetic field $\vec{B}$ oscillating along X-direction.
(b) Write two characteristics of electro-magnetic waves.

Check Solution → **[2020 • Set 55-4-3]**

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153. Identify the part of the electromagnetic spectrum used in (i) radar and (ii) eye surgery. Write their frequency range.

Check Solution → **[2019 • Set 55-1-1]**

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154. Prove that the average energy density of the oscillating electric field is equal to that of the oscillating magnetic field.

Check Solution → **[2019 • Set 55-1-1]**

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155. Name the radiation of the electromagnetic spectrum which is used for the following:
(a) Radar
(b) To photograph internal parts of human body
(c) For taking photographs of the sky during night and foggy conditions
Give the frequency range in each case.

Check Solution → **[2019 • Set 55-3-1]**

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156. When a parallel plate capacitor is being charged by a dc battery, why does a galvanometer show a momentary deflection?
Obtain the expression for the displacement current and show how it is related to conduction current in a steady state.

Check Solution → [2019 • Set 55-3-2]

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157. Write the range of the wavelength of the following electromagnetic radiations:
(a) Infrared rays
(b) Ultraviolet rays
(c) γ-rays
Write one use of each of the above.

Check Solution → [2019 • Set 55-3-3]

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158. Define displacement current. What role does it play while charging a capacitor by dc source. Is the value of displacement current same as that of the conduction current? Explain.

Check Solution → [2019 • Set 55-5-1]

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159. (a) How are electromagnetic waves produced? Explain.
(b) A plane electromagnetic wave is travelling through a medium along the +ve z-direction. Depict the electromagnetic wave showing the directions of the oscillating electric and magnetic fields.

Check Solution → [2017]

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160. (i) Identify the part of the electromagnetic spectrum which is:
(a) suitable for radar system used in aircraft navigation,
(b) produced by bombarding a metal target by high speed electrons.
(ii) Why does a galvanometer show a momentary deflection at the time of charging or discharging a capacitor? Write the necessary expression to explain this observation.

Check Solution → [2016]

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161. How are electromagnetic waves produced? What is the source of energy of these waves? Write mathematical expressions for electric and magnetic fields of an electromagnetic wave propagating along the z-axis. Write any two important properties of electromagnetic waves.

Check Solution → [2016]

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162. How are electromagnetic waves produced? What is the source of energy of these waves?
Draw a schematic sketch of the electromagnetic waves propagating along the $+x$-axis. Indicate the directions of the electric and magnetic fields. Write the relation between the velocity of propagation and the magnitudes of electric and magnetic fields.

Check Solution → [2015]

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163. Name the parts of the electromagnetic spectrum which is
(a) suitable for radar systems used in aircraft navigation.
(b) used to treat muscular strain.
(c) used as a diagnostic tool in medicine.
Write in brief, how these waves can be produced.

Check Solution → [2015]

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164. Answer the following :
(a) Name the em waves which are suitable for radar systems used in aircraft

navigation. Write the range of frequency of these waves.
(b) If the earth did not have atmosphere, would its average surface temperature be higher or lower than what it is now ? Explain.
(c) An em wave exerts pressure on the surface on which it is incident. Justify.

Check Solution → [2014]

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165. Answer the following :
(a) Name the em waves which are used for the treatment of certain forms of cancer. Write their frequency range.
(b) Thin ozone layer on top of stratosphere is crucial for human survival. Why ?
(c) Why is the amount of the momentum transferred by the em waves incident on the surface so small ?

Check Solution → [2014]

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166. Answer the following the questions :
(a) Name the em waves which are produced during radioactive decay of a nucleus. Write their frequency range.
(b) Welders wear special glass goggles while working. Why ? Explain.
(c) Why are infrared waves often called as heat waves ? Give their one application.

Check Solution → [2014]

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167. Write briefly any two factors which demonstrate the need for modulating a signal.
Draw a suitable diagram to show amplitude modulation using a sinusoidal signal as the modulating signal.

Check Solution → [2011]

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168. Draw a schematic diagram showing the (i) ground wave (ii) sky wave and (iii) space wave propagation modes for em waves.
Write the frequency range for each of the following:

(i) Standard AM broadcast
(ii) Television
(iii) Satellite communication

Check Solution → [2011 • Set 55-1-1]

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169. Write four characteristics of electromagnetic waves. Give two uses each of (i) Radio-waves (ii) Micro-waves.

Check Solution → [2007]

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170. Name the following constituent radiations of electromagnetic spectrum which
(i) produce intense heating effect.
(ii) is absorbed by the ozone layer in the atmosphere.
(iii) is used for studying crystal structure.
Write one more application for each of these radiations.

Check Solution → [2007]

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171. Draw a labelled diagram of Hertz's experimental set-up to produce electromagnetic waves. Explain the generation of electromagnetic waves using this set-up.

Check Solution → [2006]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **C**	Q9 **B**	Q17 **C**	Q25 **B**	Q33 **C**	Q41 **A**
Q2 **A**	Q10 **C**	Q18 **A**	Q26 **C**	Q34 **C**	Q42 **C**
Q3 **B**	Q11 **B**	Q19 **B**	Q27 **B**	Q35 **C**	Q48 **A**
Q4 **B**	Q12 **B**	Q20 **B**	Q28 **C**	Q36 **D**	Q49 **A**
Q5 **A**	Q13 **B**	Q21 **A**	Q29 **A**	Q37 **C**	Q50 **C**
Q6 **C**	Q14 **C**	Q22 **A**	Q30 **A**	Q38 **A**	Q51 **B**
Q7 **B**	Q15 **B**	Q23 **C**	Q31 **B**	Q39 **C**	
Q8 **A**	Q16 **C**	Q24 **D**	Q32 **C**	Q40 **D**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

An electromagnetic wave passes from vacuum into a dielectric medium with relative electrical permittivity (3/2) *and relative magnetic permeability* (8/3). *Then, its:*

(A) wavelength is doubled and frequency remains unchanged.

(B) wavelength is doubled and frequency is halved.

(C) **wavelength is halved and frequency remains unchanged.**

(D) wavelength and frequency both will remain unchanged.

[2026 • Set 55-1-1]

SOLUTION

When an electromagnetic wave passes from one medium to another, its frequency is fixed by the source and never changes. What changes is the speed and hence the wavelength, decided by the refractive index of the medium.

The refractive index of a medium in terms of its relative permittivity ε_r and relative permeability μ_r is given by:

$$n = \sqrt{\mu_r \, \varepsilon_r}$$

Here $\varepsilon_r = \frac{3}{2}$ and $\mu_r = \frac{8}{3}$.

$$n = \sqrt{\frac{8}{3} \times \frac{3}{2}}$$

$$n = \sqrt{4}$$

$$n = 2$$

Inside the medium the wavelength becomes:

$$\lambda_{med} = \frac{\lambda_{vac}}{n} = \frac{\lambda_{vac}}{2}$$

So the wavelength is halved while the frequency stays the same. This matches option (C).

Option (A) and (B) wrongly say the wavelength doubles, which would need $n = \frac{1}{2}$. Option (B) and (D) wrongly change or fix the frequency. Hence **the wavelength is halved and the frequency remains unchanged**, option (C).

COMMON MISTAKE

Frequency of an EM wave depends only on the source, not the medium. Whenever a wave enters a denser medium, only speed and wavelength drop; never change the frequency.

Final answer. The wavelength is halved and the frequency stays unchanged, so the answer is option **(C)**.

← *Back to Q1*

Q2

Electromagnetic waves used in a diagnostic tool in medicine have a wavelength range:

(A) 1 nm to 10^{-3} nm

(B) 400 nm to 1 nm

(C) 1 mm to 700 nm

(D) 0.1 m to 1 mm

[2026 • Set 55-1-1]

SOLUTION

Diagnostic imaging in medicine, such as taking radiographs to check bones and internal organs, uses **X-rays**.

X-rays lie in the wavelength range from about 1 nm down to 10^{-3} nm, that is roughly 10^{-9} *m* to 10^{-12} *m*. Their very short wavelength and high energy let them pass through soft tissue but get absorbed by bone, producing the image.

This range is exactly option (A).

The other options belong to different parts of the spectrum: option (B) 400 nm to 1 nm is the ultraviolet region, option (C) 1 mm to 700 nm covers infrared, and option (D) 0.1 *m* to 1 mm is the microwave region. None of these is used for routine diagnostic imaging.

Hence the correct wavelength range is **1 nm to 10^{-3} nm**, option (A).

Final answer. Diagnostic X-rays cover 1 nm to 10^{-3} nm, so the answer is option **(A)**.

← *Back to Q2*

Q3

Electromagnetic waves used for purification of water are:

(A) X-rays

(B) Ultraviolet rays

(C) Infrared rays

(D) Ultrasonic rays

[2026 • Set 55-1-2]

SOLUTION

The electromagnetic waves used to purify or sterilise water are **ultraviolet (UV) rays**.

Germicidal lamps in water purifiers emit short-wavelength UV-C radiation, of wavelength close to 254 nm. These high-energy UV photons damage the DNA of bacteria, viruses and other microbes present in the water, so the organisms can no longer reproduce and the water becomes safe.

This matches option (B).

Option (A) X-rays are used in medical imaging, not water treatment. Option (C) infrared rays are used for heating and remote sensing. Option (D) ultrasonic rays are sound waves, not electromagnetic waves at all, so they cannot be the answer here.

Hence water purification is done by **ultraviolet rays**, option (B).

Final answer. Water is purified using ultraviolet rays, so the answer is option **(B)**.

← *Back to Q3*

Q4

Welders wear special glass goggles or face masks with glass windows to protect their eyes from:

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Microwaves

[2026 • Set 55-1-3]

SOLUTION

A welding arc is extremely bright and emits large amounts of **ultraviolet (UV) radiation** along with intense visible and infrared light.

UV rays are the main hazard to a welder's eyes. They can cause a painful condition called photokeratitis, commonly known as 'welder's flash' or 'arc eye', and can also burn the skin.

To block this radiation, welders wear special goggles or face masks fitted with dark UV-filtering glass windows. This matches option (B).

Option (A) infrared is also emitted and is partly filtered too, but the primary eye hazard guarded against is UV. Option (C) X-rays are not produced by a welding arc. Option (D) microwaves are not emitted by welding either.

Hence welders protect their eyes mainly from **ultraviolet rays**, option (B).

Final answer. Welders shield their eyes from ultraviolet rays, so the answer is option **(B)**.

← Back to Q4

Q5

An electromagnetic wave is propagating along x-axis. At any instant, the phase difference (in radian) between the electric field ($\vec{E}$) and the magnetic field ($\vec{B}$) associated with the wave is:

(A) zero

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) π

[2026 • Set 55-2-1]

SOLUTION

In a plane electromagnetic wave travelling through space, the electric field $\vec{E}$ and the magnetic field $\vec{B}$ oscillate together. They reach their maximum value at the same instant and become zero at the same instant.

For a wave moving along the x-axis, the fields can be written as:

$$\vec{E} = E_0 \hat{j} \sin(kx - \omega t)$$

$$\vec{B} = B_0 \hat{k} \sin(kx - \omega t)$$

Both fields carry the same phase term $(kx - \omega t)$, so the phase difference between them is:

$$\Delta\phi = 0$$

This matches option (A).

The fields are perpendicular to each other in direction ($\vec{E}$ along $\hat{j}$, $\vec{B}$ along $\hat{k}$), and both are perpendicular to the direction of propagation, but this is a difference in direction, not in phase. So options (B), (C) and (D) are wrong.

Hence the phase difference between $\vec{E}$ and $\vec{B}$ is **zero**, option (A).

COMMON MISTAKE

Do not confuse 'perpendicular in direction' with 'out of phase'. E and B are at 90 ° in space but oscillate in step, so their phase difference is 0.

Final answer. The electric and magnetic fields are in phase, so the phase difference is zero, option **(A)**.

← Back to Q5

Q6

The expressions of magnetic fields associated with four electromagnetic waves are given below:

I. $\mathbf{B_1 = (4\times 10^{-7}\ T)\sin[0{\cdot}7\times 10^{3}\ x + 1{\cdot}4\times 10^{11}\ t]}$

II. $\mathbf{B_2 = (2\times 10^{-7}\ T)\sin[0{\cdot}6\times 10^{3}\ x + 1{\cdot}5\times 10^{11}\ t]}$

III. $\mathbf{B_3 = (3\times 10^{-6}\ T)\sin[0{\cdot}5\times 10^{3}\ x + 1{\cdot}5\times 10^{11}\ t]}$

IV. $\mathbf{B_4 = (5\times 10^{-7}\ T)\sin[0{\cdot}2\times 10^{4}\ x + 4{\cdot}8\times 10^{11}\ t]}$

Which wave is travelling in free space?

(A) I

(B) II

(C) **III**

(D) IV

[2026 • Set 55-2-2]

SOLUTION

A wave travels in free space only when its phase speed equals the speed of light $\mathbf{c = 3\times 10^{8}\ m\ s^{-1}}$.

For a magnetic field of the form $\mathbf{B = B_0\sin(kx + \omega\, t)}$, the phase speed is the ratio of the angular frequency (coefficient of t) to the wavenumber (coefficient of x):

$$v = \frac{\omega}{k}$$

We compute v for each of the four waves and compare with c.

Wave I: $\mathbf{\omega = 1.4\times 10^{11}}$, $\mathbf{k = 0.7\times 10^{3} = 700}$.

$$v_1 = \frac{1.4 \times 10^{11}}{700} = 2.0 \times 10^8 \; m\,s^{-1}$$

Wave II: $\omega = 1.5 \times 10^{11}$, $k = 0.6 \times 10^3 = 600$.

$$v_2 = \frac{1.5 \times 10^{11}}{600} = 2.5 \times 10^8 \; m\,s^{-1}$$

Wave III: $\omega = 1.5 \times 10^{11}$, $k = 0.5 \times 10^3 = 500$.

$$v_3 = \frac{1.5 \times 10^{11}}{500} = 3.0 \times 10^8 \; m\,s^{-1}$$

Wave IV: $\omega = 4.8 \times 10^{11}$, $k = 0.2 \times 10^4 = 2000$.

$$v_4 = \frac{4.8 \times 10^{11}}{2000} = 2.4 \times 10^8 \; m\,s^{-1}$$

Only Wave III gives $v = 3.0 \times 10^8 \; m\,s^{-1} = c$, so it is the one travelling in free space.

Waves I, II and IV all have speeds less than c, so they are travelling in media with refractive index $n = c/v$ greater than 1, not in free space.

Hence the wave travelling in free space is **Wave III**, option (C).

TIP

Always read off ω as the coefficient of t and k as the coefficient of x, then test v = ω/k against c = 3×10^8 m/s.

Final answer. Only Wave III has v = omega/k = 3.0 x 10^8 m/s = c, so it travels in free space, option **(C)**.

← *Back to Q6*

Q7

The electric field and the magnetic field components of an electromagnetic wave travelling in vacuum are represented as $E = m \sin(\alpha z - \beta t)$ *and* $B = n \sin(\alpha z - \beta t)$,

where m, n, α and β are constants characterising the wave. Then:

(A) $\frac{m}{n} = \frac{\alpha}{\beta}$

(B) $\frac{m}{n} = \frac{\beta}{\alpha}$

(C) $\frac{n}{m} = \frac{\beta}{\alpha}$

(D) $\frac{n}{m} = \alpha\beta$

[2026 • Set 55-2-3]

SOLUTION

For a plane electromagnetic wave travelling in vacuum, the amplitudes of the electric and magnetic fields are linked to the speed of light by a fixed relation.

$$\frac{E_0}{B_0} = c$$

The speed of the wave can also be written from its phase term as the ratio of angular frequency to wavenumber:

$$c = \frac{\omega}{k}$$

In the given expressions $E = m \sin(\alpha z - \beta t)$ and $B = n \sin(\alpha z - \beta t)$, the amplitudes are $E_0 = m$ and $B_0 = n$, while $\beta = \omega$ (coefficient of t) and $\alpha = k$ (coefficient of z).

Substituting these into $\frac{E_0}{B_0} = \frac{\omega}{k}$ gives:

$$\frac{m}{n} = \frac{\beta}{\alpha}$$

This matches option (B).

Option (A) inverts the frequency-to-wavenumber ratio, while options (C) and (D) wrongly invert the amplitude ratio or multiply α and β, which is dimensionally and physically incorrect.

Hence $\frac{m}{n} = \frac{\beta}{\alpha}$, option (B).

Final answer. Using $E_0/B_0 = c = \omega/k$ gives $m/n = \beta/\alpha$, so the answer is option **(B)**.

← Back to Q7

Q8

A plane electromagnetic wave travels through a medium and the magnetic field associated with it is given by $B = 5\times 10^{-8}\sin(3\times 10^{10}\,t - 150\,x)\ T$, *where x is in metres and t is in seconds. The velocity of the wave is:*

(A) $2{\cdot}0 \times 10^{8}\ m\ s^{-1}$

(B) $4{\cdot}5 \times 10^{7}\ m\ s^{-1}$

(C) $3{\cdot}5 \times 10^{7}\ m\ s^{-1}$

(D) $2{\cdot}5 \times 10^{8}\ m\ s^{-1}$

[2026 • Set 55-3-1]

SOLUTION

A plane wave written as $B = B_0\sin(\omega t - kx)$ has angular frequency ω as the coefficient of t and wavenumber k as the coefficient of x. Its phase speed is:

$$v = \frac{\omega}{k}$$

Comparing with the given field $B = 5\times 10^{-8}\sin(3\times 10^{10}\,t - 150\,x)\ T$:

$$\omega = 3\times 10^{10}\ \text{rad s}^{-1}$$

$$k = 150\ \text{m}^{-1}$$

Substituting these values:

$$v = \frac{3\times 10^{10}}{150}$$

$$v = 2.0\times 10^{8}\ m\ s^{-1}$$

This matches option (A). Since this speed is less than $c = 3\times 10^{8}\ m\ s^{-1}$, the wave is travelling in a medium of refractive index $n = \frac{c}{v} = \frac{3\times 10^{8}}{2\times 10^{8}} = 1.5$.

Hence the velocity of the wave is $2.0\times 10^{8}\ m\ s^{-1}$, option (A).

Final answer. $v = \omega/k = 2.0 \times 10^8$ m s^{-1}, so the answer is option **(A)**.

← *Back to Q8*

Q9

The ratio of amplitude of electric field to the amplitude of the magnetic field associated with an electromagnetic wave propagating in glass **($n = 1 \cdot 5$)** *is:*

(A) 3×10^8 m s^{-1}

(B) $\mathbf{2 \times 10^8}$ **m** $\mathbf{s^{-1}}$

(C) $3 \cdot 3 \times 10^{-9}$ m s^{-1}

(D) 5×10^{-9} m s^{-1}

[2026 • Set 55-4-1]

SOLUTION

For an electromagnetic wave, the ratio of the electric field amplitude to the magnetic field amplitude equals the speed of the wave in that medium.

$$\frac{E_0}{B_0} = v$$

Inside a medium of refractive index n, the speed of the wave is:

$$v = \frac{c}{n}$$

For glass, $n = 1.5$ and $c = 3 \times 10^8$ m s^{-1}. Substituting:

$$\frac{E_0}{B_0} = \frac{c}{n} = \frac{3 \times 10^8}{1.5}$$

$$\frac{E_0}{B_0} = 2 \times 10^8 \; m \; s^{-1}$$

This matches option (B). Note that the ratio carries units of speed (m s^{-1}), so the very small values in options (C) and (D) cannot be correct. Option (A) 3×10^8 m s^{-1} is the ratio in vacuum, not in glass.

Hence the ratio of field amplitudes in glass is $\mathbf{2 \times 10^8}$ $\boldsymbol{m}$ $\mathbf{s^{-1}}$, option (B).

CONCEPT

The ratio E_0/B_0 is a speed, not a pure number. In vacuum it equals c; inside a medium it drops to c/n.

Final answer. $E_0/B_0 = c/n = 2 \times 10^8$ m s^{-1} in glass, so the answer is option **(B)**.

← *Back to Q9*

Q10

The magnetic field in a plane electromagnetic wave travelling in glass **($n = 1.5$)** *is given by*

$\mathbf{B_y = (2 \times 10^{-7}\ T)\ \sin(\alpha\, x + 1.5 \times 10^{11}\, t)}$

where x is in metres and t is in seconds. The value of α *is:*

(A) 0.5×10^3 m^{-1}

(B) 6.0×10^2 m^{-1}

(C) $\mathbf{7.5 \times 10^2\ m^{-1}}$

(D) 1.5×10^3 m^{-1}

[2026 • Set 55-5-1]

SOLUTION

The coefficient of x in the wave expression is the wavenumber α, and the coefficient of t is the angular frequency ω. They are linked to the wave speed by:

$$v = \frac{\omega}{\alpha} \Rightarrow \alpha = \frac{\omega}{v}$$

The wave travels in glass of refractive index $n = 1.5$, so its speed is:

$$v = \frac{c}{n} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8\ m\ s^{-1}$$

From the given field $\mathbf{B_y = (2 \times 10^{-7}\ T)\ \sin(\alpha\, x + 1.5 \times 10^{11}\, t)}$, the angular frequency is:

$$\omega = 1.5 \times 10^{11}\ \text{rad s}^{-1}$$

Substituting into $\alpha = \frac{\omega}{v}$:

$$\alpha = \frac{1.5 \times 10^{11}}{2 \times 10^{8}}$$

$$\alpha = 750\ \text{m}^{-1}$$

$$\alpha = 7.5 \times 10^{2}\ \text{m}^{-1}$$

This matches option (C).

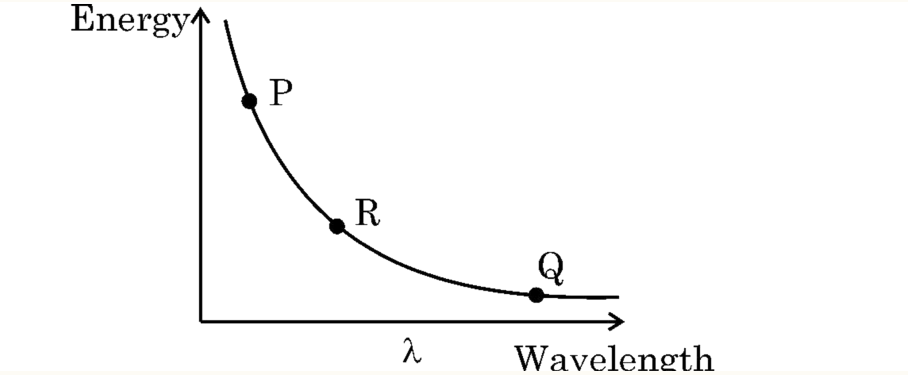
Hence the value of α is $\mathbf{7.5 \times 10^{2}\ m^{-1}}$, option (C).

Final answer. $\alpha = \omega/v = 7.5 \times 10^{2}\ \text{m}^{-1}$, so the answer is option **(C)**.

← *Back to Q10*

Q11

The given diagram exhibits the relationship between the wavelength of the electromagnetic waves and the energy of photon associated with them. The three points P, Q and R marked on the diagram may correspond respectively to:



(A) X-rays, microwaves, UV radiation

(B) X-rays, UV radiation, microwaves

(C) UV radiation, microwaves, X-rays

(D) Microwaves, UV radiation, X-rays

[2025 • Set 55-1-1]

SOLUTION

The energy carried by a photon of an electromagnetic wave is related to its wavelength by $E = h\nu = \frac{hc}{\lambda}$.

Here h is Planck's constant, c is the speed of light and λ is the wavelength, so photon energy is inversely proportional to wavelength: $E \propto \frac{1}{\lambda}$.

A point with high energy must therefore have a short wavelength, and a point with low energy must have a long wavelength. On the E versus λ graph the three points run from high energy (short λ) at P to low energy (long λ) at R.

Comparing the three radiations: X-rays have $\lambda \sim 10^{-10}\ m$ (highest energy), UV radiation has $\lambda \sim 10^{-7}\ m$ (medium energy), and microwaves have $\lambda \sim 10^{-2}\ m$ (lowest energy).

So the point of highest energy and shortest wavelength, P, is **X-rays**; the middle point Q is **UV radiation**; and the point of lowest energy and longest wavelength, R, is **microwaves**.

Option A and option D place microwaves or UV at the high-energy end, which is wrong; option C reverses the order completely. Only option B gives X-rays, UV radiation, microwaves in the correct order.

TIP

On any energy-versus-wavelength plot, always anchor your reasoning to $E \propto 1/\lambda$ first, then place the radiations by their known wavelength order.

Final answer. Correct option is B (P: X-rays, Q: UV radiation, R: microwaves).

← *Back to Q11*

Q12

Which one of the following correctly represents the change in wave characteristics (all

in vacuum) from microwaves to X-rays in electromagnetic spectrum?

(A) Speed: Remains same; Wavelength: Decreases; Frequency: Remains same

(B) **Speed: Remains same; Wavelength: Decreases; Frequency: Increases**

(C) Speed: Increases; Wavelength: Increases; Frequency: Decreases

(D) Speed: Remains same; Wavelength: Increases; Frequency: Remains same

[2025 • Set 55-1-2]

SOLUTION

All electromagnetic waves, whatever their frequency, travel through vacuum with the same constant speed $c = 3 \times 10^8\ m\ s^{-1}$, given by $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

Therefore, in moving from microwaves to X-rays the **speed remains the same** and equal to c.

The frequencies of these radiations are: microwaves ~ 10^{10} Hz and X-rays ~ 10^{18} Hz, so going from microwaves to X-rays the **frequency increases**.

Since speed is fixed, wavelength and frequency are related by $\lambda = \frac{c}{\nu}$, so $\lambda \propto \frac{1}{\nu}$.

As the frequency increases, the **wavelength decreases**.

Thus the correct combination is: speed remains the same, wavelength decreases, frequency increases. Options A and D wrongly keep the frequency unchanged, and option C wrongly makes the speed change, so they are all ruled out.

Final answer. Correct option is B (speed same, wavelength decreases, frequency increases).

← *Back to Q12*

Q13

X-rays are more harmful to human beings than ultraviolet radiations because X-rays:

(A) have frequency lower than that of ultraviolet radiations.

(B) have wavelength smaller than that of ultraviolet radiations.

(C) move faster than ultraviolet radiations in air.

(D) are mechanical waves but ultraviolet radiations are electromagnetic waves.

[2025 • Set 55-1-3]

SOLUTION

The biological damage caused by an electromagnetic wave depends on the energy of its photons, given by $E = h\nu = \frac{hc}{\lambda}$.

From this relation, a smaller wavelength means a higher frequency and hence a larger photon energy: $E \propto \frac{1}{\lambda}$.

X-rays have wavelength $\lambda \sim 10^{-10}\ m$ giving photon energy of the order of 10 keV, whereas UV radiation has $\lambda \sim 100$ nm giving photon energy of only about 10 eV.

Because X-ray photons carry far more energy, they penetrate deeper into the body and ionise biological molecules much more readily, which is why X-rays are more harmful than UV radiation.

Hence the reason is that **X-rays have a smaller wavelength** (and therefore higher energy) than UV radiation.

Option A is wrong because X-rays have a higher, not lower, frequency than UV; option C is wrong because all EM waves move at the same speed c in air or vacuum; option D is wrong because X-rays are electromagnetic waves, not mechanical waves.

Final answer. Correct option is B (X-rays have a smaller wavelength than UV radiation).

← *Back to Q13*

Q14

Assertion (A): *Out of Infrared and radio waves, the radio waves show more diffraction effect.*
Reason (R): *Radio waves have greater frequency than infrared waves.*

(C) Assertion (A) is true but Reason (R) is false.

[2025 • Set 55-2-1]

SOLUTION

Diffraction of a wave is significant only when the size of the obstacle or aperture is comparable to the wavelength; a longer wavelength produces a more pronounced diffraction effect.

Radio waves have wavelengths of the order of metres, while infrared waves have wavelengths of the order of $10^{-6}\ m$. Since radio waves have a much longer wavelength, they diffract more easily than infrared waves.

Therefore the Assertion, that radio waves show more diffraction than infrared waves, is **TRUE**.

Now examine the Reason. Wavelength and frequency are related by $\nu = \frac{c}{\lambda}$, so a longer wavelength means a lower frequency.

Radio waves (λ large) actually have a **lower** frequency than infrared waves, which is the opposite of what the Reason claims. Hence the Reason, that radio waves have greater frequency than infrared waves, is **FALSE**.

So Assertion (A) is true but Reason (R) is false, which corresponds to option C. Options A and B fail because R is false, and option D fails because A is true.

Final answer. Correct option is C (Assertion is true, Reason is false).

← *Back to Q14*

Q15

The amplitude of electric field in an electromagnetic wave in free space is 1000 $V\,m^{-1}$.

The amplitude of the magnetic field in this electromagnetic wave is :

(A) $3.0 \times 10^{-6}\ T$

(B) $\mathbf{3.33 \times 10^{-6}\ T}$

(C) $3.0 \times 10^{-11}\ T$

(D) $3.33 \times 10^{-11}\ T$

[2025 • Set 55-4-1]

SOLUTION

In an electromagnetic wave travelling in free space, the amplitudes of the electric and magnetic fields are related by $\mathbf{E_0 = c\, B_0}$, where c is the speed of light.

Rearranging for the magnetic field amplitude: $\mathbf{B_0 = \dfrac{E_0}{c}}$.

Here $\mathbf{E_0 = 1000\ V\,m^{-1}}$ and $\mathbf{c = 3 \times 10^8\ m\,s^{-1}}$.

Substituting the values: $\mathbf{B_0 = \dfrac{1000}{3 \times 10^8}\ T}$.

Calculating: $\mathbf{B_0 = 3.33 \times 10^{-6}\ T}$.

Thus the amplitude of the magnetic field is $\mathbf{3.33 \times 10^{-6}\ T}$, which is option B.

Final answer. Correct option is B ($\mathbf{B_0 = 3.33 \times 10^{-6}\ T}$).

← *Back to Q15*

Q16

The electric field in space between the plates of a parallel plate capacitor (each of area $2.5 \times 10^{-3}\ m^2$*) is changing at the rate of* $4 \times 10^6\ V\,m^{-1}s^{-1}$*. The displacement current between the plates of the capacitor is :*

(A) $1.8 \times 10^{-5}\ A$

(B) $3.47 \times 10^{-8}\ A$

(C) $\mathbf{8.85 \times 10^{-8}\ A}$

(D) $6.32 \times 10^{-4}\ A$

[2025 • Set 55-4-2]

SOLUTION

The displacement current arises from a changing electric field between the capacitor plates and is given by $\mathbf{I_d = \varepsilon_0 A \frac{dE}{dt}}$.

Here $\varepsilon_0 = 8.85 \times 10^{-12}\ F\,m^{-1}$, $A = 2.5 \times 10^{-3}\ m^2$, and $\frac{dE}{dt} = 4 \times 10^6\ V\,m^{-1}s^{-1}$.

Substituting the values: $I_d = (8.85 \times 10^{-12})(2.5 \times 10^{-3})(4 \times 10^6)\ A$.

Multiplying the coefficients: $8.85 \times 2.5 \times 4 = 88.5$.

Adding the powers of ten: $10^{-12} \times 10^{-3} \times 10^6 = 10^{-9}$.

Hence $I_d = 88.5 \times 10^{-9}\ A = 8.85 \times 10^{-8}\ A$.

Thus the displacement current between the plates is 8.85×10^{-8} A, which is option C.

Final answer. Correct option is C ($I_d = 8.85 \times 10^{-8}\ A$).

← *Back to Q16*

Q17

In an electromagnetic wave travelling in free space, the amplitude of magnetic field is 6.0×10^{-4} *T. The amplitude of its electric field is :*

(A) $2 \times 10^{-4}\ V\,m^{-1}$

(B) $1.5 \times 10^{12}\ V\,m^{-1}$

(C) $\mathbf{1.8 \times 10^5\ V\,m^{-1}}$

(D) $0.3 \times 10^4\ V\,m^{-1}$

[2025 • Set 55-4-3]

SOLUTION

For an electromagnetic wave in free space, the amplitudes of the electric and magnetic fields are linked by the relation $\frac{E_0}{B_0} = c$, so $E_0 = c\,B_0$.

Here $B_0 = 6.0 \times 10^{-4}\ T$ and $c = 3 \times 10^8\ m\,s^{-1}$.

Substituting the values: $E_0 = (3 \times 10^8)(6.0 \times 10^{-4})\ V\,m^{-1}$.

Multiplying: $3 \times 6.0 = 18$ and $10^{8} \times 10^{-4} = 10^{4}$.

Therefore $E_0 = 18 \times 10^{4}\ V\,m^{-1} = 1.8 \times 10^{5}\ V\,m^{-1}$.

Thus the amplitude of the electric field is $1.8 \times 10^{5}\ V\,m^{-1}$, which is option C.

Final answer. Correct option is C ($E_0 = 1.8 \times 10^{5}\ V\,m^{-1}$).

← *Back to Q17*

Q18

Which of the following rays coming from the Sun plays an important role in maintaining the Earth's warmth?

(A) Infrared rays

(B) γ rays

(C) UV rays

(D) Visible light rays

[2025 • Set 55-5-1]

SOLUTION

The Sun emits a broad band of electromagnetic radiation, but a large part of the thermal energy reaching the Earth lies in the infrared region of the spectrum.

Infrared rays are absorbed by the Earth's surface and by greenhouse gases such as water vapour (H_2O), carbon dioxide (CO_2) and methane (CH_4) in the atmosphere.

These gases trap the infrared radiation re-emitted by the warm surface of the Earth, producing the greenhouse effect that keeps the planet warm.

Hence **infrared rays** play the key role in maintaining the Earth's warmth.

Among the other options: γ rays from the Sun are blocked high in the atmosphere and carry negligible thermal share; UV rays form only a small fraction of the solar output and are largely absorbed by the ozone layer; visible light contributes to illumination but the dominant heating effect is due to infrared.

Final answer. Correct option is A (infrared rays).

← *Back to Q18*

Q19

The dimensions of (με), *where* ε *is permittivity and* μ *is permeability of a medium, are:*

(A) $[M^0\,L^{-1}\,T]$

(B) $[M^0\,L^{-2}\,T^2]$

(C) $[M^{-1}\,L^{-2}\,T^{-1}]$

(D) $[M^1\,L^1\,T^{-1}]$

[2025 • Set 55-5-1]

SOLUTION

The speed of an electromagnetic wave in a medium is $v = \frac{1}{\sqrt{\mu\varepsilon}}$, where μ is the permeability and ε is the permittivity.

Squaring both sides gives $v^2 = \frac{1}{\mu\varepsilon}$, so $\mu\varepsilon = \frac{1}{v^2}$.

The product με therefore has the dimensions of $\frac{1}{v^2}$.

Speed has dimensions $[v] = [L\,T^{-1}]$, so $[v^2] = [L^2\,T^{-2}]$.

Hence $[\mu\varepsilon] = \frac{1}{[L^2\,T^{-2}]} = [L^{-2}\,T^2]$.

Writing this with the mass dimension shown explicitly, $[\mu\varepsilon] = [M^0\,L^{-2}\,T^2]$, which is option B.

> **TIP**
>
> Deriving dimensions from $c = 1/\sqrt{\mu\varepsilon}$ is far safer than recalling the separate dimensions of μ and ε in the exam.

Final answer. Correct option is B ($[M^0\,L^{-2}\,T^2]$).

← Back to Q19

Q20

Assertion (A): *EM waves do not require a medium for their propagation.*

Reason (R): *EM waves are transverse waves.*

Select the correct answer:

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

[2025 • Set 55-7-1]

SOLUTION

Consider the Assertion first. Electromagnetic waves consist of mutually perpendicular oscillating electric and magnetic fields that sustain each other through the coupling $\vec{E} \leftrightarrow \vec{B}$ described by Maxwell's equations.

A changing electric field produces a magnetic field and a changing magnetic field produces an electric field, so the wave regenerates itself as it travels and needs no material medium. Hence the Assertion, that EM waves do not require a medium, is **TRUE**.

Now consider the Reason. In an EM wave the fields $\vec{E}$ and $\vec{B}$ both oscillate perpendicular to the direction of propagation, so the wave is indeed transverse. Hence the Reason is also **TRUE**.

However, being transverse is a property concerning the orientation of the fields; it is not the cause of the wave being able to travel without a medium. The real reason is the self-sustaining coupling between $\vec{E}$ and $\vec{B}$ given by Maxwell's equations.

Therefore both A and R are true, but R is **not** the correct explanation of A, which corresponds to option B. Option A is rejected because R does not explain A, and options C and D are rejected because both statements are in fact true.

Final answer. Correct option is B (both A and R true, but R is not the correct explanation of A).

← Back to Q20

Q21

In the four regions, I, II, III and IV, the electric fields are described as:

Region I: $\mathbf{E_y = E_0 \sin(kz - \omega t)}$

Region II: $\mathbf{E_y = E_0}$

Region III: $\mathbf{E_y = E_0 \sin kz}$

Region IV: $\mathbf{E_y = E_0 \cos kz}$

The displacement current will exist in the region:

(A) I

(B) IV

(C) II

(D) III

[2024 • Set 55-1-1]

SOLUTION

The displacement current is defined as $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$, where Φ_E is the electric flux. It exists only in a region where the electric field changes with time, that is, where $\frac{\partial E}{\partial t} \neq 0$.

We test each region in turn.

Region I: $\mathbf{E_y = E_0 \sin(kz - \omega t)}$. This contains the time variable t, so $\mathbf{\frac{\partial E_y}{\partial t} = -E_0 \omega \cos(kz - \omega t) \neq 0}$. A displacement current exists here.

Region II: $\mathbf{E_y = E_0}$. This is a constant field, so $\mathbf{\frac{\partial E_y}{\partial t} = 0}$. No displacement current.

Region III: $\mathbf{E_y = E_0 \sin kz}$. This depends only on position z, not on t, so $\mathbf{\frac{\partial E_y}{\partial t} = 0}$. No displacement current.

Region IV: $\mathbf{E_y = E_0 \cos kz}$. Again only space-dependent, so $\mathbf{\frac{\partial E_y}{\partial t} = 0}$. No displacement current.

Only Region I has a time-varying electric field, so the displacement current exists only there. The correct option is **(A) Region I**.

> **TIP**
>
> Displacement current depends on the time rate of change of flux. A field that varies only with position (like sin kz) gives zero displacement current; the variable t must appear.

Final answer. The displacement current exists only in Region I, so the answer is option **(A)**.

← *Back to Q21*

Q22

The electric and magnetic fields of electromagnetic waves are:

(A) In the same phase and perpendicular to each other.

(B) In the same phase and not perpendicular to each other.

(C) Not in the same phase but are perpendicular to each other.

(D) Neither in the same phase nor perpendicular to each other.

[2024 • Set 55-1-2]

SOLUTION

In a plane electromagnetic wave travelling through vacuum, the electric and magnetic fields can be written as $\vec{E} = \mathbf{E_0 sin(kz - \omega\, t)}\,\hat{x}$ and $\vec{B} = \mathbf{B_0 sin(kz - \omega\, t)}\,\hat{y}$.

Both fields carry the same sine argument $(kz - \omega t)$, so they reach their maximum, zero, and minimum values at the same instant and at the same point. This means $\vec{E}$ and $\vec{B}$ are **in the same phase**.

The field directions $\hat{x}$ (for $\vec{E}$) and $\hat{y}$ (for $\vec{B}$) are at right angles, and both are perpendicular to the direction of propagation $\hat{z}$. Hence $\vec{E}$ and $\vec{B}$ are **mutually perpendicular**.

Picture the wave moving along z: the $\vec{E}$ vector oscillates up and down along y, the $\vec{B}$ vector oscillates in and out along x, and both crest together.

Therefore the correct description is in the same phase and perpendicular to each other, option (A). Options (B), (C) and (D) are ruled out because they wrongly deny either the in-phase nature or the perpendicularity, both of which are required features of an EM wave.

Final answer. The fields are in phase and mutually perpendicular, so the answer is option **(A)**.

← *Back to Q22*

Q23

The first scientist who produced and observed electromagnetic waves of wavelengths in the range 25 mm – 5 mm *was:*

(A) J.C. Maxwell

(B) H.R. Hertz

(C) **J.C. Bose**

(D) G. Marconi

[2024 • Set 55-1-3]

SOLUTION

The wavelength range given, 25 mm down to 5 mm, lies in the short microwave / millimetre-wave region of the electromagnetic spectrum.

Jagadish Chandra Bose (J.C. Bose) was the first scientist to produce and detect electromagnetic waves of such short, millimetre-scale wavelengths. Around 1895 he demonstrated the generation, transmission and reception of these waves using compact transmitters and his own novel detectors.

The other names do not fit this specific range. J.C. Maxwell only predicted electromagnetic waves theoretically and did not produce them. H.R. Hertz first generated and detected EM waves, but of much longer, metre-scale wavelengths. G. Marconi worked on long-distance radio communication using long-wavelength radio waves.

Hence the correct answer is **J.C. Bose**, option (C).

Final answer. The millimetre-range waves were first produced by J.C. Bose, so the answer is option **(C)**.

← Back to Q23

Q24

Electromagnetic waves with wavelength 10 nm are called:

(A) Infrared waves

(B) Ultraviolet rays

(C) Gamma rays

(D) X-rays

[2024 • Set 55-2-1]

SOLUTION

We identify the part of the electromagnetic spectrum corresponding to the given wavelength.

$\lambda = 10\ \text{nm} = 10 \times 10^{-9}\ m = 10^{-8}\ m$.

The X-ray region of the spectrum spans wavelengths from about $10^{-8}\ m$ down to about $10^{-11}\ m$. A wavelength of $10^{-8}\ m$ sits at the start of this X-ray band.

For comparison, ultraviolet rays have longer wavelengths (about 10^{-7} to 410^{-7} m), infrared is longer still, and gamma rays are much shorter (below $10^{-11}\ m$). So those options are ruled out.

Therefore electromagnetic waves of wavelength 10 nm are classified as **X-rays**, option (D).

Final answer. Waves of wavelength 10 nm are X-rays, so the answer is option **(D)**.

← Back to Q24

Q25

Welders wear special glass goggles or face masks with glass windows to protect their eyes from radiations produced by welding arcs. These radiations are:

(A) X-rays

(B) Ultraviolet rays

(C) Infrared waves

(D) Gamma rays

[2024 • Set 55-2-2]

SOLUTION

A welding arc reaches very high temperatures and emits radiation across a broad band, including a large amount of ultraviolet (UV) light along with visible and infrared.

The ultraviolet component is the dangerous part for the eyes. Strong UV exposure damages the cornea and causes a painful condition called welder's flash or photokeratitis, and it can also burn exposed skin.

To protect against this, welders wear masks or goggles fitted with special glass windows that absorb and block ultraviolet radiation.

The other options do not fit. Welding arcs do not produce significant X-rays or gamma rays, which need very high-energy processes such as those in atoms or nuclei. Infrared from the arc gives heat but is not the radiation the UV-blocking glass is specifically designed to stop.

Hence the protective glass guards mainly against **ultraviolet rays**, option (B).

Final answer. The harmful welding-arc radiation is ultraviolet, so the answer is option **(B)**.

← *Back to Q25*

Q26

In four regions I, II, III and IV, the magnetic field is given by:

I. $\mathbf{B_y = B_0 \sin kz}$

II. $\mathbf{B_y = B_0 \cos kz}$

III. $\mathbf{B_y = B_0 \sin(kz - \omega t)}$

IV. $\mathbf{B_y = B_0 \sin kz + B_0 \cos kz}$

The electromagnetic wave will exist in the region:

(A) IV

(B) I

(C) **III**

(D) II

[2024 • Set 55-2-3]

SOLUTION

A genuine electromagnetic wave is a disturbance that travels through space, so its field must depend on both position and time together, in the travelling-wave form $B_0 \sin(kz - \omega t)$ (or with cosine). A field that depends only on position is a static spatial pattern, not a propagating wave.

We examine each region.

Region I: $\mathbf{B_y = B_0 \sin kz}$. This depends only on z, not on t, so it is stationary in time and is not a travelling wave.

Region II: $\mathbf{B_y = B_0 \cos kz}$. Again only position-dependent, so not a wave.

Region III: $\mathbf{B_y = B_0 \sin(kz - \omega t)}$. This has the combination $(kz - \omega t)$, which represents a wave travelling in the $+z$ direction with speed $v = \frac{\omega}{k}$. This is a true electromagnetic wave.

Region IV: $\mathbf{B_y = B_0 \sin kz + B_0 \cos kz}$. Both terms depend only on z, so this is still time-independent and not a travelling wave.

Only Region III has a field of the travelling-wave form, so the electromagnetic wave exists there. The correct option is **(C) Region III**.

Final answer. Only the travelling-wave field of Region III is an EM wave, so the answer is option **(C)**.

← *Back to Q26*

Q27

Which one of the following has the highest frequency?

(A) Infrared rays

(B) Gamma rays

(C) Radio waves

(D) Microwaves

[2024 • Set 55-3-1]

SOLUTION

We use the standard ordering of the electromagnetic spectrum by increasing frequency.

Radio waves < Microwaves < Infrared < Visible < Ultraviolet < X-rays < Gamma rays.

Among the four given options, the order is Radio waves < Microwaves < Infrared rays < Gamma rays.

Gamma rays sit at the extreme high-frequency end, with frequencies above about 310^{19} Hz, which is far greater than that of infrared, microwaves or radio waves.

Therefore the radiation with the highest frequency is **gamma rays**, option (B).

CONCEPT

Frequency rises as wavelength falls. Remember the order from radio (lowest) up to gamma (highest); the highest-frequency option is always the shortest-wavelength one.

Final answer. Gamma rays have the highest frequency, so the answer is option **(B)**.

← Back to Q27

Q28

Electromagnetic waves with frequency 1.0×10^{18} *Hz are known as:*

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Gamma rays

[2024 • Set 55-3-2]

SOLUTION

We locate the given frequency on the electromagnetic spectrum.

$\boldsymbol{\nu = 1.010^{18}\ \text{Hz}}$.

The X-ray region spans frequencies from about 310^{16} Hz to about 310^{19} Hz. The given value 10^{18} Hz lies well inside this range.

We can confirm using the photon energy $\boldsymbol{E = h\nu}$, with $\boldsymbol{h = 6.6310^{-34}\ J\,s}$.

$\boldsymbol{E = (6.6310^{-34})(1.010^{18}) = 6.6310^{-16}\ J}$.

Converting to electron-volts, $\boldsymbol{E = \dfrac{6.6310^{-16}}{1.610^{-19}} \approx 4.110^{3}\ \text{eV} \approx 4.1\ \text{keV}}$, which is a typical soft X-ray energy.

Ultraviolet has lower frequency, infrared lower still, and gamma rays much higher, so those options are ruled out.

Hence electromagnetic waves of frequency 1.010^{18} Hz are **X-rays**, option (C).

Final answer. Waves of frequency $\boldsymbol{1.010^{18}}$ Hz are X-rays, so the answer is option **(C)**.

← Back to Q28

Q29

A plane electromagnetic wave is travelling in air in +x direction. At a particular moment, its electric field $\vec{E}$ is along +y direction. At that moment, the magnetic field $\vec{B}$ is along:

(A) $+z$ direction and in phase with $\vec{E}$

(B) $-z$ direction and in phase with $\vec{E}$

(C) $+z$ direction and out of phase with $\vec{E}$

(D) $-z$ direction and out of phase with $\vec{E}$

[2024 • Set 55-3-3]

SOLUTION

For a plane electromagnetic wave, the direction of propagation is along $\hat{E}\times\hat{B}$. So $\hat{E}$, $\hat{B}$ and the propagation direction $\hat{n}$ form a right-handed set, with $\hat{\boldsymbol{n}}=\hat{\boldsymbol{E}}\times\hat{\boldsymbol{B}}$.

Here the wave travels in the $+x$ direction, so $\hat{\boldsymbol{n}}=\hat{\boldsymbol{x}}$, and $\vec{E}$ is along $+y$, so $\hat{\boldsymbol{E}}=\hat{\boldsymbol{y}}$.

We require $\hat{\boldsymbol{E}}\times\hat{\boldsymbol{B}}=\hat{\boldsymbol{n}}$, that is $\hat{\boldsymbol{y}}\times\hat{\boldsymbol{B}}=\hat{\boldsymbol{x}}$.

Using $\hat{\boldsymbol{y}}\times\hat{\boldsymbol{z}}=\hat{\boldsymbol{x}}$, we get $\hat{\boldsymbol{B}}=\hat{\boldsymbol{z}}$.

So at that moment $\vec{B}$ points along the $+z$ direction.

Also, in a plane electromagnetic wave $\vec{E}$ and $\vec{B}$ oscillate in phase, reaching their maxima and zeros together.

Hence $\vec{B}$ is along the $+z$ direction and in phase with $\vec{E}$, option (A). Options with $-z$ are wrong because the cross product fixes the sign as $+z$, and options stating out of phase are wrong because $\vec{E}$ and $\vec{B}$ are always in phase.

> **COMMON MISTAKE**
>
> *Fix the field directions with the right-hand rule from $\hat{\boldsymbol{E}}\times\hat{\boldsymbol{B}}=\hat{\boldsymbol{n}}$. Getting the order wrong (using $\hat{B}\times\hat{E}$) flips the sign and gives $-z$ by mistake.*

Final answer. $\vec{\boldsymbol{B}}$ is along $+\boldsymbol{z}$ and in phase with $\vec{\boldsymbol{E}}$, so the answer is option **(A)**.

← *Back to Q29*

Q30

The phase difference between electric field $\vec{E}$ and magnetic field $\vec{B}$ in an electromagnetic wave propagating along z-axis is

(A) zero

(B) π

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4}$

[2024 • Set 55-4-1]

SOLUTION

Consider a plane electromagnetic wave propagating along the z-axis. Its fields are written as $\vec{E} = \mathbf{E_0 sin(kz - \omega\, t)}\,\hat{x}$ and $\vec{B} = \mathbf{B_0 sin(kz - \omega\, t)}\,\hat{y}$.

Both fields have the identical phase argument (kz−ω t). So as the wave advances, $\vec{E}$ and $\vec{B}$ reach their maximum values together, become zero together, and reverse direction together.

The phase difference between them is therefore

$\mathbf{\Delta\phi = (kz - \omega\, t) - (kz - \omega\, t) = 0}$.

This means the electric and magnetic fields oscillate exactly in step; they are in phase. (They are still mutually perpendicular and both perpendicular to $\hat{z}$, but that is a spatial relation, not a phase relation.)

Hence the phase difference between $\vec{E}$ and $\vec{B}$ is **zero**, option (A).

Final answer. The phase difference between $\vec{E}$ and $\vec{B}$ is zero, so the answer is option **(A)**.

← *Back to Q30*

Q31

The electromagnetic waves used to purify water are:

(A) Infrared rays

(B) Ultraviolet rays

(C) X-rays

(D) Gamma rays

[2024 • Set 55-5-1]

SOLUTION

Water purifiers disinfect water using the germicidal action of the ultraviolet (UV) band of the electromagnetic spectrum.

The lamp emits short-wavelength UV-C radiation, typically around $\lambda \approx 254$ **nm**, which is strongly absorbed by the nucleic acids of micro-organisms.

This radiation disrupts the DNA and RNA of bacteria, viruses and protozoa, so they can no longer reproduce and are effectively inactivated.

Ruling out the others: infrared (A) only heats the water, X-rays (C) and gamma rays (D) are far too energetic and dangerous for domestic use, and neither is the standard germicidal source.

Hence the electromagnetic waves used to purify water are **ultraviolet rays**, option (B).

Final answer. The waves used to purify water are ultraviolet rays, option **(B)**.

← *Back to Q31*

Q32

In the process of charging of a capacitor, the current produced between the plates of

the capacitor is (where symbols have their usual meanings):

(A) $\mu_0 \dfrac{d\phi_E}{dt}$

(B) $\dfrac{1}{\mu_0} \dfrac{d\phi_E}{dt}$

(C) $\boldsymbol{\varepsilon_0 \dfrac{d\phi_E}{dt}}$

(D) $\dfrac{1}{\varepsilon_0} \dfrac{d\phi_E}{dt}$

[2023 • Set 55-1-1]

SOLUTION

Between the plates of a charging capacitor there are no free charges, so no conduction current can flow there. To keep Ampere's circuital law consistent, Maxwell introduced the displacement current.

The displacement current is produced by the changing electric flux ϕ_E between the plates and is given by

$$I_d = \varepsilon_0 \frac{d\phi_E}{dt}$$

where ε_0 is the permittivity of free space and $\dfrac{d\phi_E}{dt}$ is the rate of change of electric flux.

The constant in front must be ε_0, not μ_0 or their reciprocals, so options (A), (B) and (D) are dimensionally and physically wrong.

Hence the current between the plates is $\boldsymbol{I_d = \varepsilon_0 \dfrac{d\phi_E}{dt}}$, **option (C)**.

CONCEPT

Remember that the displacement current between the plates exactly equals the conduction current in the connecting wires, which keeps the total current continuous.

Final answer. The current between the plates is $\boldsymbol{\varepsilon_0 \dfrac{d\phi_E}{dt}}$, option **(C)**.

← *Back to Q32*

Q33

$\vec{E}$ and $\vec{B}$ represent the electric and the magnetic field of an electromagnetic wave respectively. The direction of propagation of the wave is along

(A) $\vec{B}$

(B) $\vec{E}$

(C) $\vec{E}\times\vec{B}$

(D) $\vec{B}\times\vec{E}$

[2023 • Set 55-2-1]

SOLUTION

In a plane electromagnetic wave the electric field $\vec{E}$, the magnetic field $\vec{B}$, and the direction of propagation are mutually perpendicular, forming a right-handed set.

The direction of energy flow, and hence of propagation, is given by the Poynting vector

$$\vec{S} = \frac{1}{\mu_0}\,\vec{E}\times\vec{B}$$

so the wave travels along $\vec{E}\times\vec{B}$, which is perpendicular to both fields.

Options (A) and (B) are wrong because the wave does not travel along $\vec{E}$ or $\vec{B}$ themselves, and (D) gives $\vec{B}\times\vec{E}$, which points opposite to the actual direction of propagation.

Hence the direction of propagation is along $\vec{E}\times\vec{B}$, **option (C)**.

Final answer. The wave propagates along $\vec{E}\times\vec{B}$, option **(C)**.

← *Back to Q33*

Q34

The ratio of the magnitudes of the electric field and magnetic field of a plane

electromagnetic wave is

(A) 1

(B) $\frac{1}{c}$

(C) c

(D) c^2

[2023 • Set 55-2-2]

SOLUTION

For a plane electromagnetic wave travelling in vacuum, the magnitudes of the electric and magnetic fields are related at every instant by

$$E = c\,B$$

where c is the speed of light in vacuum.

This follows from Maxwell's equations, which give the field amplitudes the relation $\frac{E_0}{B_0} = c$.

Therefore the ratio of the magnitudes is

$$\frac{|E|}{|B|} = c$$

Since $c \approx 3 10^8\ m\ s^{-1}$, this ratio is large; options (A), (B) and (D) give the wrong value.

Hence the ratio of the electric to magnetic field magnitudes is c, **option (C)**.

Final answer. The ratio $|E|/|B| = c$, option **(C)**.

← *Back to Q34*

Q35

An electromagnetic wave is produced by a charge

(A) moving with a constant velocity

(B) moving with a constant speed parallel to a magnetic field

(C) **moving with an acceleration**

(D) at rest

[2023 • Set 55-2-3]

SOLUTION

An electromagnetic wave is radiated only when a charge is accelerated, that is, when its velocity changes in magnitude or direction (for example an oscillating charge).

A charge at rest (D) produces only a static electric field and radiates no energy.

A charge moving with constant velocity (A), or with constant speed parallel to a magnetic field (B), produces steady electric and magnetic fields but does not radiate, because there is no acceleration.

An accelerating charge sets up time-varying electric and magnetic fields that regenerate each other and propagate outward as an electromagnetic wave.

Hence an electromagnetic wave is produced by a charge **moving with an acceleration**, option (C).

Final answer. An EM wave is produced by an accelerating charge, option **(C)**.

← *Back to Q35*

Q36

The electromagnetic radiations used to kill germs in water purifiers are called:

(A) Infrared waves

(B) X-rays

(C) Gamma rays

(D) Ultraviolet rays

[2023 • Set 55-3-1]

SOLUTION

The radiation used to kill germs in water purifiers belongs to the ultraviolet (UV) part of the electromagnetic spectrum.

Short-wavelength UV-C light, around $\lambda \approx 254$ **nm**, is strongly germicidal.

It damages the DNA of bacteria, viruses and other pathogens, so they cannot multiply and are destroyed.

Among the options, infrared (A) only heats, while X-rays (B) and gamma rays (C) are far too penetrating and hazardous for domestic purifiers, leaving UV as the standard sterilising radiation.

Hence the radiation used to kill germs is **ultraviolet rays**, option (D).

Final answer. Germs are killed by ultraviolet rays, option **(D)**.

← *Back to Q36*

Q37

Electromagnetic waves of wavelength of the order of a few meters were first produced and detected in the laboratory by:

(A) J.C. Maxwell

(B) J.C. Bose

(C) H.R. Hertz

(D) G. Marconi

[2023 • Set 55-3-2]

SOLUTION

Maxwell predicted electromagnetic waves theoretically in 1865, but he did not produce them in the laboratory.

Electromagnetic waves with wavelengths of the order of a few metres were first generated and detected experimentally by Heinrich Rudolf Hertz in 1887.

He used a spark-gap oscillator as the source and a small loop with a gap as the detector, producing waves of frequency about 60 MHz, corresponding to wavelengths of a few metres.

J.C. Bose (B) later worked at much shorter, millimetre wavelengths, and G. Marconi (D) developed long-distance wireless communication, so neither was the first to produce these waves.

Hence the waves were first produced and detected by **H.R. Hertz**, option (C).

Final answer. These waves were first produced and detected by H.R. Hertz, option **(C)**.

← *Back to Q37*

Q38

A capacitor is charged by a battery. Between the plates, during process of charging:

(A) only displacement current exists.

(B) only conduction current exists.

(C) both displacement current and conduction current exist.

(D) no current exists.

[2023 • Set 55-3-3]

SOLUTION

While a capacitor is being charged, the region between its plates contains no free charges, so no conduction current can exist there.

However, the charge on the plates increases, so the electric field and hence the electric flux Φ_E between the plates change with time.

This changing flux gives rise to a displacement current

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

which equals the conduction current flowing in the connecting wires, keeping the current continuous in the circuit.

So between the plates only this displacement current exists; the conduction current is present in the wires, not in the gap, which rules out (B), (C) and (D).

Hence, between the plates during charging, **only displacement current exists**, option (A).

Final answer. Between the plates, only displacement current exists, option **(A)**.

← *Back to Q38*

Q39

The electromagnetic waves used in radar systems are:

(A) Infrared waves

(B) Ultraviolet rays

(C) **Microwaves**

(D) X-rays

[2023 • Set 55-4-1]

SOLUTION

Radar (Radio Detection and Ranging) systems use microwaves, the part of the spectrum lying between radio waves and infrared.

These typically have frequencies in the gigahertz range, about 1 to 100 GHz, with wavelengths from roughly 3 mm to 30 cm.

Microwaves are ideal for radar because their short wavelength lets them be focused into narrow, well-directed beams, they reflect strongly from metallic targets such as aircraft, and they travel through clouds and rain with little loss.

Infrared (A) and ultraviolet (B) are absorbed or scattered too easily, and X-rays (D) are unsuitable for ranging.

Hence the electromagnetic waves used in radar systems are **microwaves**, option (C).

Final answer. Radar systems use microwaves, option **(C)**.

← *Back to Q39*

Q40

Name the electromagnetic waves also known as 'heat waves'.

(A) Radio waves

(B) Microwaves

(C) X-rays

(D) Infrared waves

[2023 • Set 55-4-3]

SOLUTION

Electromagnetic waves known as 'heat waves' are infrared radiations.

Warm bodies at ordinary temperatures emit most of their thermal radiation in the infrared region of the spectrum.

By Wien's displacement law, a body near room temperature ($\approx 300\ K$) radiates most strongly at $\lambda_{max} \approx 10\ \mu m$, which lies in the infrared band.

Because infrared radiation is readily absorbed and produces a sensation of heat, it is commonly called heat radiation.

Radio waves (A) and microwaves (B) are of longer wavelength and X-rays (C) are highly energetic, so none of these is the heat band.

Hence the heat waves are **infrared waves**, option (D).

Final answer. The heat waves are infrared waves, option **(D)**.

← *Back to Q40*

Q41

Which one of the following electromagnetic radiation has the least wavelength?

(A) Gamma rays

(B) Microwaves

(C) Visible light

(D) X-rays

[2023 • Set 55-5-1]

SOLUTION

Wavelength and frequency are related by $c = \nu\lambda$, so $\lambda = \frac{c}{\nu}$.

The radiation with the highest frequency has the smallest wavelength.

In the electromagnetic spectrum, the order of frequency (highest to lowest) is: gamma rays > X-rays > ultraviolet > visible > infrared > microwaves > radio waves.

Among the given options, gamma rays have the highest frequency, and therefore the least wavelength (typically $\lambda < 10^{-12}$ m).

Microwaves, visible light and X-rays all have longer wavelengths than gamma rays, so options (B), (C) and (D) are ruled out.

Hence the correct answer is **gamma rays**, option (A).

Final answer. The least wavelength is for gamma rays, option **(A)**.

← *Back to Q41*

Q42

Choose the correct option related to wavelengths (λ) of different parts of

electromagnetic spectrum.

(A) $\lambda_{X\text{–rays}} < \lambda_{\text{micro waves}} < \lambda_{\text{visible}}$

(B) $\lambda_{\text{visible}} > \lambda_{X\text{–rays}} > \lambda_{\text{radio waves}}$

(C) $\lambda_{\text{radio waves}} > \lambda_{\text{visible}} > \lambda_{X\text{–rays}}$

(D) $\lambda_{\text{visible}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{X\text{–rays}}$

[2023 • Set 55-5-2]

SOLUTION

Wavelength is inversely related to frequency through $\lambda = \frac{c}{\nu}$, so the part of the spectrum with the lowest frequency has the longest wavelength.

Arranging the spectrum in order of decreasing wavelength (longest to shortest): radio waves > microwaves > infrared > visible > ultraviolet > X-rays > gamma rays.

From this order, radio waves have a longer wavelength than visible light, and visible light has a longer wavelength than X-rays.

Therefore $\lambda_{\text{radio waves}} > \lambda_{\text{visible}} > \lambda_{X\text{–rays}}$.

Options (A), (B) and (D) place these in the wrong order and are incorrect.

Hence the correct answer is **option (C)**.

Final answer. Correct order is $\lambda_{\text{radio}} > \lambda_{\text{visible}} > \lambda_{X\text{–rays}}$, option **(C)**.

← *Back to Q42*

Q43

What oscillates in an electromagnetic wave of frequency 10 MHz*?*

[2021]

SOLUTION

An electromagnetic wave consists of mutually perpendicular electric and magnetic fields that vary with time and space.

In such a wave, what oscillates are the electric field $\vec{E}$ and the magnetic field $\vec{B}$.

Both fields oscillate sinusoidally and are perpendicular to each other and to the direction of propagation, so the wave is transverse.

For a wave travelling along the z-axis, the fields may be written as $\vec{E}(z,t) = \mathbf{E_0 \sin(kz - \omega t)}\,\hat{x}$ and $\vec{B}(z,t) = \mathbf{B_0 \sin(kz - \omega t)}\,\hat{y}$.

Here both $\vec{E}$ and $\vec{B}$ oscillate at the same frequency as the wave, namely $\nu = 10$ **MHz** $= 10^7$ **Hz**, with $\omega = 2\pi\nu = 2\pi \times 10^7$ **rad/s**.

No material particles move during propagation; only the fields oscillate.

Hence **the electric field $\vec{E}$ and the magnetic field $\vec{B}$ oscillate at 10 MHz**.

> **COMMON MISTAKE**
>
> *A common slip is to say 'charges' or 'particles' oscillate. In an EM wave it is the fields, not matter, that oscillate.*

Final answer. The electric field $\vec{E}$ and magnetic field $\vec{B}$ oscillate, both at **10 MHz**.

← *Back to Q43*

Q44

Name the electromagnetic radiation used for killing germs in water purifiers.

[2021]

SOLUTION

Water purifiers kill germs using radiation that damages the genetic material of microorganisms.

The electromagnetic radiation used for this purpose is **ultraviolet (UV) radiation**, specifically UV-C of wavelength about 254 nm.

UV-C photons disrupt the DNA and RNA of bacteria, viruses and protozoa, so the microorganisms can no longer reproduce and are effectively destroyed.

In practice, a low-pressure mercury lamp emitting near 254 nm is used, and the water flows past the lamp for a short time to be sterilized.

Hence the radiation used for killing germs in water purifiers is **ultraviolet (UV) rays**.

Final answer. Ultraviolet (UV-C) radiation is used for killing germs in water purifiers.

← *Back to Q44*

Q45

Write the mathematical form of Ampere-Maxwell circuital law.

[2020 • Set 55-1-1]

SOLUTION

The Ampere-Maxwell circuital law generalises Ampere's law by including the displacement current, so that it holds even for time-varying electric fields.

Its mathematical (integral) form is:

$$\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \, (I_c + I_d)$$

$$\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \, I_c + \mu_0 \varepsilon_0 \, \frac{\mathrm{d}\Phi_E}{\mathrm{dt}}$$

Here I_c is the conduction current, $\mathbf{I_d = \varepsilon_0 \, \frac{d\Phi_E}{dt}}$ is the displacement current, and Φ_E is the electric flux through any surface bounded by the closed loop.

μ_0 is the permeability and ε_0 the permittivity of free space.

The displacement-current term, added by Maxwell, makes the law consistent for situations such as a charging capacitor and predicts electromagnetic waves travelling at $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$.

Final answer. The law is $\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \, I_c + \mu_0 \varepsilon_0 \, \frac{\mathrm{d}\Phi_E}{\mathrm{dt}}$.

← *Back to Q45*

Q46

How is displacement current produced between the plates of a parallel plate capacitor during charging?

[2020 • Set 55-1-2]

SOLUTION

When a parallel plate capacitor is charged, free charge accumulates on its plates, and the electric field $\vec{E}$ in the gap between the plates grows with time.

As the charge increases, the electric flux $\Phi_E = E \cdot A$ through any surface bounded by the gap also increases with time.

This changing electric flux is equivalent to a current, called the displacement current, given by:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

No actual charge crosses the gap between the plates; the displacement current arises purely from the time-varying electric field there.

This displacement current in the gap equals the conduction current in the connecting wires, so the current is continuous throughout the circuit.

Hence, **the changing electric flux between the plates during charging produces the displacement current** $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$.

Final answer. The changing electric flux between the plates produces $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$.

← *Back to Q46*

Q47

Differentiate between conduction current and displacement current.

[2020 • Set 55-1-3]

SOLUTION

Both conduction current and displacement current appear together in the Ampere-Maxwell law, but they arise from completely different causes.

Conduction current (I_c): It is due to the actual flow of free charges (such as electrons) through a conductor.

Its current density is $\vec{j} = ne\vec{v}_d$, where n is the number density of carriers and $\vec{v}_d$ is their drift velocity.

It requires a material medium containing mobile charges and obeys Ohm's law in a conductor.

Displacement current (I_d): It arises from a time-varying electric flux, not from any flow of charge.

Its magnitude is $I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$.

It can exist even in vacuum or in an insulator (for example, between the plates of a charging capacitor), where there is no actual movement of charge.

Both currents produce a magnetic field and both appear in the generalised Ampere-Maxwell law $\oint \vec{B} \cdot d\vec{l} = \mu_0(I_c + I_d)$.

Final answer. Conduction current is due to actual charge flow ($\vec{j}=ne\vec{v}_d$), while displacement current is due to changing electric flux ($I_d=\varepsilon_0 \frac{d\Phi_E}{dt}$).

← Back to Q47

Q48

Displacement current exists only when

(A) electric field is changing.

(B) magnetic field is changing.

(C) electric field is not changing.

(D) magnetic field is not changing.

[2020 • Set 55-2-1]

SOLUTION

The displacement current is defined through the rate of change of electric flux:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Since the electric flux Φ_E depends on the electric field, I_d is non-zero only when the electric field (and hence the electric flux) is changing with time.

If the electric field is constant, then $\frac{d\Phi_E}{dt} = 0$, so $I_d = 0$, which rules out options (C) and (D).

A changing magnetic field gives rise to an induced electric field (Faraday's law), not to a displacement current directly, so option (B) is incorrect.

Hence the displacement current exists only when **the electric field is changing**, option (A).

Final answer. Displacement current exists only when the electric field is changing, option **(A)**.

← *Back to Q48*

Q49

Electromagnetic waves used as a diagnostic tool in medicine are

(A) X-rays.

(B) ultraviolet rays.

(C) infrared radiation.

(D) ultrasonic waves.

[2020 • Set 55-2-1]

SOLUTION

A diagnostic tool in medicine must be an electromagnetic wave that can penetrate the body and reveal internal structures.

X-rays are used for this purpose, in radiography, fluoroscopy and CT scanning.

The image is formed because different body tissues absorb X-rays to different extents: dense tissue like bone absorbs strongly, while soft tissue absorbs

weakly, giving good contrast.

Ultrasonic waves (option D) are mechanical sound waves, not electromagnetic waves, so they are excluded by the question.

Ultraviolet rays (B) and infrared radiation (C) cannot penetrate deep into the body and are not used for such imaging.

Hence the correct answer is **X-rays**, option (A).

Final answer. X-rays are the EM diagnostic tool, option **(A)**.

← *Back to Q49*

Q50

A welder wears special glasses to protect his eyes mostly from the harmful effect of

(A) very intense visible light.

(B) infrared radiation.

(C) **ultraviolet rays.**

(D) microwaves.

[2020 • Set 55-2-2]

SOLUTION

An electric welding arc emits radiation across a broad range, including visible, infrared and ultraviolet.

The most harmful component for the eyes is the **ultraviolet (UV) radiation**.

Intense UV exposure causes 'welder's flash' (photokeratitis), a painful inflammation of the cornea, and repeated exposure can lead to cataracts.

A welder's special glasses are made of glass that strongly absorbs UV (and also IR), while transmitting just enough visible light to see the work, so very intense visible light (A) and microwaves (D) are not the main concern.

Infrared (B) mainly produces heating and is far less damaging to the eye than UV.

Hence the glasses chiefly protect against **ultraviolet rays**, option (C).

Final answer. Welder's glasses mainly protect against ultraviolet rays, option **(C)**.

← *Back to Q50*

Q51

In a travelling plane electromagnetic wave, which of the following have zero average value?

(A) Magnetic energy and electric energy

(B) Magnetic field and electric field

(C) Magnetic energy and magnetic field

(D) Electric energy and electric field

[2020 • Set 55-2-3]

SOLUTION

In a travelling plane electromagnetic wave, the electric field and the magnetic field both vary sinusoidally with position and time.

The electric field is $E = E_0 \sin(kx - \omega t)$ and the magnetic field is $B = B_0 \sin(kx - \omega t)$.

Over one complete cycle a sine function takes equal positive and negative values, so the average of both $\vec{E}$ and $\vec{B}$ over a full period is zero.

Now consider the energy densities. The electric energy density is $u_E = \frac{1}{2}\varepsilon_0 E^2$ and the magnetic energy density is $u_B = \frac{B^2}{2\mu_0}$.

Because both E^2 and B^2 are squared quantities, they are never negative. Their average over a cycle is positive and equal to $u_E = \frac{1}{4}\varepsilon_0 {E_0}^2$ and $u_B = \frac{{B_0}^2}{4\mu_0}$, which are non-zero.

Hence the quantities with zero average value are the magnetic field and the electric field, while the energy densities have non-zero averages.

This rules out options (A), (C) and (D), each of which wrongly includes an energy term that has a non-zero average. So the correct choice is **option (B)**.

Final answer. The correct answer is option **(B)**: magnetic field and electric field.

← *Back to Q51*

Q52

Mention the contribution of Indian physicist J.C. Bose in the production of electromagnetic waves.

[2020 • Set 55-3-1]

SOLUTION

Around the year 1895, the Indian physicist J.C. Bose (Jagadish Chandra Bose) successfully produced and detected electromagnetic waves in the millimetre and centimetre wavelength range, which lies in the microwave region of the spectrum.

Using simple apparatus built by himself, he demonstrated that these waves showed the standard properties of light, namely reflection, refraction and polarisation.

His experiments were among the earliest practical confirmations of Maxwell's electromagnetic theory and were an important precursor to modern wireless communication.

Thus, **J.C. Bose produced and studied short-wavelength microwaves and verified their wave nature, laying the foundation for wireless technology.**

Final answer. J.C. Bose produced and detected microwaves (millimetre/centimetre EM waves) and demonstrated their reflection, refraction and polarisation.

← *Back to Q52*

Q53

Write one use of the electromagnetic waves of frequency range from 10^{16} Hz *to* 10^{20} Hz.

[2020 • Set 55-3-1]

SOLUTION

The frequency range 10^{16} Hz to 10^{20} Hz corresponds to the X-ray region of the electromagnetic spectrum (extending into gamma rays at the very high end).

X-rays have very short wavelengths and high penetrating power, and different body tissues absorb them by different amounts.

One important use of these waves is in medical diagnostic imaging, such as taking X-ray photographs of bones and the chest, and in CT scans, where the differential absorption reveals internal structure.

Thus, **electromagnetic waves of this frequency range (X-rays) are used in medical imaging to study bones and internal organs.**

Final answer. X-rays of this frequency range are used in medical diagnostic imaging (for example, X-rays of bones).

← *Back to Q53*

Q54

Which physical quantity associated with an electromagnetic wave does not change when it propagates from one medium into another medium?

[2020 • Set 55-3-2]

SOLUTION

When an electromagnetic wave passes from one medium into another, its frequency does not change because the frequency is fixed by the source that produces the wave.

In a medium of refractive index n, the speed of the wave becomes $v = \frac{c}{n}$, so the speed changes.

Since $v = \nu\lambda$, and the frequency ν stays constant while v changes, the wavelength $\lambda = \frac{v}{\nu}$ must also change.

Therefore, as the wave enters a denser medium both the speed and the wavelength decrease, but the frequency stays the same.

Hence, **the frequency of an electromagnetic wave remains unchanged when it propagates from one medium into another.**

> **COMMON MISTAKE**
>
> *A very common slip is to write that wavelength stays constant. It is the frequency that is fixed by the source; the wavelength changes because the speed changes.*

Final answer. The frequency of the electromagnetic wave does not change when it changes medium.

← *Back to Q54*

Q55

Name the electromagnetic radiations which may cause sunburn. Write their wavelength range.

[2020 • Set 55-3-3]

SOLUTION

Sunburn is caused by ultraviolet (UV) radiation present in sunlight, mainly the UV-B component.

The wavelength range of ultraviolet radiation is approximately $\lambda \approx$ **100 nm** to 400 nm.

Within this band, UV-A lies roughly between 315 nm and 400 nm, UV-B between 280 nm and 315 nm, and UV-C between 100 nm and 280 nm. UV-B reaches the outer skin layer and damages cells, producing the inflammation we call sunburn.

Hence, **the radiation causing sunburn is ultraviolet (UV) radiation, with wavelength range about 100 nm to 400 nm.**

Final answer. Ultraviolet (UV) radiation causes sunburn; its wavelength range is about 100 nm to 400 nm.

← *Back to Q55*

Q56

An A.C. source with variable frequency is connected to a parallel plate capacitor. How will the displacement current be affected with the decrease in frequency of the source?

[2020 • Set 55-4-1]

SOLUTION

The displacement current in a parallel plate capacitor is given by $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 A \frac{dE}{dt}}$, where Φ_E is the electric flux and A is the plate area.

For an AC source of frequency f, the voltage across the capacitor, and hence the electric field between the plates, varies as $\boldsymbol{E \propto \sin(2\pi f t)}$.

Differentiating, $\mathbf{\frac{dE}{dt} \propto 2\pi f \cos(2\pi f t)}$, so the peak rate of change of the field is proportional to f.

Therefore the displacement current is directly proportional to the frequency, $\mathbf{I_d \propto f}$.

Hence, when the frequency of the source decreases, the displacement current also **decreases** (in direct proportion to f).

Final answer. As the frequency decreases, the displacement current decreases, since $\mathbf{I_d \propto f}$.

← *Back to Q56*

Q57

Depict the fields diagram of an electromagnetic wave propagating along positive X-axis with its electric field along Y-axis.

[2020 • Set 55-5-1]

SOLUTION

For an electromagnetic wave, the electric field $\vec{E}$, the magnetic field $\vec{B}$ and the direction of propagation are mutually perpendicular, and the direction of propagation is along $\vec{E} \times \vec{B}$.

Here the wave travels along the positive X-axis and the electric field is along the Y-axis, so the propagation direction obeys $\hat{E} \times \hat{B} = \hat{x}$.

Substituting $\hat{E} = \hat{y}$, we get $\hat{y} \times \hat{B} = \hat{x}$, which gives $\hat{B} = \hat{z}$.

So the magnetic field oscillates along the positive Z-axis.

Diagram (described): Draw three axes from a common origin, the X-axis pointing right (direction of propagation), the Y-axis pointing up, and the Z-axis pointing out toward the reader. Show the electric field $\vec{E}$ as a sinusoidal curve oscillating in the X-Y plane along the Y-axis, labelled $\vec{E}$ (or E_y). Show the magnetic field $\vec{B}$ as a sinusoidal curve oscillating in the X-Z plane along the Z-axis, labelled $\vec{B}$ (or B_z). Both curves have the same wavelength and frequency and are in phase, that is, they reach their maxima and zeros at the same points along the X-axis. Mark the propagation direction with an arrow along $+X$, labelled c.

Hence, **$\vec{E}$ oscillates along $+Y$, $\vec{B}$ oscillates along $+Z$, and the wave moves along $+X$, with $\vec{E}$ and $\vec{B}$ in phase and mutually perpendicular.**

Final answer. The wave moves along $+X$ with $\vec{E}$ oscillating along $+Y$ and $\vec{B}$ oscillating along $+Z$, both in phase.

← *Back to Q57*

Q58

Which part of the electromagnetic spectrum is used in RADAR? Give its frequency range.

OR

How are electromagnetic waves produced by accelerating charges?

[2019 • Set 55-2-1]

SOLUTION

Part 1 (RADAR): RADAR uses the microwave part of the electromagnetic spectrum.

The frequency range of microwaves is about 10^9 Hz to 10^{11} Hz (roughly 1 GHz to 100 GHz), corresponding to wavelengths from about 1 mm to 30 cm.

Their short wavelength allows narrow, well-directed beams, and they reflect efficiently from metallic targets such as aircraft, which makes them ideal for RADAR.

Hence, **RADAR uses microwaves of frequency about 10^9 to 10^{11} Hz.**

OR (Production by accelerating charges): Any charge that is accelerated radiates electromagnetic waves.

Consider a charge oscillating with frequency f. As it accelerates back and forth, it produces a time-varying electric field $\vec{E}$ at points around it.

By Maxwell's equations, this changing electric field sets up a changing magnetic field $\vec{B}$, and the changing $\vec{B}$ in turn regenerates a changing $\vec{E}$.

These mutually sustaining fields propagate outward as an electromagnetic wave of the same frequency f, travelling at the speed of light c.

This is exactly how an antenna works: an alternating current of frequency f in the antenna makes charges oscillate, radiating EM waves of frequency f.

Final answer. RADAR uses microwaves, frequency about $\mathbf{10^9}$ Hz to $\mathbf{10^{11}}$ Hz (1 to 100 GHz).

← *Back to Q58*

Q59

The small ozone layer on top of the stratosphere is crucial for human survival. Why?

OR

Illustrate by giving suitable examples, how you can show that electromagnetic waves carry both energy and momentum.

[2019 • Set 55-4-1]

SOLUTION

Part 1 (Ozone layer): The ozone layer at the top of the stratosphere absorbs most of the harmful ultraviolet radiation (UV-B and UV-C) coming from the Sun.

Without this protective layer, intense UV radiation would reach the Earth's surface and cause serious harm. It damages DNA in living cells, leading to skin cancers and cataracts, kills phytoplankton in the oceans (disturbing food chains), and inhibits photosynthesis in plants.

Therefore, **the ozone layer is crucial for survival because it shields life on Earth from the Sun's harmful ultraviolet radiation.**

OR (Energy and momentum of EM waves): An electromagnetic wave carries energy. This is clear from everyday experience, since sunlight warms our skin and is used to heat water in solar heaters, showing that energy is transported by the wave.

An electromagnetic wave also carries momentum. A wave carrying energy E transports momentum $p = \frac{E}{c}$, where c is the speed of light. When the wave is absorbed by or reflected from a surface, it exerts a radiation pressure on that surface.

Examples of momentum transfer: a solar sail in space is pushed forward by the radiation pressure of sunlight, and the tail of a comet always points away from the Sun because solar radiation pushes the gas and dust outward.

Hence, **electromagnetic waves carry both energy (they heat objects) and momentum (they exert radiation pressure), with $p = E/c$.**

Final answer. The ozone layer absorbs the Sun's harmful UV radiation, protecting life on Earth.

← Back to Q59

Q60

Name the electromagnetic radiations used for (a) water purification, and (b) eye surgery.

[2018]

SOLUTION

(a) Water purification: Ultraviolet (UV) radiation is used to purify drinking water.

In particular, UV-C radiation of wavelength around 254 nm destroys bacteria, viruses and other micro-organisms in the water, making it safe to drink.

(b) Eye surgery: Laser radiation, which lies in the infrared or visible region, is used in eye surgery.

For example, lasers are used in LASIK surgery and in repairing the retina, because a laser beam can make very fine, precise cuts and seal tissue.

Hence, **(a) ultraviolet (UV) radiation is used for water purification, and (b) laser (infrared/visible) radiation is used for eye surgery.**

Final answer. (a) Ultraviolet (UV) radiation for water purification; (b) laser (infrared/visible) radiation for eye surgery.

← Back to Q60

Q61

How is the speed of em-waves in vacuum determined by the electric and magnetic fields?

[2017]

SOLUTION

From Maxwell's equations, an electromagnetic wave travelling in free space has its speed fixed entirely by the electric permittivity ε_0 and magnetic permeability μ_0 of vacuum.

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

Here μ_0 is the constant linking magnetic effects to current and ε_0 is the constant linking electric field to charge.

Equivalently, in any plane electromagnetic wave the speed equals the ratio of the field amplitudes:

$$c = \frac{E_0}{B_0}$$

Putting $\mu_0 = 4\pi \times 10^{-7}\ T\ m\ A^{-1}$ and $\varepsilon_0 = 8.85 \times 10^{-12}\ F\ m^{-1}$,

$$c = \frac{1}{\sqrt{(4\pi \times 10^{-7})(8.85 \times 10^{-12})}} \approx 3 \times 10^8\ m\ s^{-1}$$

So the speed of electromagnetic waves in vacuum is determined by the **electric and magnetic constants of free space**, and it comes out as $c = 3 \times 10^8\ m\ s^{-1}$.

Final answer. The speed is $c = 1/\sqrt{\mu_0 \varepsilon_0} = E_0/B_0 = 3 \times 10^8\ m\ s^{-1}$.

← *Back to Q61*

Q62

Write the expression for speed of electromagnetic waves in a medium of electrical permittivity ε *and magnetic permeability* μ.

[2017]

SOLUTION

In a linear medium of electrical permittivity ε and magnetic permeability μ, Maxwell's equations give a wave that travels with speed

$$v = \frac{1}{\sqrt{\mu \varepsilon}}$$

For free space we put $\mu = \mu_0$ and $\varepsilon = \varepsilon_0$, which gives the vacuum speed

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3\times 10^8\ m\ s^{-1}$$

Since a medium has $\mu > \mu_0$ and $\varepsilon > \varepsilon_0$, the speed v is always **less than** c, which is why light slows down inside a material.

Hence the required expression is $\mathbf{v = \frac{1}{\sqrt{\mu\varepsilon}}}$.

Final answer. The speed in the medium is $\boldsymbol{v = \frac{1}{\sqrt{\mu\varepsilon}}}$.

← *Back to Q62*

Q63

Do electromagnetic waves carry energy and momentum?

[2017]

SOLUTION

Yes, electromagnetic waves carry **both energy and momentum**.

The energy is carried by the oscillating electric and magnetic fields. The rate of flow of energy per unit area is given by the Poynting vector,

$$S = \frac{1}{\mu_0} E\times B$$

which points along the direction of propagation of the wave.

The wave also carries momentum. The momentum density (momentum per unit volume) is related to the energy flow by

$$p = \frac{S}{c^2}$$

When the wave is absorbed by a surface, it delivers energy (which heats the surface) and momentum (which produces a tiny force called **radiation pressure**).

Therefore electromagnetic waves transport energy as well as momentum.

Final answer. Yes, electromagnetic waves carry both energy and momentum.

← Back to Q63

Q64

Write the relation for the speed of electromagnetic waves in terms of the amplitudes of electric and magnetic fields.

[2017]

SOLUTION

In a plane electromagnetic wave travelling in vacuum, the amplitude of the electric field and the amplitude of the magnetic field are not independent. They are linked through the speed of light.

The required relation is

$$c = \frac{E_0}{B_0}$$

where E_0 is the amplitude of the electric field, B_0 is the amplitude of the magnetic field, and c is the speed of the wave.

This follows directly from Maxwell's equations (Faraday's law together with the Ampere-Maxwell law), which force the plane-wave solution to satisfy $\boldsymbol{E} = \mathbf{cB}$ at every point and at every instant.

Hence the speed in terms of the field amplitudes is $\mathbf{c = \frac{E_0}{B_0}}$.

Final answer. The relation is $\boldsymbol{c = \frac{E_0}{B_0}}$.

← Back to Q64

Q65

In which directions do the electric and magnetic field vectors oscillate in an

electromagnetic wave propagating along the x-axis?

[2017]

SOLUTION

In a plane electromagnetic wave, both the electric field and the magnetic field oscillate at right angles to the direction in which the wave moves. They are also at right angles to each other.

For a wave travelling along the $+x$-axis, the propagation direction is x, so the fields cannot have any component along x.

Taking the standard convention, the electric field oscillates along the y-axis:

$$E = \mathrm{E}_0\, y$$

and the magnetic field oscillates along the z-axis:

$$B = \mathrm{B}_0\, z$$

This arrangement makes $E \times B$ point along x, the direction of propagation, since $y \times z = x$.

So E oscillates along the y-direction and B along the z-direction (the pair may be swapped, but they are always **mutually perpendicular and perpendicular to the x-axis**).

COMMON MISTAKE

A common mistake is to give E or B a component along the propagation axis. In an EM wave both fields are strictly transverse, so neither has any x component when the wave travels along x.

Final answer. E oscillates along the y-axis and B along the z-axis (both perpendicular to the x-axis of propagation).

← Back to Q65

Q66

In which situation is there a displacement current but no conduction current?

[2016]

SOLUTION

A displacement current arises whenever the electric flux through a region is changing, even if no charges actually move there. By definition,

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

The classic situation is the region **between the plates of a charging (or discharging) capacitor** when the gap is vacuum or air.

In this gap no charge actually flows across, so the conduction current is zero:

$$I_c = 0$$

But while the capacitor is charging, the electric field between the plates keeps increasing, so the electric flux changes with time:

$$\frac{d\Phi_E}{dt} \neq 0 \Rightarrow I_d \neq 0$$

This is exactly what Maxwell's modified Ampere's law describes:

$$\oint B \cdot dl = \mu_0 (I_c + I_d)$$

So **in the gap of a charging capacitor** there is a displacement current but no conduction current.

Final answer. In the gap between the plates of a charging capacitor: $\mathbf{I_c = 0}$ but $\mathbf{I_d \neq 0}$.

← *Back to Q66*

Q67

To which part of the electromagnetic spectrum does a wave of frequency 5×10^{19} *Hz belong ?*

[2014]

SOLUTION

The part of the electromagnetic spectrum is identified by its frequency. The given frequency is

$$\nu = 5\times 10^{19}\ \text{Hz}$$

The approximate frequency ranges are: X-rays from about 3×10^{16} Hz to about 3×10^{19} Hz, and gamma rays above about 3×10^{19} Hz.

Since

$$5\times 10^{19}\ \text{Hz} > 3\times 10^{19}\ \text{Hz}$$

the frequency lies just above the X-ray limit, in the **gamma-ray region** of the spectrum.

Hence a wave of frequency 5×10^{19} Hz belongs to the **gamma rays** (it may also be described as very hard X-rays at the boundary).

Final answer. It belongs to the gamma-ray region of the electromagnetic spectrum.

← *Back to Q67*

Q68

A capacitor has been charged by a dc source. What are the magnitudes of conduction and displacement currents, when it is fully charged?

[2013]

SOLUTION

Once a capacitor has been fully charged by a DC source, it reaches a steady state: the charge on its plates no longer changes and no charge flows in the connecting wires.

The conduction current is the rate of flow of charge in the wire:

$$I_c = \frac{dq}{dt}$$

Since q is now constant,

$$I_c = 0$$

The displacement current depends on the changing electric flux between the plates:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

With the plate voltage steady, the electric field and hence the flux between the plates is also constant, so

$$\frac{d\Phi_E}{dt} = 0 \Rightarrow I_d = 0$$

Therefore, in the fully-charged steady state, **both the conduction current and the displacement current are zero**, and $\mathbf{I_c = I_d = 0}$.

Final answer. When fully charged, $\mathbf{I_c = 0}$ and $\mathbf{I_d = 0}$.

← *Back to Q68*

Q69

Name the part of electromagnetic spectrum whose wavelength lies in the range of 10^{-10} *m. Give its one use.*

[2012]

SOLUTION

The region of the spectrum is fixed by the wavelength. Here

$$\lambda = 10^{-10}\ m = 0.1\ \text{nm}$$

X-rays span wavelengths from about $10^{-8}\ m$ down to about $10^{-11}\ m$. Since $10^{-10}\ m$ lies inside this band, the wave belongs to the **X-ray** part of the electromagnetic spectrum.

One important use of X-rays is in **medical diagnosis**, such as taking radiographs of bones and the chest to detect fractures or disease.

Hence the wave of wavelength $10^{-10}\ m$ is an **X-ray**, used in medical imaging.

Final answer. It is an X-ray; one use is medical diagnosis (bone and chest radiography).

← *Back to Q69*

Q70

What are the directions of electric and magnetic field vectors relative to each other and relative to the direction of propagation of electromagnetic waves?

[2012]

SOLUTION

In a plane electromagnetic wave the three directions are fixed relative to one another.

The electric field vector E and the magnetic field vector B are **mutually perpendicular** to each other.

Both E and B are also **perpendicular to the direction of propagation** k of the wave, so the wave is transverse.

The three together form a right-handed set, with the direction of propagation given by

$$k \parallel E \times B$$

For example, for a wave moving along z one may write

$$E = \mathrm{E}_0\, x\, \sin(\mathrm{kz} - \omega\, t),\ B = \mathrm{B}_0\, y\, \sin(\mathrm{kz} - \omega\, t)$$

so that $x \times y = z$, the propagation direction. The fields oscillate **in phase**, each perpendicular to the other and to k.

Final answer. E and B are mutually perpendicular and both are perpendicular to the direction of propagation, with $k \parallel E \times B$.

← Back to Q70

Q71

Name the physical quantity which remains same for microwaves of wavelength 1 mm *and UV radiations of* 1600 $\circ A$ *in vacuum.*

[2012]

SOLUTION

Both microwaves and ultraviolet radiation are electromagnetic waves. A characteristic property of all electromagnetic waves is that they travel through free space (vacuum) with the same constant speed, independent of their wavelength or frequency.

This common speed is the speed of light in vacuum,

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \, m/s$$

The microwave of wavelength 1 mm and the UV radiation of wavelength 1600 $\circ A$ have different frequencies and different wavelengths, but both satisfy $c = \nu\lambda$ with the same value of c.

Therefore the physical quantity that remains the same for both is the **speed in vacuum**, $c = 3 \times 10^8 \, m/s$.

Final answer. The speed in vacuum (c = 3 x 10^8 m/s) is the same for both.

← Back to Q71

Q72

Name the electromagnetic waves, which (i) maintain the Earth's warmth and (ii) are used in aircraft navigation.

[2012]

SOLUTION

(i) The electromagnetic waves that maintain the Earth's warmth are **infrared (IR) radiations**.

Infrared radiation from the Sun is absorbed by the Earth's surface and by atmospheric gases such as water vapour and carbon dioxide. These gases re-emit infrared radiation and trap it near the surface (the greenhouse effect), keeping the Earth warm.

(ii) The electromagnetic waves used in aircraft navigation are **microwaves**.

Microwaves are used in RADAR systems. Because they travel in straight lines and are reflected by metallic objects, the reflected microwave signal from an aircraft gives its position and speed, which is used for air-traffic control and navigation.

Final answer. (i) Infrared radiation; (ii) Microwaves (RADAR).

← *Back to Q72*

Q73

A plane electromagnetic wave travels in vacuum along z-direction. What can you say about the direction of electric and magnetic field vectors?

[2011 • Set 55-1-1]

SOLUTION

An electromagnetic wave is a transverse wave. This means the electric field vector $\vec{E}$, the magnetic field vector $\vec{B}$, and the direction of propagation are all mutually perpendicular to one another.

Here the wave travels along the z-direction. Therefore both $\vec{E}$ and $\vec{B}$ must lie in a plane perpendicular to the z-axis, that is, in the xy-plane.

The two fields are also perpendicular to each other, and they oscillate in the same phase. If the electric field points along the x-axis, then the magnetic field points along the y-axis, since the propagation direction is along $\vec{E} \times \vec{B}$.

$$\vec{E} \parallel \hat{x}, \vec{B} \parallel \hat{y}, \vec{E} \times \vec{B} \parallel \hat{z}$$

Thus $\vec{E}$ and $\vec{B}$ are mutually perpendicular, and both are perpendicular to the direction of propagation (the z-axis).

Final answer. E and B are mutually perpendicular and both lie in the xy-plane, perpendicular to the z-direction of propagation.

← *Back to Q73*

Q74

Which part of electromagnetic spectrum has largest penetrating power?

[2010]

SOLUTION

The penetrating power of an electromagnetic wave increases with the energy of its photons, given by $\boldsymbol{E = h\nu}$. Higher the frequency ν, greater the photon energy, and hence greater the penetrating power.

Among all parts of the electromagnetic spectrum, **gamma rays** have the highest frequency (greater than about 3×10^{19} Hz) and therefore the highest photon energy.

The order of penetrating power, highest first, is:

Gamma rays > X-rays > Ultraviolet > Visible > Infrared > Microwaves > Radio waves

Hence gamma rays have the largest penetrating power. They can pass through thick metal sheets, and dense materials such as thick lead are needed to shield against them.

Final answer. Gamma rays have the largest penetrating power.

← *Back to Q74*

Q75

Write the following radiations in ascending order in respect of their frequencies: X-rays, microwaves, UV rays and radio waves.

[2009]

SOLUTION

In the electromagnetic spectrum, the frequency increases in the following general order:

Radio waves < Microwaves < Infrared < Visible < Ultraviolet < X-rays < Gamma rays

Taking the four given radiations and their approximate frequencies:

Radio waves ~ 10^6 Hz

Microwaves ~ 10^9 Hz

UV rays ~ 10^{15} to 10^{17} Hz

X-rays ~ 10^{17} to 10^{19} Hz

Arranging in ascending order of frequency:

Radio waves < Microwaves < UV rays < X-rays

Final answer. Radio waves < Microwaves < UV rays < X-rays.

← *Back to Q75*

Q76

Name the EM waves used for studying crystal structure of solids. What is its frequency range?

[2009]

SOLUTION

The electromagnetic waves used for studying the crystal structure of solids are **X-rays**.

X-rays have wavelengths of the order of 0.1 nm (~ 10^{-10} m), which is comparable to the spacing between atoms in a crystal lattice. Because of this, X-rays are diffracted by the regular arrangement of atoms in a crystal. This diffraction pattern is analysed (using Bragg's law) to determine the crystal structure, a technique called X-ray crystallography.

The frequency range of X-rays is approximately

$$\nu \sim 3 \times 10^{16} \text{ Hz to } 3 \times 10^{19} \text{ Hz}$$

corresponding to wavelengths from about 10^{-8} m to 10^{-11} m.

Final answer. X-rays; frequency range about 3 x 10^16 Hz to 3 x 10^19 Hz.

← *Back to Q76*

Q77

Name the part of the electromagnetic spectrum of wavelength 10^{-2} m *and mention its one application.*

[2008]

SOLUTION

The given wavelength is

$$\lambda = 10^{-2} \, m = 1 \text{ cm}$$

This wavelength lies in the **microwave** region of the electromagnetic spectrum, which extends roughly from 1 mm to about 30 cm.

One application of microwaves is in **RADAR** (radio detection and ranging), which is used for tracking and navigation of aircraft and ships, and for weather monitoring. Microwaves are also used in microwave ovens and in mobile and satellite communication.

Hence the radiation of wavelength 10^{-2} m belongs to the microwave region and is used in RADAR.

Final answer. Microwave region; used in RADAR.

← *Back to Q77*

Q78

Identify the part of the electromagnetic spectrum to which the following wavelengths belong: (i) 1 mm *(ii)* 10^{-11} m.

[2008]

SOLUTION

(i) Wavelength $\lambda = 1\ \text{mm} = 10^{-3}\ m$.

This wavelength lies in the **microwave** region of the electromagnetic spectrum (microwaves extend from about 1 mm to 30 cm). It lies near the boundary of the far-infrared region.

(ii) Wavelength $\lambda = 10^{-11}\ m = 10\ \text{pm}$.

This very small wavelength lies in the **gamma-ray** region (it may also be regarded as the very hard X-ray region), since gamma rays have wavelengths of the order of 10^{-11} m and shorter.

Hence: (i) microwave region, and (ii) gamma-ray region.

Final answer. (i) Microwave region; (ii) Gamma-ray region.

← *Back to Q78*

Q79

What is the name given to that part of electromagnetic spectrum which is used for taking photographs of earth under foggy conditions from great heights?

[2004]

SOLUTION

The part of the electromagnetic spectrum used for taking photographs of the Earth through fog and from great heights is **infrared (IR) radiation**.

Infrared radiation has a longer wavelength than visible light. Since the amount of scattering by small particles (such as fog and dust) varies as $\frac{1}{\lambda^4}$ (Rayleigh scattering), infrared radiation is scattered much less than visible light. As a result, infrared radiation can penetrate fog, haze and mist far better than ordinary light.

Using infrared-sensitive cameras and films, clear photographs of the Earth's surface can therefore be taken even under foggy conditions and from satellites at great heights.

CONCEPT

Remember the scattering link: less scattering (1/lambda^4) is exactly why the longer-wavelength IR cuts through fog, the same reason the daytime sky is blue.

Final answer. Infrared (IR) radiation.

← *Back to Q79*

Q80

Consider an induced magnetic field due to changing electric field and an induced electric field due to changing magnetic field. Which one is more easily observed? Justify your answer.

[2023 • Set 55-1-1]

SOLUTION

An **induced electric field produced by a changing magnetic field** is much more easily observed than an induced magnetic field produced by a changing electric field.

The induced electric field follows Faraday's law of electromagnetic induction:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

This effect is large and is the working principle of everyday devices such as the AC generator, the transformer and the induction stove. The induced EMF it produces is easily detected by ordinary instruments.

The induced magnetic field due to a changing electric field comes from the displacement-current term in the Ampere-Maxwell law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

This term is multiplied by the very small constant $\mu_0 \varepsilon_0$ (with $\boldsymbol{\varepsilon_0 \approx 8.85 \times 10^{-12}}$ $\mathbf{C^2N^{-1}m^{-2}}$), so the induced magnetic field is extremely weak and needs very sensitive instruments to detect.

Hence the induced electric field due to a changing magnetic field is the one that is more easily observed.

Final answer. The induced electric field due to a changing magnetic field is more easily observed, because the magnetic effect is weighted by the very small constant mu_0*epsilon_0.

← *Back to Q80*

Q81

A plane electromagnetic wave propagates in vacuum along x-axis.
(a) Give the direction of electric field and magnetic field vectors.
(b) E_0 *and* B_0 *are the magnitudes of electric and magnetic fields respectively in a plane electromagnetic wave. Find the ratio of energy densities of electric field and magnetic field.*

[2023 • Set 55-1-2]

SOLUTION

(a) An electromagnetic wave is a transverse wave in which the electric field $\vec{E}$, the magnetic field $\vec{B}$, and the direction of propagation are mutually perpendicular.

Since the wave propagates along the x-axis, both $\vec{E}$ and $\vec{B}$ lie in the y-z plane, perpendicular to $\hat{x}$ and to each other.

Taking $\vec{E}$ along the y-axis and $\vec{B}$ along the z-axis satisfies $\hat{y}\times\hat{z}=\hat{x}$, so that $\vec{E}\times\vec{B}$ points along the direction of propagation $\hat{x}$.

(b) The energy density of the electric field is $\mathbf{u_E=\frac{1}{2}\varepsilon_0 E_0^2}$.

The energy density of the magnetic field is $\mathbf{u_B=\frac{B_0^2}{2\mu_0}}$.

In an electromagnetic wave the magnitudes are related by $\mathbf{\frac{E_0}{B_0}=c\Rightarrow E_0=cB_0}$.

$$u_E=\frac{1}{2}\varepsilon_0 (cB_0)^2=\frac{1}{2}\varepsilon_0 c^2B_0^2$$

Using $\mathbf{c^2=\frac{1}{\mu_0\varepsilon_0}}$,

$$u_E=\frac{1}{2}\varepsilon_0\cdot\frac{1}{\mu_0\varepsilon_0}B_0^2=\frac{B_0^2}{2\mu_0}=u_B$$

Therefore $\mathbf{\frac{u_E}{u_B}=1}$. The **electric and magnetic energy densities are equal** in an electromagnetic wave.

Final answer. Direction: E along y-axis, B along z-axis; and the ratio of energy densities $\mathbf{u_E:u_B = 1}$.

← *Back to Q81*

Q82

Identify the part of electromagnetic spectrum which is:
(a) next to the lowest frequency end of the visible part of electromagnetic spectrum, and
(b) produced by bombarding a metal target by high energy electrons.
Give one use of each of them.

[2023 • Set 55-1-3]

SOLUTION

(a) The visible spectrum lies roughly between 400 nm (violet) and 700 nm (red). The lowest-frequency end of the visible region is the red end, which has the longest wavelength.

The part of the electromagnetic spectrum just beyond the red (lowest-frequency) end, having still longer wavelength and lower frequency, is the **infrared** region.

Use of infrared: in thermal imaging and remote sensing, or in TV and appliance remote controls (also for physiotherapy heating).

(b) When a metal target is bombarded by high-energy electrons, the electrons decelerate sharply and knock out inner-shell electrons of the target atoms. This produces **X-rays**.

Use of X-rays: in medical diagnosis to image bones and internal organs (radiography), or in detecting the structure of crystals through X-ray diffraction.

Final answer. (a) Infrared (used in remote controls / thermal imaging); (b) X-rays (used in medical imaging).

← *Back to Q82*

Q83

How are infrared waves produced? Why are these waves referred to as heat waves? Give any two uses of infrared waves.

[2023 • Set 55-2-1]

SOLUTION

Production of infrared waves: Infrared waves are produced by hot bodies and molecules. Any object whose temperature is above absolute zero has vibrating and rotating molecules, and these molecular vibrations and rotations radiate infrared waves.

Why they are called heat waves: When infrared waves fall on a body, they are readily absorbed, increasing the vibrational and kinetic energy of the molecules of the body. This raises the temperature of the body. Because they

are so effective at producing heating on absorption, infrared waves are called **heat waves**.

Two uses of infrared waves:

(1) In physiotherapy, to provide heat for treating muscular strains and joint pain.

(2) In remote controls of electronic appliances, and in night-vision devices and thermal imaging cameras.

Final answer. IR is produced by hot bodies and molecular vibrations; it is called heat waves because it heats bodies on absorption; uses include physiotherapy and remote controls.

← *Back to Q83*

Q84

How are X-rays produced? Give any two uses of these.

[2023 • Set 55-2-1]

SOLUTION

Production of X-rays: X-rays are produced when fast-moving electrons, accelerated through a high potential difference, strike a metal target of high atomic number such as tungsten.

On striking the target, the high-speed electrons are suddenly stopped (decelerated). The rapid deceleration of these electrons radiates high-frequency electromagnetic waves known as **X-rays**. (Inner-shell electron transitions in the target atoms also give characteristic X-rays.)

Two uses of X-rays:

(1) In medical diagnosis, to take radiographs of bones and to detect fractures and study internal organs.

(2) In the study of crystal structure through X-ray diffraction, and in detecting flaws and cracks in metals and in airport security checks.

Final answer. X-rays are produced when fast electrons strike a metal target; uses include medical radiography and crystal-structure analysis.

← *Back to Q84*

Q85

What is meant by the term 'displacement current'? Briefly explain how this current is different from a conduction current.

[2023 • Set 55-3-1]

SOLUTION

Displacement current: The displacement current is the current that arises in a region due to a time-varying electric field (changing electric flux). It is defined as

$$I_d=\varepsilon_0 \frac{d\Phi_E}{dt}$$

where Φ_E is the electric flux and ε_0 is the permittivity of free space. Although no charge actually flows, the changing electric flux produces a magnetic field just like an ordinary current.

Difference from conduction current:

Conduction current I_c is due to the actual flow of free charges (electrons) through a conductor. It exists only where there is a conducting medium carrying charges.

Displacement current I_d is due to a time-varying electric field and exists even in vacuum or in the gap of a capacitor, where no free charges flow.

Together, the total current I_c+I_d remains continuous throughout a circuit (for example, across the plates of a charging capacitor) and obeys the **Ampere-Maxwell law**.

COMMON MISTAKE

A common error is to say displacement current needs charges to flow; stress that it exists in vacuum/capacitor gap with no charge movement.

Final answer. Displacement current is $I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$ due to a changing electric flux, whereas conduction current is the actual flow of charges in a conductor.

← Back to Q85

Q86

(a) What is a displacement current? How is it different from a conduction current?

OR

(b) Write any two characteristics of an electromagnetic wave. Why are microwaves used in radar systems?

[2023 • Set 55-4-1]

SOLUTION

(a) Displacement current is the current produced by a time-varying electric field (changing electric flux), given by

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Difference: Conduction current I_c is due to the actual flow of free charges through a conductor, whereas displacement current I_d is due to a changing electric flux and exists even in vacuum (for example in the gap between capacitor plates) where no real charges move.

OR

(b) Two characteristics of an electromagnetic wave:

(1) Electromagnetic waves are transverse waves in which $\vec{E}$, $\vec{B}$ and the direction of propagation are mutually perpendicular, i.e. $\vec{E} \perp \vec{B} \perp$ propagation.

(2) They travel through vacuum with a constant speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8\ m/s$, the same for all wavelengths.

Microwaves in radar: Microwaves are used in radar because they travel in straight lines and are strongly reflected by metallic objects such as aircraft and ships. Their short wavelength allows sharp, narrow beams, making them ideal for detecting and ranging distant objects accurately.

Final answer. (a) Displacement current $\mathbf{I_d=\varepsilon_0\, d\Phi_E/dt}$ vs charge-flow conduction current; OR (b) EM waves are transverse and travel at speed c; microwaves suit radar as they go straight and reflect off metals.

← *Back to Q86*

Q87

Identify the electromagnetic wave whose wavelengths range is from about
(a) 10^{-12} *m to about* 10^{-8} *m.*
(b) 10^{-3} *m to about* 10^{-1} *m.*
Write one use of each.

[2023 • Set 55-5-1]

SOLUTION

(a) Wavelength range 10^{-12} *m* to 10^{-8} *m*:

This range (about 0.001 nm to 10 nm) corresponds to **X-rays**.

Use: in medical diagnosis to take radiographs of bones and to detect fractures, since X-rays are absorbed differently by bone and soft tissue.

(b) Wavelength range 10^{-3} *m* to 10^{-1} *m*:

This range (about 1 mm to 10 cm) corresponds to **microwaves**.

Use: in radar systems for aircraft navigation and speed detection, and in microwave ovens for cooking food.

Final answer. (a) X-rays (used in medical imaging); (b) Microwaves (used in radar / microwave ovens).

← *Back to Q87*

Q88

Identify the electromagnetic radiation and write its wavelength range, which is used

to kill germs in water purifier. Name the two sources of these radiations.

[2023 • Set 55-5-2]

SOLUTION

Identification: The radiation used to kill germs in water purifiers is **ultraviolet (UV)** radiation. UV light has enough energy to damage the DNA of micro-organisms and so destroys germs and bacteria in water.

Wavelength range: Ultraviolet radiation spans about λ = **100 nm** to 400 nm (that is 10^{-7} m to 410^{-7} m); the germicidal UV-C used in purifiers is around 254 nm.

Two sources of UV radiation:

(1) The Sun, which is the main natural source of ultraviolet radiation.

(2) Special mercury-vapour lamps (UV discharge lamps), which emit strong UV-C, and also electric arcs and sparks.

Final answer. Ultraviolet radiation (range about 100 nm to 400 nm) kills germs; its sources are the Sun and mercury-vapour (UV) lamps.

← *Back to Q88*

Q89

How are electromagnetic waves produced? Write their two characteristics.

[2023 • Set 55-5-3]

SOLUTION

Production of electromagnetic waves: Electromagnetic waves are produced by accelerating (or oscillating) electric charges. An oscillating charge produces an oscillating electric field, which in turn produces an oscillating magnetic field, and these mutually sustaining fields propagate outward as an electromagnetic wave.

The frequency of the wave is equal to the frequency of oscillation of the charge, and the energy carried by the wave comes from the energy of the oscillating charge.

Two characteristics of electromagnetic waves:

(1) They are transverse waves in which the electric field $\vec{E}$, the magnetic field $\vec{B}$, and the direction of propagation are mutually perpendicular, i.e. $\vec{E} \perp \vec{B} \perp$ propagation.

(2) They travel through vacuum with the same speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8\ m/s$ for all wavelengths, and they do not require any material medium to propagate.

Final answer. EM waves are produced by accelerating charges; they are transverse and travel through vacuum at speed $c = 3 \times 10^8$ m/s.

← *Back to Q89*

Q90

(a) The electric field of an electromagnetic wave is represented as $\mathbf{E_x = E_0 \sin(\omega t + kz)}$.
(i) In which direction is the wave propagating?
(ii) In which direction does the magnetic field oscillate?
(b) Write two characteristics of electromagnetic waves.

[2021]

SOLUTION

(a)(i) The electric field is given as $\mathbf{E_x = E_0 \sin(\omega t + kz)}$.

The phase of this wave is $(\omega t + kz)$. A wave whose phase has the form $(\omega t + kz)$, with the same sign on both terms, travels in the **negative z-direction**, i.e. along $-\hat{z}$.

(a)(ii) Here the electric field oscillates along the x-axis $(\hat{x})$ and the wave moves along $-\hat{z}$. The magnetic field must be perpendicular to both $\vec{E}$ and the direction of propagation.

The magnetic field therefore oscillates along the y-axis (specifically along $-\hat{y}$, so that $\vec{E} \times \vec{B}$ points along the propagation direction $-\hat{z}$).

(b) Two characteristics of electromagnetic waves:

(1) They are transverse waves in which $\vec{E}$, $\vec{B}$ and the direction of propagation are mutually perpendicular.

(2) They travel through vacuum with a constant speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8\ m/s$ for all wavelengths.

TIP

Sign rule: a phase like $(\omega t + kz)$ means motion along $-z$; only $(\omega t - kz)$ gives motion along $+z$.

Final answer. (a)(i) The wave propagates along $-z$; (ii) the magnetic field oscillates along the y-axis; (b) EM waves are transverse and travel at speed c in vacuum.

← *Back to Q90*

Q91

Which of the following electromagnetic waves has (a) minimum wavelength, and (b) minimum frequency? Write one use of each of these two waves.
Infrared waves, Microwaves, γ-rays and X-rays

[2020 • Set 55-2-1]

SOLUTION

The electromagnetic spectrum, arranged in increasing order of frequency (and decreasing order of wavelength), is: Microwaves, Infrared waves, X-rays, γ-rays.

Since wavelength and frequency are related by $c = \nu\lambda$ (where c is the constant speed of light), a higher frequency means a shorter wavelength.

(a) The wave with the **minimum wavelength** is the **γ-ray**, because it has the highest frequency among the four.

Use of γ-rays: They are used in radiotherapy for the treatment of cancer (using a ^{60}Co source) and for the sterilisation of surgical instruments.

(b) The wave with the **minimum frequency** is the **Microwave**, because it has the longest wavelength among the four.

Use of Microwaves: They are used in RADAR systems for aircraft navigation and in microwave ovens for cooking and heating food.

Thus, γ-rays have the minimum wavelength and microwaves have the minimum frequency.

Final answer. Minimum wavelength: gamma rays (used in cancer radiotherapy); minimum frequency: microwaves (used in RADAR / microwave ovens).

← *Back to Q91*

Q92

How does an oscillating charge radiate an electromagnetic wave? Give the relation between the frequency of radiated wave and the frequency of oscillating charge.
OR
(a) Explain briefly the fact that electromagnetic waves carry energy.
(b) Why do we not feel the pressure due to sunshine?

[2020 • Set 55-3-1]

SOLUTION

When a charge oscillates with some frequency, it experiences acceleration, since its velocity changes continuously in both magnitude and direction.

An accelerating (oscillating) charge produces a time-varying electric field $\vec{E}$ in the surrounding space.

By Maxwell's idea of displacement current, this changing electric field produces a time-varying magnetic field $\vec{B}$.

In turn, this changing magnetic field produces a changing electric field, and so on. These mutually perpendicular $\vec{E}$ and $\vec{B}$ fields regenerate each other and propagate outward through space as an electromagnetic wave.

The frequency of the radiated electromagnetic wave is exactly equal to the frequency of oscillation of the charge:

$$\nu_{\text{EM wave}} = \nu_{\text{oscillating charge}}$$

OR

(a) An electromagnetic wave consists of oscillating electric and magnetic fields. Both of these fields store energy. The energy density of the electric field is $u_E = \frac{1}{2}\varepsilon_0 E^2$ and that of the magnetic field is $u_B = \frac{B^2}{2\mu_0}$.

As the wave propagates, it carries this stored energy with it. The rate of flow of energy per unit area is given by the Poynting vector $\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}$. Hence, electromagnetic waves carry energy.

(b) Sunlight does exert a radiation pressure when it is absorbed or reflected. However, this pressure is extremely small, of the order of a few μPa. It is far too feeble for our skin to detect, which is why we do not feel any pressure due to sunshine.

Final answer. An oscillating charge radiates an EM wave through mutually regenerating time-varying fields, with $\nu_{wave} = \nu_{charge}$.

← *Back to Q92*

Q93

Gamma rays and radio waves travel with the same velocity in free space. Distinguish between them in terms of their origin and the main application.

[2020 • Set 55-5-1]

SOLUTION

Although gamma rays and radio waves both travel with the same speed $c = 310^8$ $m\ s^{-1}$ in free space, they differ greatly in origin and application because they lie at opposite ends of the electromagnetic spectrum.

Gamma rays:

Origin: They are produced during nuclear reactions and by transitions within the nuclei of radioactive atoms (de-excitation of excited nuclei).

Application: They are used in the treatment of cancer (gamma knife / radiotherapy) and for sterilising medical equipment.

Radio waves:

Origin: They are produced by the acceleration of electrons in conducting wires and antennas, that is, by oscillating electric circuits.

Application: They are used in radio and television communication and broadcasting (AM and FM), and in mobile telephony.

Thus, gamma rays originate from the nucleus and are used in medicine, while radio waves originate from oscillating circuits and are used in communication.

Final answer. Gamma rays come from nuclei (used in cancer therapy); radio waves come from oscillating circuits/antennas (used in communication).

← *Back to Q93*

Q94

A capacitor made of two parallel plates, each of area 'A' and separation 'd' is charged by an external dc source. Show that during charging, the displacement current inside the capacitor is the same as the current charging the capacitor.

[2019 • Set 55-2-1]

SOLUTION

Consider a parallel-plate capacitor with each plate of area A and separation d, charged by an external DC source. Let q be the instantaneous charge on the plates at any time during charging.

The surface charge density on a plate is $\sigma = \frac{q}{A}$.

The uniform electric field between the plates is

$$E = \frac{\sigma}{\varepsilon_0} = \frac{q}{\varepsilon_0 A}$$

The electric flux through the area A between the plates is

$$\Phi_E = E\,A = \frac{q}{\varepsilon_0 A} \times A = \frac{q}{\varepsilon_0}$$

The displacement current between the plates is defined as

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Substituting $\Phi_E = \frac{q}{\varepsilon_0}$,

$$I_d = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right) = \frac{dq}{dt}$$

But $\frac{dq}{dt}$ is precisely the rate at which charge is supplied to the plates, that is, the conduction current I_c charging the capacitor.

Therefore $\mathbf{I_d = I_c}$, which proves that the **displacement current inside the capacitor is equal to the conduction current charging it.**

Final answer. Since $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \frac{dq}{dt} = I_c}$, the displacement current equals the charging current.

← *Back to Q94*

Q95

How is the equation for Ampere's circuital law modified in the presence of displacement current? Explain.

[2019 • Set 55-2-2]

SOLUTION

The original Ampere's circuital law states that the line integral of the magnetic field around a closed loop equals μ_0 times the conduction current enclosed:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_c$$

This law gives inconsistent results for a circuit containing a charging capacitor. Consider an Amperian loop around the connecting wire. If we take a flat surface pierced by the wire, the enclosed conduction current is I_c. But if we take a balloon-shaped surface passing between the capacitor plates (bounded by the same loop), no conduction current flows through it, since no charge crosses the gap, giving $\mathbf{I_c = 0}$. The same loop thus gives two different answers, which is contradictory.

To remove this inconsistency, Maxwell pointed out that the changing electric field between the plates is associated with a current called the displacement current,

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Adding this term modifies the law to the Ampere-Maxwell law:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0(I_c + I_d) = \mu_0(I_c + \varepsilon_0 \frac{d\Phi_E}{dt})$$

Now the total current $(I_c + I_d)$ is continuous across the capacitor gap, so the law gives the same result for both surfaces. This modification also predicts the existence of electromagnetic waves.

COMMON MISTAKE

A common mistake is to write only the displacement-current term. The full Ampere-Maxwell law keeps BOTH the conduction current I_c and the displacement current I_d.

Final answer. Ampere's law is modified to $\oint \vec{B} \cdot d\vec{s} = \mu_0(I_c + \varepsilon_0 \frac{d\Phi_E}{dt})$ by adding the displacement current.

← Back to Q95

Q96

How are electromagnetic waves produced by oscillating charges? What is the source of the energy associated with the em waves?

[2019 • Set 55-2-3]

SOLUTION

When a charge oscillates with a certain frequency, it is continuously accelerated, because its velocity keeps changing in magnitude and direction.

This accelerating charge sets up a time-varying electric field $\vec{E}$ around it.

According to Maxwell, a changing electric field produces a changing (time-varying) magnetic field $\vec{B}$ through the displacement-current effect.

This changing magnetic field, in turn, produces a changing electric field. The two fields, being mutually perpendicular to each other and to the direction of travel, keep regenerating one another and propagate through space as an electromagnetic wave, moving at the speed of light.

The frequency of the emitted electromagnetic wave equals the frequency of oscillation of the charge.

Source of energy: The energy carried by the electromagnetic wave comes from the energy of the oscillating (accelerated) charge. This is the energy supplied by the external source that drives the oscillation of the charge, for example, an external EMF or oscillator.

Thus, **the oscillating charge produces the wave, and the energy of the wave comes from the source maintaining the oscillation of the charge.**

Final answer. Oscillating charges produce EM waves via mutually regenerating fields; the wave energy comes from the energy of the oscillating charge (its driving source).

← *Back to Q96*

Q97

A parallel plate capacitor of plate area A each and separation d, is being charged by an ac source. Show that the displacement current inside the capacitor is the same as the current charging the capacitor.

[2019 • Set 55-4-1]

SOLUTION

Consider a parallel-plate capacitor with each plate of area A and separation d, connected to an AC source. Let the instantaneous charge on the plates be $q(t)$.

The surface charge density on a plate is $\sigma = \frac{q}{A}$, so the uniform electric field between the plates is

$$E = \frac{\sigma}{\varepsilon_0} = \frac{q}{\varepsilon_0 A}$$

The electric flux through the region between the plates (over area A) is

$$\Phi_E = E\,A = \frac{q}{\varepsilon_0 A} \times A = \frac{q}{\varepsilon_0}$$

The displacement current between the plates is

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Substituting $\Phi_E = \frac{q}{\varepsilon_0}$,

$$I_d = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right) = \frac{dq}{dt}$$

Now $\frac{dq}{dt}$ is the rate of flow of charge in the connecting wires, that is, the conduction current I_c charging the capacitor.

Hence $\mathbf{I_d = I_c}$, proving that **the displacement current inside the capacitor is exactly equal to the conduction current charging it.** This holds at every instant, even with an AC source, so the total current is continuous across the capacitor gap.

Final answer. Since $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \frac{dq}{dt} = I_c}$, the displacement current inside the capacitor equals the charging current.

← Back to Q97

Q98

(a) Why are infra-red waves often called heat waves ? Explain.
(b) What do you understand by the statement, "Electromagnetic waves transport momentum" ?

[2018]

SOLUTION

(a) Infrared waves are called heat waves because they are readily absorbed by most substances, including water and the molecules of our skin.

When a substance absorbs infrared radiation, the energy goes into increasing the vibrational and rotational motion of its molecules. This increase in molecular kinetic energy raises the temperature of the substance, that is, it produces a heating effect.

Since infrared waves produce this strong heating on absorption, they are commonly known as heat waves.

(b) The statement "electromagnetic waves transport momentum" means that an electromagnetic wave carries linear momentum along with its energy.

If a wave carrying total energy U is completely absorbed by a surface, it delivers a momentum

$$p = \frac{U}{c}$$

to that surface, where c is the speed of light. Because of this transfer of momentum, the wave exerts a force on the surface, called radiation pressure.

Thus, when sunlight or any electromagnetic radiation falls on a surface and is absorbed or reflected, it pushes on the surface, demonstrating that the wave carries momentum.

Final answer. (a) IR is absorbed and raises molecular motion, causing heating, so it is called a heat wave. (b) EM waves carry momentum $\boldsymbol{p = U/c}$, exerting radiation pressure on a surface.

← *Back to Q98*

Q99

How does Ampere-Maxwell law explain the flow of current through a capacitor when it is being charged by a battery? Write the expression for the displacement current in terms of the rate of change of electric flux.

[2017]

SOLUTION

When a capacitor is being charged by a battery, conduction current I_c flows in the connecting wires, but no conduction current flows in the gap between the

plates, since no charge crosses it.

However, as the capacitor charges, the charge q on the plates increases, so the electric field and hence the electric flux Φ_E between the plates also change with time.

This changing electric flux is equivalent to a current, called the displacement current, given by

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

According to the Ampere-Maxwell law,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 (I_c + \varepsilon_0 \frac{d\Phi_E}{dt})$$

For an Amperian loop encircling the wire, the conduction current I_c is enclosed. For the same loop with a surface passing between the plates, the conduction current is zero, but the displacement current I_d flows there.

Since it can be shown that $\mathbf{I_d = I_c}$ at every instant, the current does not break at the capacitor gap. The displacement current between the plates continues the conduction current of the wire, so current flows continuously through the circuit even though no charge crosses the gap.

The expression for the displacement current in terms of the rate of change of electric flux is

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Final answer. The changing electric flux between the plates gives a displacement current $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$ that continues the conduction current across the gap.

← Back to Q99

Q100

Identify the electromagnetic waves whose wavelengths vary as
(a) $\mathbf{10^{-12}\ m < \lambda < 10^{-8}\ m}$
(b) $\mathbf{10^{-3}\ m < \lambda < 10^{-1}\ m}$
Write one use for each.

[2017]

SOLUTION

The type of electromagnetic wave is identified by the range of the electromagnetic spectrum in which its wavelength lies.

(a) A wavelength of about 10^{-12} *m* lies in the **gamma-ray / X-ray** region of the spectrum. Wavelengths around 10^{-12} *m* correspond to gamma rays (with X-rays just above this, near 10^{-10} *m*).

Use: Gamma rays are used in the treatment of cancer (radiotherapy); X-rays of this short wavelength are used in medical imaging (radiography) and in X-ray crystallography to study atomic structure.

(b) A wavelength of about 10^{-3} *m* (that is, 1 mm) lies in the **microwave** region of the spectrum.

Use: Microwaves are used in RADAR systems, in microwave ovens for cooking, and in satellite and mobile communication.

Thus, 10^{-12} *m* corresponds to gamma rays / X-rays and 10^{-3} *m* corresponds to microwaves.

Final answer. (a) $\mathbf{10^{-12}\ m}$ is a gamma ray / X-ray (used in radiotherapy / imaging); (b) $\mathbf{10^{-3}\ m}$ is a microwave (used in RADAR / ovens).

← *Back to Q100*

Q101

(i) How are infrared waves produced? Write their one important use.
(ii) The thin Ozone layer on top of the stratosphere is crucial for human survival.

Why?

[2016]

SOLUTION

(i) Production of infrared waves: Infrared waves are produced by the vibration of atoms and molecules in hot bodies. Because these vibrations are thermal in origin, infrared waves are also called heat waves.

Common sources are the Sun, hot filaments, heated objects, and infrared LEDs.

One important use: Infrared waves are used in physiotherapy and in remote controls of TV and other electronic appliances. They are also used in night-vision cameras and weather forecasting through Earth satellites.

(ii) Importance of the ozone layer: The thin ozone (O_3) layer present at the top of the stratosphere absorbs most of the harmful ultraviolet (UV) radiation coming from the Sun.

UV radiation is very energetic and, if it reached the Earth's surface in full, it would cause skin cancer, damage to the eyes, and harm to plant and animal life.

By absorbing this UV radiation, the ozone layer acts as a protective shield, which is why it is **crucial for human survival**.

Final answer. Infrared waves are produced by thermal vibrations of atoms and molecules in hot bodies (used in remote controls/physiotherapy), and the ozone layer protects life by absorbing harmful solar UV radiation.

← *Back to Q101*

Q102

State two properties of electromagnetic waves. How can we show that em waves carry momentum?

[2016]

SOLUTION

Two properties of electromagnetic waves:

1. Electromagnetic waves are transverse in nature: the electric field $\vec{E}$, the magnetic field $\vec{B}$, and the direction of propagation are mutually perpendicular to one another.

2. They travel through vacuum with the same speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3\times 10^8\ m/s$, independent of their frequency.

Electromagnetic waves carry momentum: An electromagnetic wave carries energy as well as momentum. If an electromagnetic wave transports total energy U to a surface in a given time, the momentum delivered to that surface is

$$p = \frac{U}{c}$$

When the wave is incident on a surface and is absorbed (or reflected), this momentum is transferred to the surface. As a result the wave exerts a pressure on the surface, called **radiation pressure**.

The fact that a beam of light falling on a surface produces a small but measurable force (as demonstrated experimentally) shows that electromagnetic waves carry momentum.

Final answer. EM waves are transverse and travel at speed c in vacuum; they carry momentum $p = U/c$, shown by the radiation pressure they exert on a surface.

← *Back to Q102*

Q103

Considering the case of a parallel plate capacitor being charged, show how one is required to generalize Ampere's circuital law to include the term due to displacement current.

[2014]

SOLUTION

Consider a parallel plate capacitor being charged by a current I. Apply Ampere's circuital law to a plane circular loop surrounding the connecting wire. The law in its original form is

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_c$$

where I_c is the conduction current passing through any surface bounded by the loop.

Take Surface 1 as a flat disc that the connecting wire pierces. The conduction current I_c passes through it, so the law gives $\oint \vec{B} \cdot d\vec{s} = \mu_0 I_c$.

Now take Surface 2 as a balloon-shaped surface bounded by the same loop but passing between the two capacitor plates. No conduction current crosses the gap between the plates, so $\oint \vec{B} \cdot d\vec{s} = 0$.

The same loop gives two different answers, so Ampere's law in its original form is inconsistent.

To remove this inconsistency, Maxwell noted that during charging the electric field, and hence the electric flux Φ_E, between the plates is changing with time. He introduced an additional current called the displacement current:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

This I_d has exactly the same magnitude as I_c, so the gap is bridged. The generalized (Ampere-Maxwell) law becomes

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 (I_c + \varepsilon_0 \frac{d\Phi_E}{dt})$$

which now gives the **same result for any surface** bounded by the loop.

> **COMMON MISTAKE**
>
> *A very common slip is to forget the* ε_0 *in front of* $d\Phi_E/dt$*; the displacement current is* $\varepsilon_0 d\Phi_E/dt$*, not just* $d\Phi_E/dt$.

Final answer. Ampere's law is generalized to $\oint \vec{B} \cdot d\vec{s} = \mu_0 (I_c + \varepsilon_0 \frac{d\Phi_E}{dt})$ by adding the displacement current $I_d = \varepsilon_0 \, d\Phi_E/dt$.

← Back to Q103

Q104

(a) An em wave is travelling in a medium with a velocity $\vec{v} = v\,\hat{i}$*. Draw a sketch showing the propagation of the em wave, indicating the direction of the oscillating electric and magnetic fields.*
(b) How are the magnitudes of the electric and magnetic fields related to the velocity of the em wave?

[2013]

SOLUTION

(a) Sketch of the propagating electromagnetic wave: Let the wave travel along the $+x$ direction, so $\vec{v} = v\,\hat{i}$.

Take the electric field oscillating along the y-axis and the magnetic field oscillating along the z-axis:

$$\vec{E} = E_0 \sin(kx - \omega t)\,\hat{j}$$

$$\vec{B} = B_0 \sin(kx - \omega t)\,\hat{k}$$

In the sketch, draw the x-axis horizontally as the propagation direction. Show $\vec{E}$ as a sinusoidal curve oscillating up and down in the xy-plane (red), and $\vec{B}$ as a sinusoidal curve oscillating in and out in the xz-plane (blue), drawn perpendicular to $\vec{E}$. Both curves have their crests, troughs, and zeros at the same points along x, showing the two fields are in phase. Mark $\vec{E} \times \vec{B}$ pointing along $+\hat{i}$ (the direction of propagation) by the right-hand rule.

(b) Relation between the magnitudes and the wave velocity: The magnitudes of the electric and magnetic fields are related to the speed of the wave by

$$v = \frac{E_0}{B_0} = \frac{|\vec{E}|}{|\vec{B}|}$$

In vacuum this speed equals $c = \dfrac{1}{\sqrt{\mu_0 \varepsilon_0}}$, so that $\boldsymbol{c = E_0/B_0}$.

Final answer. $\vec{E}$ along $\hat{j}$ and $\vec{B}$ along $\hat{k}$ oscillate in phase with propagation along $\hat{i}$, and their magnitudes obey $v = E_0/B_0$ (equal to c in vacuum).

← Back to Q104

Q105

Draw a sketch of a plane electromagnetic wave propagating along the z-direction. Depict clearly the directions of electric and magnetic fields varying sinusoidally with z.

[2012]

SOLUTION

Plane electromagnetic wave propagating along the z-axis: For a plane wave travelling in the $+z$ direction, take the electric field oscillating sinusoidally along the x-axis and the magnetic field oscillating sinusoidally along the y-axis:

$$\vec{E} = E_0 \sin(kz - \omega t)\,\hat{x}$$

$$\vec{B} = B_0 \sin(kz - \omega t)\,\hat{y}$$

Description of the sketch: Draw the z-axis horizontally as the direction of propagation.

Show $\vec{E}$ as a sinusoidal curve in the xz-plane (drawn vertically up and down, in red), and $\vec{B}$ as a sinusoidal curve in the yz-plane (drawn into and out of the page, in blue).

The two curves are perpendicular to each other and to the z-axis, and their crests, troughs, and zeros line up at the same values of z, showing the fields are in phase.

Mark an arrow along $+z$ for the direction of propagation, noting that $\hat{z} = \hat{x} \times \hat{y}$, i.e. the wave advances along $\vec{E} \times \vec{B}$.

This shows the wave is **transverse**, with $\vec{E}$, $\vec{B}$ and the propagation direction mutually perpendicular.

Final answer. $\vec{E} = E_0 \sin(kz - \omega t)\hat{x}$ and $\vec{B} = B_0 \sin(kz - \omega t)\hat{y}$ oscillate in phase, mutually perpendicular, propagating along $+z$.

← *Back to Q105*

Q106

A capacitor of capacitance 'C' is being charged by connecting it across a dc source along with an ammeter. Will the ammeter show a momentary deflection during the process of charging? If so, how would you explain this momentary deflection and the resulting continuity of current in the circuit? Write the expression for the current inside the capacitor.

[2012]

SOLUTION

Yes, the ammeter shows a momentary deflection during the charging of the capacitor.

Explanation: While the capacitor is being charged, a conduction current I_c flows in the connecting wires. This charging current lasts only for a short time (until the capacitor is fully charged), so the ammeter deflects only momentarily.

In the gap between the plates, no charge actually crosses; instead, the electric field E (and hence the electric flux Φ_E) between the plates increases with time. This changing flux is equivalent to a current called the displacement current:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

During charging, the displacement current between the plates is exactly equal to the conduction current in the wires:

$$I_c = I_d$$

So the current is continuous throughout the circuit: it flows as conduction current in the wires and continues as displacement current in the gap. This is why the ammeter records the same current and the circuit current remains continuous.

Expression for the current inside the capacitor (the displacement current): since $\Phi_E = E\,A$ for a uniform field over plate area A,

$$I_d = \varepsilon_0 A \frac{dE}{dt}$$

Final answer. Yes; the momentary deflection is the charging conduction current, matched between the plates by the displacement current $\mathbf{I_d = \varepsilon_0 A \frac{dE}{dt}}$, keeping the circuit current continuous.

← *Back to Q106*

Q107

When an ideal capacitor is charged by a dc battery, no current flows. However, when an ac source is used, the current flows continuously. How does one explain this, based on the concept of displacement current?

[2012]

SOLUTION

The current in such a circuit is the displacement current between the capacitor plates, given by

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

where Φ_E is the electric flux between the plates.

Case of a dc battery: When the capacitor is connected to a dc source, it charges up in a very short time and then becomes fully charged. After this, the voltage across the plates is steady, so the electric field E between the plates is constant.

$$\frac{d\Phi_E}{dt} = 0 \Rightarrow I_d = 0$$

Hence no current flows once the capacitor is charged.

Case of an ac source: When the capacitor is connected to an ac source, the applied voltage changes continuously with time. Therefore the electric field, and hence the electric flux Φ_E between the plates, also changes continuously.

$\frac{d\Phi_E}{dt} \neq 0 \Rightarrow I_d \neq 0$ at all times

This continuously varying displacement current between the plates is equal to the conduction current in the connecting wires, so **current flows continuously** in the ac circuit.

Final answer. With dc the field becomes steady ($d\Phi_E/dt = 0$, so $I_d = 0$), but with ac the field varies continuously ($d\Phi_E/dt \neq 0$), so a continuous displacement current flows.

← *Back to Q107*

Q108

Explain briefly how electromagnetic waves are produced by an oscillating charge. How is the frequency of the em waves produced related to that of the oscillating charge?

[2012]

SOLUTION

Production of electromagnetic waves by an oscillating charge: A charge at rest produces only a static electric field, and a charge moving with constant velocity produces a steady magnetic field as well. Neither of these radiates energy.

When a charge oscillates (accelerates), it produces an electric field that changes with time. By Maxwell's theory, this time-varying electric field $\vec{E}$ produces a time-varying magnetic field $\vec{B}$ in the surrounding space.

This changing magnetic field $\vec{B}$ in turn produces a changing electric field $\vec{E}$, and so on. The two fields keep regenerating each other, and this disturbance travels outward through space as an **electromagnetic wave**.

Relation of frequencies: The electric and magnetic fields oscillate with the same frequency as the source charge. Therefore the frequency of the electromagnetic wave produced is equal to the frequency of oscillation of the charge:

$$\nu_{\text{wave}} = \nu_{\text{charge}}$$

CONCEPT

Only an accelerating charge radiates EM waves; a charge at rest or in uniform motion does not, so always stress the oscillation/acceleration in the answer.

Final answer. An oscillating (accelerating) charge produces mutually regenerating time-varying $\vec{E}$ and $\vec{B}$ fields that travel out as an EM wave, of frequency equal to that of the oscillating charge ($\nu_{\text{wave}} = \nu_{\text{charge}}$).

← *Back to Q108*

Q109

What is sky wave communication? Why is this mode of propagation restricted to the frequencies only upto few MHz?

[2011]

SOLUTION

Sky wave communication: In sky wave (or ionospheric) propagation, radio waves sent up from a transmitting antenna are reflected back to the Earth by the ionosphere, the layer of charged particles in the upper atmosphere.

Because the waves are bounced back from the ionosphere, they can reach receivers far beyond the horizon, allowing long-distance communication. This mode is used in the frequency range of about 3 MHz to 30 MHz (the high-frequency or short-wave band).

Why it is limited to a few MHz: The ionosphere can reflect radio waves only up to a certain maximum frequency (the critical frequency). For frequencies up to a few MHz, the waves are reflected back to Earth.

For frequencies higher than this limit (above about 30 MHz), the ionosphere is no longer able to reflect the waves; instead the waves penetrate through the ionosphere and escape into outer space. Hence sky wave propagation **cannot be used for frequencies higher than a few MHz**.

Final answer. Sky wave communication uses ionospheric reflection of radio waves for long-distance transmission; it is limited to a few MHz because higher

frequencies penetrate the ionosphere and escape into space instead of being reflected.

← Back to Q109

Q110

A parallel plate capacitor is being charged by a time varying current. Explain briefly how Ampere's circuital law is generalized to incorporate the effect due to the displacement current.

[2011]

SOLUTION

Consider a parallel plate capacitor being charged by a time-varying current I. Apply Ampere's circuital law to a circular loop that encircles the connecting wire. In its original form,

$$\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \, \mathrm{I}_c$$

where I_c is the conduction current through any surface bounded by the loop.

The inconsistency: Choose Surface 1 as a flat disc through which the wire passes. The full conduction current I_c crosses it, giving $\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \, \mathrm{I}_c$.

Now choose Surface 2 as a pot-shaped surface bounded by the same loop but passing through the gap between the capacitor plates. No conduction current flows across this gap, so $\oint \vec{B} \cdot \mathrm{d}\vec{l} = 0$.

The same loop gives two different results, which is a contradiction. So Ampere's law in its original form is incomplete.

Maxwell's generalization: While charging, the electric field between the plates increases with time, so the electric flux Φ_E through Surface 2 changes with time. Maxwell associated this changing flux with a displacement current:

$$\mathrm{I}_d = \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{dt}}$$

whose magnitude equals the conduction current I_c in the wires. Including this term, the generalized Ampere-Maxwell law is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0(I_c + I_d) = \mu_0\left(I_c + \varepsilon_0 \frac{d\Phi_E}{dt}\right)$$

which now gives the **same result for every surface** bounded by the loop, removing the inconsistency.

Final answer. Adding the displacement current $\mathbf{I_d = \varepsilon_0\, d\Phi_E/dt}$ generalizes Ampere's law to $\mathbf{\oint \vec{B} \cdot d\vec{l} = \mu_0(I_c + \varepsilon_0\, d\Phi_E/dt)}$, making it consistent for any surface bounded by the loop.

← *Back to Q110*

Q111

How are infrared waves produced? Why are these referred to as heat waves? Write their one important use.

[2011 • Set 55-1-1]

SOLUTION

Infrared waves are produced by the vibration of atoms and molecules in hot bodies. When the atoms and molecules of a warm object oscillate, they make transitions between their vibrational and rotational energy levels and emit radiation in the infrared region of the spectrum.

Common sources include hot bodies such as the Sun, incandescent lamps, a stove or any heated object, as well as molecules vibrating about their mean positions.

These waves are called heat waves because they are readily absorbed by water molecules and most other substances. When a body absorbs infrared radiation, the absorbed energy increases the thermal motion of its molecules, so the temperature of the body rises. Since infrared waves produce a strong heating effect on absorption, they are referred to as **heat waves**.

One important use: infrared waves are used in physiotherapy to provide gentle deep heating of muscles and to relieve pain. (They are also used in remote controls and in night-vision/thermal-imaging devices.)

Final answer. Infrared waves are produced by vibrating atoms and molecules of hot bodies; they cause strong heating on absorption (heat waves) and are used in physiotherapy.

← *Back to Q111*

Q112

Show how the equation for Ampere's circuital law, viz.
$\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 I$
is modified in the presence of displacement current.

[2011 • Set 55-2-1]

SOLUTION

Ampere's circuital law in its original form states that the line integral of the magnetic field around a closed loop equals μ_0 times the conduction current passing through the loop.

$$\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \, \mathrm{I}_c$$

Maxwell pointed out a flaw in this law while considering a capacitor being charged. Consider a closed loop around the wire feeding charge to the capacitor. If we take a flat surface that the wire pierces, a conduction current I_c passes through it. But if we take a bulging (pot-shaped) surface bounded by the same loop, passing through the gap between the capacitor plates, no conduction current crosses it because there is no charge flow in the gap.

Thus Ampere's law gives two different values of $\oint \vec{B} \cdot \mathrm{d}\vec{l}$ for the same loop, which is a contradiction.

To remove this inconsistency, Maxwell introduced an additional current, called the **displacement current** I_d, due to the changing electric field (and hence changing electric flux Φ_E) between the plates.

$$\mathrm{I}_d = \varepsilon_0 \frac{\mathrm{d}\Phi_E}{\mathrm{dt}}$$

The total current is the sum of the conduction current and the displacement current, and this total current is continuous across any surface bounded by the

loop. Hence the modified law, called the Ampere-Maxwell law, is:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I_c + \varepsilon_0 \frac{d\Phi_E}{dt} \right)$$

This is the required modified form of Ampere's circuital law in the presence of displacement current.

> **COMMON MISTAKE**
>
> *A frequent slip is writing only I_d in the modified law. The bracket must contain BOTH the conduction current I_c and the displacement current $\varepsilon_0 \, d\Phi_E/dt$.*

Final answer. The modified law is $\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + \varepsilon_0 \frac{d\Phi_E}{dt})$, the displacement current $\varepsilon_0 \, d\Phi_E/dt$ being added to the conduction current.

← *Back to Q112*

Q113

Arrange the following electromagnetic radiations in ascending order of their frequencies:
(i) Microwave
(ii) Radio wave
(iii) X-rays
(iv) Gamma rays
Write two uses of any one of these.

[2010]

SOLUTION

The frequency of an electromagnetic wave increases as we move from radio waves to gamma rays in the spectrum. The approximate frequency ranges are:

Radio waves ~ 10^4 to 10^9 Hz

Microwaves ~ 10^9 to 10^{11} Hz

X-rays ~ 10^{16} to 10^{19} Hz

Gamma rays > $\mathbf{10^{19}}$ **Hz**

Hence, arranging the given radiations in ascending order of frequency:

Radio waves < Microwaves < X-rays < Gamma rays

Two uses of X-rays:

(i) In medical diagnosis, X-rays are used to detect bone fractures and to image internal organs (chest X-rays, CT scans, dental imaging), since bones absorb X-rays more than soft tissue.

(ii) In the study of crystal structure, X-ray diffraction is used to determine the arrangement of atoms in crystals.

Final answer. Ascending order of frequency: Radio waves < Microwaves < X-rays < Gamma rays.

← *Back to Q113*

Q114

How does a charge q oscillating at certain frequency produce electromagnetic waves? Sketch a schematic diagram depicting electric and magnetic fields for an electromagnetic wave propagating along the Z-direction.

[2009]

SOLUTION

An oscillating charge has an acceleration that changes with time. As the charge oscillates with a certain frequency ν, it sets up a time-varying electric field $\vec{E}$ in the surrounding space.

This time-varying electric field produces a time-varying magnetic field $\vec{B}$ (through the displacement current term $\varepsilon_0 \, d\Phi_E/dt$). The time-varying magnetic field, in turn, produces a time-varying electric field, and so on.

In this way the mutually perpendicular, time-varying $\vec{E}$ and $\vec{B}$ fields regenerate each other and travel outward through space as an electromagnetic wave. The frequency of the emitted electromagnetic wave is equal to the frequency ν of oscillation of the charge.

Schematic diagram for a wave travelling along the $+Z$ direction:

Draw three mutually perpendicular axes X, Y and Z. The wave propagates along the $+Z$ axis. The electric field $\vec{E}$ is drawn as a sinusoidal curve oscillating in the X-Z plane (that is, $\vec{E}$ points along the $\hat{x}$ direction), labelled $\vec{E} = \mathbf{E_0 \sin(kz - \omega t)\,\hat{x}}$. The magnetic field $\vec{B}$ is drawn as a sinusoidal curve oscillating in the Y-Z plane (that is, $\vec{B}$ points along the $\hat{y}$ direction), labelled $\vec{B} = \mathbf{B_0 \sin(kz - \omega t)\,\hat{y}}$. Both sinusoids are in phase (their crests and troughs occur at the same points along Z), are perpendicular to each other, and are perpendicular to the direction of propagation $\hat{z}$.

Final answer. An oscillating charge produces time-varying $\vec{E}$ and $\vec{B}$ fields that regenerate each other and propagate as an EM wave of the same frequency ν; for a wave along $+Z$, $\vec{E} \parallel \hat{x}$ and $\vec{B} \parallel \hat{y}$, mutually perpendicular and in phase.

← *Back to Q114*

Q115

Answer the following questions:
(a) Optical and radio telescopes are built on the ground while X-ray astronomy is possible only from satellites orbiting the Earth. Why?
(b) The small ozone layer on top of the stratosphere is crucial for human survival. Why?

[2009]

SOLUTION

(a) X-rays coming from astronomical sources are almost completely absorbed by the gases (mainly oxygen and nitrogen) of the Earth's atmosphere, so they cannot reach the ground. Therefore, to observe these X-rays, the detecting equipment must be placed above the atmosphere, on satellites orbiting the Earth.

On the other hand, the atmosphere has windows that allow visible light and radio waves to pass through to the ground with little absorption. Hence optical and radio telescopes can be built on the Earth's surface, while X-ray astronomy is possible only from satellites.

(b) The ozone layer in the upper part of the stratosphere absorbs most of the harmful ultraviolet (UV-B and UV-C) radiation coming from the Sun. This high-energy ultraviolet radiation can damage living cells and cause skin cancer, cataracts and genetic damage (damage to DNA).

By absorbing this ultraviolet radiation, the thin ozone layer shields life on Earth from its harmful effects. That is why this small ozone layer is crucial for human survival.

Final answer. (a) The atmosphere absorbs X-rays (but is transparent to visible and radio waves), so X-ray telescopes must orbit above it. (b) The ozone layer absorbs harmful solar UV radiation, protecting life on Earth.

← *Back to Q115*

Q116

Define magnetic susceptibility of a material. Name two elements, one having positive susceptibility and the other having negative susceptibility. What does negative susceptibility signify?

[2008]

SOLUTION

Magnetic susceptibility (χ_m) of a material is defined as the ratio of the magnetization M (magnetic moment per unit volume) produced in the material to the magnetic intensity H that produces it.

$$\chi_m = \frac{M}{H}$$

It is a dimensionless quantity that measures how easily a material can be magnetized when placed in a magnetizing field.

Element with positive susceptibility (paramagnetic): Aluminium (Al). Such a material has small positive χ_m.

Element with negative susceptibility (diamagnetic): Bismuth (Bi). Such a material has small negative χ_m.

Significance of negative susceptibility: A negative value of χ_m means that the magnetization $\vec{M}$ produced in the material is directed opposite to the applied magnetic intensity $\vec{H}$. Such materials are weakly **magnetized in the opposite direction** to the field and are therefore feebly repelled by a magnet. This behaviour is characteristic of **diamagnetic** substances.

Final answer. Magnetic susceptibility $\chi_m = M/H$; aluminium has positive χ_m (paramagnetic), bismuth has negative χ_m (diamagnetic), and negative χ_m signifies magnetization opposite to the applied field, i.e. diamagnetism.

← *Back to Q116*

Q117

What is meant by the transverse nature of electromagnetic waves? Draw a diagram showing the propagation of an electromagnetic wave along the x-direction, indicating clearly the directions of the oscillating electric and magnetic fields associated with it.

[2008]

SOLUTION

Transverse nature of electromagnetic waves means that in an electromagnetic wave the oscillating electric field $\vec{E}$ and magnetic field $\vec{B}$ are both perpendicular to the direction of propagation of the wave (and also perpendicular to each other). No part of the wave oscillates along the direction of propagation. Because the field vectors vibrate at right angles to the direction in which the wave travels, electromagnetic waves are transverse waves.

Diagram for propagation along the $+X$ direction:

Draw three mutually perpendicular axes X, Y and Z, with the wave travelling along the $+X$ axis. Show the electric field $\vec{E}$ as a sinusoidal curve oscillating in the X-Y plane (along the $\hat{y}$ direction), labelled $\vec{E} = \mathbf{E_0 \sin(kx - \omega t)}\,\hat{y}$. Show the magnetic field $\vec{B}$ as a sinusoidal curve oscillating in the X-Z plane (along the $\hat{z}$ direction), labelled $\vec{B} = \mathbf{B_0 \sin(kx - \omega t)}\,\hat{z}$.

The two sinusoids are in phase, are mutually perpendicular, and are both perpendicular to the $+X$ propagation direction, with the direction of propagation given by $\hat{x} = \hat{E} \times \hat{B}$.

Final answer. EM waves are transverse because $\vec{E}$ and $\vec{B}$ oscillate perpendicular to the direction of propagation and to each other; for propagation along $+X$, $\vec{E} \parallel \hat{y}$ and $\vec{B} \parallel \hat{z}$.

← Back to Q117

Q118

Find the wavelength of electromagnetic waves of frequency 5×10^{19} Hz *in free space. Give its two applications.*

[2004]

SOLUTION

In free space an electromagnetic wave travels with the speed of light c, so its wavelength is related to its frequency by:

$$\lambda = \frac{c}{\nu}$$

Here $c = 3 \times 10^8\ m\ s^{-1}$ and $\nu = 5 \times 10^{19}$ Hz.

$$\lambda = \frac{3 \times 10^8}{5 \times 10^{19}}$$

$$\lambda = 0.6 \times 10^{-11}\ m$$

$$\lambda = 6 \times 10^{-12}\ m$$

This wavelength lies in the gamma-ray region of the electromagnetic spectrum.

Two applications of these waves (gamma rays):

(i) In medicine, gamma rays are used in radiotherapy for the treatment of cancer, as they destroy malignant cells.

(ii) They are used for sterilization of surgical instruments and food, since they kill bacteria and other micro-organisms.

COMMON MISTAKE

Keep track of powers of ten: $\frac{10^{8}}{10^{19}} = 10^{-11}$, *so* $\frac{3}{5} \times 10^{-11} = 6 \times 10^{-12}$ ***m***, *not* 6×10^{-11} *m.*

Final answer. The wavelength is $\lambda = 6 \times 10^{-12}$ ***m*** (gamma-ray region), used in cancer radiotherapy and in sterilization of medical equipment and food.

← *Back to Q118*

Q119

Name the electromagnetic waves which are used for:
(i) detection of fractures in bones
(ii) physiotherapy
(iii) radar systems
Also write their wavelength range.

[2026 • Set 55-2-1]

SOLUTION

(i) Detection of fractures in bones: **X-rays**. X-rays pass through soft tissue but are strongly absorbed by bones, so they reveal fractures clearly. Their wavelength range is about 10^{-8} *m* to 10^{-11} *m* (roughly 1 nm down to 1 pm).

(ii) Physiotherapy: **Infrared waves**. Infrared lamps provide deep heating that relaxes muscles and improves blood circulation. Their wavelength range is about 7×10^{-7} *m* (700 nm) to 1×10^{-3} *m* (1 mm).

(iii) Radar systems: **Microwaves**. Their short wavelength makes them suitable for radar (RAdio Detection And Ranging) and speed detection. Their wavelength range is about 1 mm to 0.3 *m* (30 cm).

Final answer. X-rays (10^{-8} to 10^{-11} m) for bone fractures, infrared (700 nm to 1 mm) for physiotherapy, and microwaves (1 mm to 0.3 m) for radar.

← *Back to Q119*

Q120

Name the electromagnetic waves which are used:
(i) as a diagnostic tool in medicine
(ii) in remote switches for TV sets
(iii) in water purifiers
Also write their wavelength range.

[2026 • Set 55-2-2]

SOLUTION

(i) Diagnostic tool in medicine: **X-rays**. They are used for chest X-rays, CT scans and dental imaging because bones and soft tissues absorb them differently, giving a clear image. Their wavelength range is about $10^{-8}\ m$ to $10^{-11}\ m$ (roughly 1 nm to 1 pm).

(ii) Remote switches for TV sets: **Infrared waves**. TV remote controls use infrared LEDs (typically around 940 nm) to send signals. The wavelength range of infrared is about $7 \times 10^{-7}\ m$ (700 nm) to $1 \times 10^{-3}\ m$ (1 mm).

(iii) Water purifiers: **Ultraviolet rays**. UV radiation (germicidal UV-C near 254 nm) kills bacteria and micro-organisms, so it is used to disinfect water. The wavelength range of ultraviolet is about $4 \times 10^{-7}\ m$ (400 nm) down to $6 \times 10^{-10}\ m$ (about $10^{-9}\ m$).

Final answer. X-rays ($\mathbf{10^{-8}}$ to $\mathbf{10^{-11}}$ m) as a diagnostic tool, infrared (**700** nm to **1** mm) in TV remotes, and ultraviolet (**400** nm to about $\mathbf{10^{-9}}$ m) in water purifiers.

← *Back to Q120*

Q121

Name the electromagnetic waves which are (i) also known as 'heat waves', (ii) used in LASIK eye surgery, (iii) used to destroy cancer cells. Also write their wavelength range.

[2026 • Set 55-2-3]

SOLUTION

Each part asks for the electromagnetic wave that performs the stated function, together with its wavelength range.

(i) The electromagnetic waves known as **heat waves** are **infrared (IR) waves**. They are emitted by all warm bodies and are strongly absorbed by water molecules, raising the temperature of the absorbing body, which is why they are felt as heat.

Wavelength range of infrared: $\lambda \approx$ **700 nm** to 1 mm, i.e. $710^{-7}\ m$ to $10^{-3}\ m$.

(ii) The electromagnetic waves used in **LASIK eye surgery** are **ultraviolet (UV) waves** (produced by an argon-fluoride excimer laser of wavelength about 193 nm). The UV beam precisely reshapes the cornea.

Wavelength range of ultraviolet: $\lambda \approx$ **1 nm** to 400 nm, i.e. $10^{-9}\ m$ to $410^{-7}\ m$.

(iii) The electromagnetic waves used to **destroy cancer cells** (radiotherapy) are **gamma rays**. Being very high in energy, they damage and kill the rapidly dividing malignant cells.

Wavelength range of gamma rays: $\lambda <$ **10^{-12} m** (i.e. less than about 0.01 nm).

Thus the three waves are **infrared (heat), ultraviolet (LASIK) and gamma rays (cancer therapy)**, with the wavelength ranges given above.

> **COMMON MISTAKE**
>
> *Examiners deduct marks if the wavelength range is omitted; always quote both the name AND its range.*

Final answer. Heat waves are infrared (700 nm to 1 mm), LASIK uses ultraviolet (1 to 400 nm), and cancer cells are destroyed by gamma rays (less than 10^{-12} m).

← *Back to Q121*

Q122

(a) Depict the variation of electric field ($\vec{E}$) and magnetic field ($\vec{B}$) with respect to the direction of propagation of an electromagnetic wave. Write their two important

characteristics.

(b) Show that $\frac{1}{\sqrt{\varepsilon_0 \mu_0}}$ *gives the velocity of an electromagnetic wave in free space.*

[2026 • Set 55-3-1]

SOLUTION

(a) Consider an electromagnetic wave travelling along the $+x$ axis. The electric field $\vec{E}$ oscillates along the y axis and the magnetic field $\vec{B}$ oscillates along the z axis.

Description of the diagram: draw the x axis horizontally as the direction of propagation. Along it draw a sinusoidal curve in the vertical (xy) plane labelled $\vec{E}$ (along y), and a second sinusoidal curve in the horizontal (xz) plane labelled $\vec{B}$ (along z). The two curves have the same wavelength, reach their maxima and zeros at the same points (they are in phase), and $\vec{E}$, $\vec{B}$, and the propagation direction x are mutually perpendicular.

The field components vary as $\mathbf{E_y = E_0 \sin(kx - \omega t)}$ and $\mathbf{B_z = B_0 \sin(kx - \omega t)}$.

Two important characteristics: (1) Electromagnetic waves are **transverse**, since $\vec{E}$ and $\vec{B}$ are both perpendicular to the direction of propagation and to each other. (2) They travel through vacuum with the constant speed $\mathbf{c = 3 10^8\ m\ s^{-1}}$, and at every instant $\mathbf{E_0 = cB_0}$.

(b) Maxwell's equations for a region free of charges and currents lead to the wave equation for the electric field:

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E}{\partial t^2}$$

The standard one-dimensional wave equation for a disturbance moving with speed v is:

$$\frac{\partial^2 E}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 E}{\partial t^2}$$

Comparing the two equations term by term:

$$\frac{1}{v^2} = \mu_0 \varepsilon_0$$

$$\Rightarrow v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

Substituting $\mu_0 = 4\pi 10^{-7}\ T\,m\,A^{-1}$ and $\varepsilon_0 = 8.8510^{-12}\ C^2N^{-1}m^{-2}$:

$$v = \frac{1}{\sqrt{(4\pi 10^{-7})(8.8510^{-12})}} = 310^8\ m\ s^{-1} = c$$

Hence $\frac{1}{\sqrt{\varepsilon_0\,\mu_0}}$ equals the **speed of an electromagnetic wave in free space**.

Final answer. The wave is transverse with E ⊥ B ⊥ propagation, and $1/\sqrt{(\varepsilon_0\mu_0)} = 3\times10^8$ m s^{-1} = c, the speed of light in vacuum.

← *Back to Q122*

Q123

What are microwaves? How are they produced? Give any two uses of microwaves.

[2026 • Set 55-4-2]

SOLUTION

Microwaves are electromagnetic waves whose wavelength lies between about 1 mm and 1 *m*, corresponding to a frequency range of roughly 300 MHz to 300 GHz (gigahertz).

$\lambda \approx 10^{-3}\ m$ to $1\ m$; $\nu \approx 310^8$ Hz to 310^{11} Hz

Production: Microwaves are produced by special electronic devices, namely **klystrons, magnetrons and Gunn diodes (solid-state oscillators)**. In these devices electrons are made to oscillate at very high frequencies in resonant cavities, and the resulting high-frequency oscillating currents radiate microwaves through an antenna. In short, microwaves are generated by oscillating electric currents of very high frequency.

Two uses of microwaves:

(1) In **RADAR systems** for aircraft navigation, speed detection of vehicles, and weather forecasting, because their short wavelength allows them to be sent in narrow beams.

(2) In **microwave ovens**, where their frequency matches the natural frequency of water molecules, so the food (which contains water) absorbs the energy and heats up uniformly.

Thus microwaves are short-wavelength radio-type waves, produced by klystrons/magnetrons, and are used in radar and microwave ovens.

Final answer. Microwaves ($\lambda \approx 1$ mm to 1 m) are produced by klystrons, magnetrons and Gunn diodes, and are used in radar and microwave ovens.

← *Back to Q123*

Q124

What is displacement current (i_d)? Considering the case of charging of a capacitor, show that $\mathbf{i_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$. *What is the value of i_d for a conductor across which a constant voltage is applied?*

[2026 • Set 55-5-1]

SOLUTION

Displacement current is the current that arises in a region wherever the electric flux (and hence the electric field) is changing with time. It is not due to the motion of real charges; it is defined as:

$$i_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

where Φ_E is the electric flux through the area considered. Maxwell introduced it so that Ampere's circuital law would remain valid even in the gap of a charging capacitor.

Derivation for a charging capacitor: Let a parallel plate capacitor of plate area A carry charge q at an instant. If E is the uniform electric field between the plates, then:

$$E = \frac{q}{\varepsilon_0 A}$$

The electric flux through the area A between the plates is:

$$\Phi_E = E\,A = \frac{q}{\varepsilon_0 A} \times A = \frac{q}{\varepsilon_0}$$

The displacement current is therefore:

$$i_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right)$$

$$i_d = \frac{dq}{dt} = i_c$$

This shows that the displacement current between the plates is exactly equal to the conduction current i_c flowing in the connecting wires, and it equals $\varepsilon_0 \frac{d\Phi_E}{dt}$ as required.

Value for a conductor with a constant applied voltage: When a steady (constant) voltage is applied across a conductor, the electric field inside it does not change with time, so $\frac{d\Phi_E}{dt} = 0$.

$$i_d = \varepsilon_0 \frac{d\Phi_E}{dt} = 0$$

Hence the **displacement current is zero** in such a conductor; only the steady conduction current flows.

Final answer. Displacement current is $i_e = \varepsilon_0$ dΦ_E/dt; for a charging capacitor it equals the conduction current, and for a conductor under constant voltage it is zero.

← Back to Q124

Q125

Differentiate between conduction current and displacement current. A capacitor is connected across a source providing time-dependent current. Explain how the total current at an instant is the sum of conduction current and displacement current in the circuit at that instant.

[2026 • Set 55-5-2]

SOLUTION

Difference between conduction current and displacement current:

Conduction current (I_c) is the current produced by the actual flow of free charges (electrons) through a conductor. It obeys Ohm's law and exists only

where there is a material medium carrying free charges.

Displacement current (I_d) is the current associated with a time-varying electric field; it is given by $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$. No real charges move; it exists even in vacuum (for example, between the plates of a capacitor) wherever the electric flux changes with time.

Explanation for a capacitor fed by a time-dependent current: Consider a capacitor connected to a source supplying a time-dependent current. In the connecting wires charges actually flow, so a conduction current I_c exists there, while between the plates no charge crosses the gap, so $\mathbf{I_c = 0}$ in the gap.

However, as the charge q on the plates changes, the field $E = \frac{q}{\varepsilon_0 A}$ and hence the flux Φ_E between the plates change with time, giving rise to a displacement current there:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right) = \frac{dq}{dt} = I_c$$

According to the **Ampere-Maxwell law**, the total current at any point is the sum of the conduction and displacement currents:

$$I = I_c + I_d$$

In the wires $\mathbf{I_d = 0}$, so the total current is I_c; in the gap $\mathbf{I_c = 0}$, so the total current is I_d, and since $\mathbf{I_d = I_c}$, the total current $I = \mathbf{I_c + I_d}$ has the **same value at every point of the circuit**. The displacement current thus restores the continuity of current through the capacitor gap.

Final answer. Conduction current is the flow of real charges, displacement current is ε₀dΦ_E/dt from a changing field, and at every instant their sum I_c + I_d is continuous all around the circuit.

← *Back to Q125*

Q126

(a) *A parallel plate capacitor is charged by an ac source. Show that the sum of conduction current (I_c) and the displacement current (I_d) has the same value at all*

points of the circuit.

***(b)** In case (a) above, is Kirchhoff's first rule (junction rule) valid at each plate of the capacitor? Explain.*

[2025 • Set 55-1-1]

SOLUTION

(a) Let a parallel plate capacitor of plate area A be charged by an ac source, so that at an instant it carries charge q. The electric field between the plates is:

$$E = \frac{q}{\varepsilon_0 A}$$

In the connecting wires, real charges flow, giving a conduction current:

$$I_c = \frac{dq}{dt}$$

Between the plates no charge flows, but the changing field produces a changing electric flux $\Phi_E = E\,A = \frac{q}{\varepsilon_0}$, giving a displacement current:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 A \frac{dE}{dt}$$

Since $E = \frac{q}{\varepsilon_0 A}$, differentiating gives $\frac{dE}{dt} = \frac{1}{\varepsilon_0 A}\frac{dq}{dt}$. Substituting:

$$I_d = \varepsilon_0 A \times \frac{1}{\varepsilon_0 A}\frac{dq}{dt} = \frac{dq}{dt} = I_c$$

Therefore $\mathbf{I_d = I_c}$, and the total current $I_c + I_d$ has the **same value at all points of the circuit**: in the wires it is I_c (with $\mathbf{I_d = 0}$), and in the gap it is I_d (with $\mathbf{I_c = 0}$), and the two are equal in magnitude.

(b) **Yes**, Kirchhoff's first rule (junction rule) is valid at each plate, provided the displacement current is included along with the conduction current.

At a plate, the conduction current I_c arrives through the wire and the equal displacement current I_d leaves into the gap (or vice versa). Since $\mathbf{I_c = I_d}$, the total current entering the plate equals the total current leaving it. Thus the junction rule, which expresses conservation of charge, holds at every plate once the displacement current is taken into account.

CONCEPT

Junction rule fails if you count only the conduction current at a plate; it is restored only by adding the displacement current.

Final answer. Since I_d = I_c, the total current I_c + I_d is the same at all points, and Kirchhoff's junction rule holds at each plate when the displacement current is included.

← *Back to Q126*

Q127

(a) The electric field $\vec{E}$ of an electromagnetic wave propagating in north direction is oscillating in up and down direction. Describe the direction of magnetic field $\vec{B}$ of the wave.
(b) Are the wave length of radio waves and microwaves longer or shorter than those detectable by human eyes?
(c) Write main use of each of the following in human life:
(i) Infrared waves (ii) Gamma rays

[2025 • Set 55-2-1]

SOLUTION

(a) For an electromagnetic wave, $\vec{E}$, $\vec{B}$, and the direction of propagation are mutually perpendicular, and the wave travels along the direction of $\vec{E}\times\vec{B}$.

Here the wave propagates towards the **north** and $\vec{E}$ oscillates **vertically (up and down)**. Taking north as the propagation direction and the vertical as the direction of $\vec{E}$, the magnetic field must be perpendicular to both. Hence $\vec{B}$ oscillates in the **horizontal east-west direction** (perpendicular to both the vertical $\vec{E}$ and the northward propagation), so that $\vec{E}\times\vec{B}$ points north.

(b) The visible region detectable by the human eye has wavelengths of about 400 nm to 700 nm. Radio waves and microwaves have wavelengths much greater than this; therefore their wavelengths are **longer** than those detectable by the human eye.

(c) Main uses in human life:

(i) **Infrared waves** are used in remote controls, in thermal imaging and night-vision devices, and in physiotherapy for muscle relaxation (heat therapy).

(ii) **Gamma rays** are used in the treatment of cancer (radiotherapy) by destroying malignant tissue, and in medical sterilisation of equipment.

Final answer. B oscillates horizontally in the east-west direction; radio and microwaves are longer than visible light; infrared is used in remote controls/thermal imaging and gamma rays in cancer therapy.

← *Back to Q127*

Q128

(a) How is an electromagnetic wave produced?
(b) An electromagnetic wave is travelling in vertically upward direction. At an instant, its electric field vector points in west direction. In which direction does the magnetic field vector point at that instant?
(c) Estimate the ratio of shortest wave length of radio waves to the longest wave length of gamma waves.

[2025 • Set 55-2-2]

SOLUTION

(a) An electromagnetic wave is produced by an **accelerating (oscillating) electric charge**. An oscillating charge produces an oscillating electric field, which in turn produces an oscillating magnetic field, and these two mutually perpendicular fields regenerate each other and propagate through space as an electromagnetic wave. The frequency of the wave equals the frequency of oscillation of the charge.

(b) The wave travels vertically **upward**, so the propagation direction is $+\hat{z}$. At the given instant the electric field points **west**, i.e. $\vec{E}$ is along $-\hat{x}$ (taking east as $+\hat{x}$, north as $+\hat{y}$, up as $+\hat{z}$). The propagation direction is along $\vec{E}\times\vec{B}$:

$$\vec{E}\times\vec{B} \text{ is along } +\hat{z}$$

$$(-\hat{x})\times\hat{B} = +\hat{z}$$

This is satisfied when $\hat{B} = -\hat{y}$, because $(-\hat{x})\times(-\hat{y}) = +\hat{z}$. The direction $-\hat{y}$ is **south**.

Hence the magnetic field points towards the **south**.

(c) Shortest wavelength of radio waves $\approx 10^{-1}\ m$ and longest wavelength of gamma rays $\approx 10^{-12}\ m$. The required ratio is:

$$\frac{\lambda_{\text{radio,min}}}{\lambda_{\text{gamma,max}}} = \frac{10^{-1}}{10^{-12}} = 10^{11}$$

Thus the ratio is of the order of 10^{11}.

Final answer. The wave is produced by accelerating charges, the magnetic field points south, and the ratio of shortest radio to longest gamma wavelength is about 10^{11}.

← *Back to Q128*

Q129

(a) The amplitude of a light wave becomes n times. This results in intensity of the wave becoming m times. What is the relation between n and m?
(b) White light is incident on three identical surfaces - a black surface, a yellow surface and a white surface, one by one. For which surface, the pressure exerted on the surface by the incident light will be (i) maximum (ii) minimum? Justify your answer.

[2025 • Set 55-2-3]

SOLUTION

(a) The intensity of a light wave is proportional to the square of its amplitude:

$$I \propto \mathrm{A}^2$$

If the amplitude becomes n times, i.e. $A \rightarrow \mathbf{nA}$, the new intensity is:

$$I' \propto (\mathrm{nA})^2 = \mathrm{n}^2\mathrm{A}^2$$

So the intensity becomes n^2 times the original. Since it is given that the intensity becomes m times:

$$m = \mathrm{n}^2$$

Hence the required relation is $m = n^2$.

(b) The radiation pressure depends on whether light is absorbed or reflected. For a perfectly absorbing (black) surface the momentum delivered per unit area gives a pressure $P = \frac{I}{c}$, while for a perfectly reflecting (white) surface the momentum reverses, doubling the transfer, so $P = \frac{2I}{c}$.

(i) **Maximum pressure** is on the **white surface**, because it reflects almost all the incident light. The light returns with reversed momentum, so the change in momentum, and hence the pressure, is the greatest ($P \approx \frac{2I}{c}$).

(ii) **Minimum pressure** is on the **black surface**, because it absorbs almost all the incident light. The light only delivers its forward momentum without reflection, giving the least pressure ($P \approx \frac{I}{c}$). The yellow surface, which reflects only part of the light, lies in between.

Final answer. m = n^2; the white surface experiences maximum radiation pressure (2I/c) and the black surface minimum (I/c).

← Back to Q129

Q130

(a) *State any three characteristics of electromagnetic waves.*
(b) *Briefly explain how and where the displacement current exists during the charging of a capacitor.*

[2025 • Set 55-4-1]

SOLUTION

(a) Three characteristics of electromagnetic waves:

(1) They are **transverse waves**: the electric field $\vec{E}$, the magnetic field $\vec{B}$, and the direction of propagation are mutually perpendicular, with $\vec{E}$ and $\vec{B}$ oscillating in phase.

(2) They travel through **vacuum with the constant speed** $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 10^8\ m\ s^{-1}$, requiring **no material medium** for propagation.

(3) They **carry energy and momentum**, and at every instant the field magnitudes are related by $\mathbf{E_0 = c\, B_0}$. (They are also not deflected by electric or magnetic fields, being uncharged.)

(b) During the charging of a parallel plate capacitor, charge accumulates on the plates, so the electric field E between the plates increases with time. This produces a continuously changing electric flux Φ_E in the region between the plates.

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

Therefore the **displacement current exists in the space between the plates** of the capacitor (where there is no conduction current), and at every instant it is equal in magnitude to the conduction current flowing in the connecting wires. It is this displacement current that maintains the continuity of current across the gap of the capacitor.

Final answer. EM waves are transverse, travel at $c = 3\times10^8$ m s^{-1} in vacuum and carry energy and momentum; the displacement current exists between the capacitor plates due to the changing electric flux there.

← *Back to Q130*

Q131

Name the electromagnetic wave used (i) in radar, (ii) in eye surgery and (iii) as diagnostic tool in medicine. Write their wavelength range also.

[2025 • Set 55-6-1]

SOLUTION

Different electromagnetic waves are used for different purposes depending on their wavelength and penetrating power.

(i) The electromagnetic wave used in radar is the microwave.

Its wavelength range is about 1 mm to 30 cm, corresponding to frequencies of roughly 1 GHz to 300 GHz.

(ii) The electromagnetic wave used in eye surgery (LASIK) is the ultraviolet wave, produced by an excimer laser.

Its wavelength range is about 10 nm to 400 nm, that is $10^{-8}\ m$ to $410^{-7}\ m$.

(iii) The electromagnetic wave used as a diagnostic tool in medicine is the X–ray.

Its wavelength range is about $10^{-11}\ m$ to $10^{-8}\ m$, that is roughly 1 pm to 10 nm.

Thus microwaves serve radar, ultraviolet rays serve eye surgery, and X-rays serve medical diagnosis.

TIP

In one-mark naming questions CBSE awards full marks only when BOTH the wave name AND its wavelength (or frequency) range are written. Never omit the range.

Final answer. Radar: microwaves (**1 mm** to **30 cm**); eye surgery: UV (**10–400 nm**); diagnosis: X-rays ($\mathbf{10^{-11}}$–$\mathbf{10^{-8}\ m}$).

← *Back to Q131*

Q132

(a) *Differentiate between 'conduction current' and 'displacement current', giving one similarity and one dissimilarity between them.*
(b) *Explain the existence of electromagnetic waves in free space, using the concept of displacement current.*

[2025 • Set 55-7-1]

SOLUTION

(a) Conduction current is the current that flows in a conductor due to the actual motion of free charges (electrons), and is given by $\mathbf{I_c = \frac{dq}{dt}}$.

Displacement current is the current that arises in a region due to a time-varying electric field (changing electric flux), and is given by $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$.

Similarity: Both currents produce a magnetic field around them and both have the same SI unit, the ampere.

Dissimilarity: Conduction current requires an actual flow of charges through a conductor, whereas displacement current involves no flow of charge and exists even in free space or vacuum where the electric field is changing.

(b) By the Ampere-Maxwell law, a changing electric field sets up a magnetic field:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

By Faraday's law of electromagnetic induction, a changing magnetic field sets up an electric field:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

So in free space, a time-varying electric field $\vec{E}$ produces a time-varying magnetic field $\vec{B}$, and this time-varying $\vec{B}$ in turn produces a time-varying $\vec{E}$.

This mutual and continuous regeneration of $\vec{E}$ and $\vec{B}$ lets the disturbance sustain itself and propagate as an electromagnetic wave through free space at speed $\boldsymbol{c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 310^8\ m/s}$, needing no material medium.

Final answer. Both currents make a magnetic field; conduction needs moving charge, displacement does not. Mutual coupling of changing $\vec{E}$ and $\vec{B}$ sustains EM waves in free space.

← *Back to Q132*

Q133

(a) *How are electromagnetic waves produced?*

(b) *Write the wavelength range and one use of:*

(i) Microwaves, and
(ii) Ultraviolet waves.

[2025 • Set 55-7-2]

SOLUTION

(a) Electromagnetic waves are produced by accelerated (oscillating) electric charges.

When a charge oscillates with a frequency f, it produces a time-varying electric field $\vec{E}$ around it.

This changing electric field produces a time-varying magnetic field $\vec{B}$ (by the Ampere-Maxwell law), and the changing $\vec{B}$ in turn produces a changing $\vec{E}$ (by Faraday's law).

These mutually perpendicular, self-sustaining fields propagate outward as an electromagnetic wave of the same frequency f as the oscillating charge, travelling at $c = 310^8\ m/s$ in vacuum.

(b)(i) Microwaves have a wavelength range of about 1 mm to 30 cm.

Use: They are used in RADAR systems for aircraft navigation, speed detection and weather monitoring (and in microwave ovens).

(b)(ii) Ultraviolet waves have a wavelength range of about 10 nm to 400 nm.

Use: They are used to sterilize medical instruments and to purify drinking water (UV-C kills bacteria).

Final answer. EM waves come from accelerated charges; microwaves (**1 mm–30 cm**) are used in RADAR, UV waves (**10–400 nm**) for sterilization.

← *Back to Q133*

Q134

Name the electromagnetic radiations and write their frequency range:
(a) *Used to take the photograph of the bones*

(b) *Produced by hot bodies*

(c) *Used in television communication systems*

[2025 • Set 55-7-3]

SOLUTION

The required electromagnetic radiations and their frequency ranges are as follows.

(a) To take photographs of bones, *X*–rays are used.

Their frequency range is about 310^{16} Hz to 310^{19} Hz.

(b) The radiation produced by hot bodies is infrared (IR) radiation.

Its frequency range is about 310^{11} Hz to 410^{14} Hz.

(c) The radiation used in television communication systems is the radio wave (VHF and UHF bands).

Its frequency range is about 510^{7} Hz to 10^{9} Hz (overall radio waves span below a few 10^{9} Hz).

Thus X-rays photograph bones, infrared is emitted by hot bodies, and radio waves carry television signals.

Final answer. Bones: X-rays (**310^{16}–310^{19} Hz**); hot bodies: infrared (**310^{11}–410^{14} Hz**); TV: radio waves (**10^{8}–10^{9} Hz**).

← *Back to Q134*

Q135

(a) "The wavelength of the electromagnetic wave is often correlated with the characteristic size of the system that radiates." Give two examples to justify this statement.

(b) (i) Long distance radio broadcasts use short-wave bands. Why?

(ii) Optical and radio telescopes are built on the ground, but X-ray astronomy is possible only from satellites orbiting the Earth. Why?

[2024 • Set 55-1-1]

SOLUTION

(a) The statement means that a system radiates most efficiently when its physical size is comparable to the wavelength of the wave it emits. Two examples:

(i) An antenna radiates radio waves most efficiently when its length is of the order of half the wavelength, $L \sim \frac{\lambda}{2}$. Hence broadcast antennas for MHz frequencies are several metres long, matching the wave's metre-scale wavelength.

(ii) An atom has a size of about $10^{-10}\ m$, and the light it emits (visible region) has a wavelength of the same order, around $510^{-7}\ m$, while a nucleus of size $10^{-14}\ m$ emits gamma rays of correspondingly tiny wavelength.

(b)(i) Long distance radio broadcasts use short-wave (high-frequency) bands because these waves are reflected back to the Earth by the ionosphere.

This sky-wave reflection lets the signal reach far-off places beyond the line of sight, which is not possible with ground-hugging long waves.

(b)(ii) X-rays are strongly absorbed by the Earth's atmosphere, so X-rays from celestial sources cannot reach the ground.

Therefore X-ray astronomy must be done from satellites orbiting above the atmosphere. In contrast, visible light and radio waves pass easily through the atmosphere, so optical and radio telescopes work well on the ground.

Final answer. Radiated wavelength matches system size; short-wave broadcasts reflect off the ionosphere; X-rays are absorbed by the atmosphere so X-ray astronomy needs satellites.

← *Back to Q135*

Q136

(a) On what factors does the speed of an electromagnetic wave in a medium depend?
(b) How is an electromagnetic wave produced?
(c) Sketch a schematic diagram depicting the electric and magnetic fields for an electromagnetic wave propagating along z-axis.

[2024 • Set 55-2-1]

SOLUTION

(a) The speed of an electromagnetic wave in a medium is given by:

$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$$

So the speed depends on the permittivity ε and the permeability μ of the medium (equivalently on its relative permittivity ε_r and relative permeability μ_r). It does not depend on the amplitude or intensity of the wave.

(b) An electromagnetic wave is produced by an accelerated (oscillating) electric charge.

The oscillating charge sets up a time-varying electric field, which produces a time-varying magnetic field, and the two fields continuously regenerate each other and propagate outward as an EM wave of the same frequency as the oscillation.

(c) Schematic diagram for a wave travelling along the $+z$-axis:

Draw three mutually perpendicular axes. The wave moves along the horizontal z-axis (direction of propagation).

The electric field $\vec{E}$ is shown as a sine curve oscillating in the vertical x-direction (up and down), labelled E_x, with arrows of varying length along $\pm\hat{x}$.

The magnetic field $\vec{B}$ is shown as a sine curve oscillating in the y-direction (into and out of the page), labelled B_y, with arrows along $\pm\hat{y}$.

Both $\vec{E}$ and $\vec{B}$ are sinusoidal, in the same phase (peaks and zeros coincide), and are mutually perpendicular as well as perpendicular to the direction of propagation $\hat{z}$. The relation $\vec{E}\times\vec{B}$ points along $+\hat{z}$.

Final answer. Speed depends on $\boldsymbol{\varepsilon}$ and $\boldsymbol{\mu}$ of the medium; EM waves come from accelerated charges; $\vec{\boldsymbol{E}}$ (along $\boldsymbol{x}$) and $\vec{\boldsymbol{B}}$ (along $\boldsymbol{y}$) oscillate in phase, perpendicular to each other and to $\hat{\boldsymbol{z}}$.

← *Back to Q136*

Q137

The electric field in an electromagnetic wave in vacuum is given by:

$\vec{E} = (6.3\ N/C)\ [\cos((1.5\ \text{rad}/m)\ y + (4.5 \times 10^8\ \text{rad}/s)\ t)]\ \hat{i}$

(a) Find the wavelength and frequency of the wave.

(b) What is the amplitude of the magnetic field of the wave?

(c) Write an expression for the magnetic field of this wave.

[2024 • Set 55-3-1]

SOLUTION

The given field is $\vec{E} = (6.3\ N/C)\cos((1.5\ \text{rad}/m)\ y + (4.5 10^8\ \text{rad}/s)\ t)\ \hat{i}$.

Comparing with $\vec{E} = \mathrm{E_0}\cos(\mathrm{k}y + \omega\ t)\hat{i}$, the wave parameters are $\mathrm{E_0} = 6.3\ N/C$, $k = 1.5\ \text{rad}/m$ and $\omega = 4.5 10^8\ \text{rad}/s$.

(a) The wavelength is found from $\lambda = \dfrac{2\pi}{k}$.

$$\lambda = \frac{2\pi}{1.5} = 4.19\ m$$

The frequency is found from $\nu = \dfrac{\omega}{2\pi}$.

$$\nu = \frac{4.5 10^8}{2\pi} = 7.16 10^7\ \text{Hz} \approx 71.6\ \text{MHz}$$

(b) The amplitude of the magnetic field is $\mathrm{B_0} = \dfrac{\mathrm{E_0}}{c}$.

$$\mathrm{B_0} = \frac{6.3}{310^8} = 2.1 10^{-8}\ T$$

(c) Since the argument is (ky + ω *t*), the wave travels along the −*y* direction. The fields must satisfy $\vec{E} \times \vec{B}$ pointing along the direction of propagation $-\hat{j}$.

With $\vec{E}$ along $+\hat{i}$, we need $\hat{i} \times \hat{B} = -\hat{j}$, which gives $\vec{B}$ along $+\hat{k}$ (since $\hat{i} \times \hat{k} = -\hat{j}$).

Hence the magnetic field of the wave is:

$$\vec{B} = (2.1 10^{-8}\ T)\cos((1.5\ \text{rad}/m)\ y + (4.5 10^8\ \text{rad}/s)\ t)\ \hat{k}$$

COMMON MISTAKE

A sign of +ω t next to +ky means the wave moves in the negative axis direction. Fix the $\vec{B}$ direction by forcing $\vec{E}\times\vec{B}$ to point along propagation, not by guessing.

Final answer. $\lambda = 4.19\ m$, $\nu \approx 71.6$ MHz, $B_0 = 2.110^{-8}\ T$, and $\vec{B} = (2.110^{-8}\ T)\cos(1.5y + 4.510^{8}\ t)\ \hat{k}$.

← *Back to Q137*

Q138

Name the electromagnetic waves with their wavelength range which are used for
(i) FM radio broadcast
(ii) detection of fracture in bones
(iii) treatment of muscular strain

[2024 • Set 55-4-1]

SOLUTION

The required electromagnetic waves with their wavelength ranges are given below.

(i) For FM radio broadcast, radio waves are used.

Their wavelength range here is about 2.8 m to 3.4 m (FM band 88–108 MHz), and radio waves in general span from about 0.1 m up to several hundred metres.

(ii) For detection of a fracture in bones, X–rays are used.

Their wavelength range is about $10^{-11}\ m$ to $10^{-8}\ m$.

(iii) For treatment of muscular strain, infrared (IR) waves are used.

Their wavelength range is about $710^{-7}\ m$ (700 nm) to $10^{-3}\ m$ (1 mm); the heat they carry penetrates the skin and relaxes strained muscles.

Thus radio waves serve FM broadcast, X-rays detect bone fractures, and infrared treats muscular strain.

Final answer. FM: radio waves (**1–10 *m***); bone fracture: X-rays (**10^{-11}–10^{-8} *m***); muscular strain: infrared (**700 nm–1 mm**).

← *Back to Q138*

Q139

What is meant by displacement current? How is this current different from the conduction current? A capacitor is being charged by a source of emf. Justify the continuity of current in the circuit.

[2024 • Set 55-4-2]

SOLUTION

Displacement current is the current that arises in a region due to a time-varying electric field (changing electric flux), even when no charge actually flows there. It is given by:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$$

where Φ_E is the electric flux through the region.

Difference from conduction current: Conduction current I_c is due to the actual motion of free charges through a conductor and is given by $\mathbf{I_c = \frac{dq}{dt}}$.

Displacement current I_d involves no flow of charge; it exists even in vacuum or in the insulating gap of a capacitor, purely because the electric field is changing with time.

Continuity of current in *a* charging capacitor: While the capacitor charges, a conduction current I_c flows in the connecting wires, but no charge crosses the empty gap between the plates. So if we apply Ampere's circuital law to a surface passing through the gap, it seems to give zero current, breaking continuity.

As charge q builds up on the plates of area A, the electric field between them is $\boldsymbol{E = \frac{q}{\varepsilon_0 A}}$, so the electric flux is $\boldsymbol{\Phi_E = EA = \frac{q}{\varepsilon_0}}$.

The displacement current in the gap is therefore:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 \cdot \frac{1}{\varepsilon_0} \frac{dq}{dt} = \frac{dq}{dt} = I_c$$

Since $\mathbf{I_d = I_c}$, the displacement current in the gap exactly equals the conduction current in the wires.

Hence the total current (conduction plus displacement) is the same everywhere, and the continuity of current throughout the circuit is maintained.

Final answer. Displacement current $\mathbf{I_d = \varepsilon_0\, d\Phi_E/dt}$ in the capacitor gap exactly equals the conduction current $\mathbf{I_c}$ in the wires, so total current stays continuous.

← *Back to Q139*

Q140

Explain the following giving reasons:
(i) 'Electromagnetic waves differ considerably in their mode of interaction with matter'.
(ii) 'Food items to be heated in microwave oven must contain water'.
(iii) 'Welders wear face mask with glasses during welding'.

[2024 • Set 55-4-3]

SOLUTION

(i) Electromagnetic waves differ considerably in their mode of interaction with matter because they span an enormous range of wavelengths and hence photon energies (over 14 orders of magnitude).

Low-energy radio waves drive collective oscillations of free electrons (currents in antennas), infrared waves set molecules into vibration, visible and ultraviolet light cause electronic transitions in atoms, while high-energy X-rays and gamma rays ionise atoms and penetrate deep into matter. Because the energy delivered per photon differs so widely, each type interacts with matter in a distinctly different way.

(ii) Food items to be heated in a microwave oven must contain water because the microwave frequency (about 2.45 GHz) is tuned to match the natural rotational/dipolar response of water molecules.

The microwaves transfer energy efficiently to these polar water molecules, setting them into rapid rotation; this molecular agitation generates heat that cooks the food. Dry items lacking water absorb very little microwave energy and so are not heated effectively.

(iii) Welders wear a face mask fitted with dark glasses because the welding arc emits intense ultraviolet radiation along with very bright visible light.

Ultraviolet rays can seriously damage the eyes (causing corneal burns) and skin, while the glaring visible light can harm the retina. The dark glass of the mask absorbs the harmful UV and cuts down the intense visible light, protecting the welder's eyes and face.

Final answer. EM waves interact differently because their photon energies vary hugely; microwave ovens need water (its molecules absorb **2.45 GHz** microwaves); welders' masks block the arc's harmful UV and intense light.

← Back to Q140

Q141

(a) Name the parts of the electromagnetic spectrum which are (i) also known as 'heat waves' and (ii) absorbed by ozone layer in the atmosphere.
(b) Write briefly one method each, of the production and detection of these radiations.

[2024 • Set 55-5-1]

SOLUTION

(a)(i) The part of the electromagnetic spectrum known as 'heat waves' is **infrared (IR) radiation**.

Its wavelength range is roughly from 700 nm (just beyond the red end of visible light) to about 1 mm.

It is called 'heat waves' because it is readily absorbed by water molecules and most surfaces, raising their temperature, and it is itself emitted strongly by all warm bodies.

(a)(ii) The radiation absorbed by the ozone layer in the upper atmosphere is **ultraviolet (UV) radiation**.

Its wavelength range is about 400 nm down to 10 nm. The ozone layer mainly absorbs the harmful UV-B and UV-C parts, protecting life on Earth.

(b) Production and detection.

Infrared rays are produced by hot bodies and by molecules when they vibrate or rotate (for example, the heated filament of an IR lamp). They are detected by a thermopile, a bolometer, or a semiconductor (photo) detector that responds to a rise in temperature or photo-conductivity.

Ultraviolet rays are produced by very hot bodies and by special lamps such as mercury-vapour and arc lamps, and by electronic transitions in atoms. They are detected by photographic plates, by photocells (photoelectric effect), and by the fluorescence they cause on certain screens.

TIP

In the board exam, always pin down the wavelength range when you name a radiation, even if the question does not ask for it, because the marking scheme often gives a separate mark for it.

Final answer. (a)(i) Infrared; (ii) Ultraviolet; production and detection as described above.

← *Back to Q141*

Q142

Name the part of the electromagnetic spectrum which are:
(i) stopped by face mask worn by welders.
(ii) used in detectors in Earth satellites.
(iii) used in 'short-wave band' in communication.
Also write the order of wavelengths, in each case.

[2024 • Set 55-5-2]

SOLUTION

(i) The radiation stopped by the face mask (visor) worn by welders is **ultraviolet (UV) radiation**, because the bright welding arc emits intense UV that damages the eyes.

Its wavelength is of the order of 10 nm to 400 nm.

(ii) The radiation used in the detectors of Earth satellites (for remote sensing and weather monitoring) is **microwaves**.

Its wavelength is of the order of 1 mm to 1 m.

(iii) The radiation used in the 'short-wave band' for long-distance communication is **radio waves**.

Its wavelength is of the order of 10 m to 100 m, corresponding to the high-frequency band of about 3 MHz to 30 MHz.

Final answer. (i) Ultraviolet (**10** nm to **400** nm); (ii) Microwaves (**1** mm to **1** m); (iii) Radio waves (**10** m to **100** m).

← *Back to Q142*

Q143

(i) Depict a plane electromagnetic wave propagating along the x-axis. Write the expressions for its oscillating electric and magnetic fields.
(ii) Write three characteristics of electromagnetic waves.

[2022 • Set 55-2-1]

SOLUTION

(i) Consider a plane electromagnetic wave travelling along the $+x$ axis.

The diagram shows three mutually perpendicular axes meeting at the origin. The electric field $\vec{E}$ oscillates as a sine curve in the y direction (vertical), the magnetic field $\vec{B}$ oscillates as a sine curve in the z direction (horizontal, into-and-out of the page plane), and the wave advances along the x axis. At every point the two fields are in the same phase, reaching their maxima and zeros together.

The oscillating fields are written as:

$$\vec{E} = \mathrm{E}_0 \sin(\mathrm{kx} - \omega t)\,\hat{j}$$

$$\vec{B} = \mathrm{B}_0 \sin(\mathrm{kx} - \omega t)\,\hat{k}$$

Here $k = \frac{2\pi}{\lambda}$ is the propagation constant and $\omega = 2\pi\nu$ is the angular frequency. The direction of propagation $\hat{i}$ is along $\vec{E} \times \vec{B}$.

(ii) Three characteristics of electromagnetic waves:

1. They are **transverse** waves, with $\vec{E}$, $\vec{B}$ and the direction of propagation mutually perpendicular.

2. They travel through vacuum at the same speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 310^8\ m/s$, requiring no material medium.

3. They carry energy and momentum, and the ratio of the field amplitudes is fixed: $\frac{E_0}{B_0} = c$.

Final answer. $\vec{E} = E_0 \sin(kx - \omega t)\hat{j}$ and $\vec{B} = B_0 \sin(kx - \omega t)\hat{k}$, with the three listed characteristics.

← *Back to Q143*

Q144

Name the electromagnetic waves which are produced by the following:
(i) Radioactive decays of nucleus
(ii) Welding arcs
(iii) Hot bodies
Write one use each of these waves.

[2022 • Set 55-2-1]

SOLUTION

(i) The radioactive decay of a nucleus produces **gamma rays** (γ-rays).

They are emitted when an excited nucleus (for example ^{60}Co) drops to a lower energy state.

Use: in radiotherapy to destroy cancerous tumours, and for sterilising surgical instruments.

(ii) Welding arcs produce **ultraviolet (UV)** rays (together with intense visible light).

The very high temperature of the electric arc makes it emit a broad spectrum that is rich in UV.

Use: for sterilisation, since UV kills bacteria and viruses by damaging their DNA; also used in 'black-light' fluorescent detection.

(iii) Hot bodies produce **infrared (IR)** radiation.

A body at ordinary temperatures (around 300 K) radiates mainly in the infrared, with the peak wavelength near 10 μm, in agreement with Wien's law.

Use: in thermal-imaging cameras for night vision and in infrared remote controls.

Final answer. (i) Gamma rays (radiotherapy); (ii) Ultraviolet (sterilisation); (iii) Infrared (thermal imaging / remote controls).

← *Back to Q144*

Q145

Electromagnetic waves of wavelengths λ_1, λ_2 and λ_3 are used in radar systems, in water purifiers and in remote switches of TV, respectively.
(i) Identify the electromagnetic waves, and
(ii) Write one source of each of them.

[2022 • Set 55-3-1]

SOLUTION

(i) Identification of the waves:

λ_1, used in radar systems, is **microwave** radiation (wavelength of the order of a few centimetres).

λ_2, used in water purifiers, is **ultraviolet (UV)** radiation (around 254 nm, the germicidal UV-C band), because UV kills germs in the water.

λ_3, used in the remote switch of a TV, is **infrared (IR)** radiation (around 940 nm).

(ii) One source of each:

Microwaves are produced by special electronic devices such as a klystron, a magnetron or a Gunn diode.

Ultraviolet rays are produced by a low-pressure mercury-vapour lamp (and by arc lamps and the Sun).

Infrared rays are produced by hot bodies and by infrared-emitting LEDs (such as the LED inside a TV remote).

Final answer. (i) Microwaves, UV, IR; (ii) sources: magnetron/klystron, mercury-vapour lamp, IR LED respectively.

← *Back to Q145*

Q146

Identify electromagnetic waves which:
(i) are used in radar system.
(ii) affect a photographic plate.
(iii) are used in surgery.
Write their frequency range.

[2022 • Set 55-4-1]

SOLUTION

(i) The waves used in a radar system are **microwaves**.

Their frequency range is about 10^9 Hz to 10^{11} Hz (that is, about 1 GHz to 100 GHz), corresponding to wavelengths from about 1 mm to 30 cm.

(ii) The waves that strongly affect a photographic plate are **ultraviolet (UV)** rays.

Their frequency range is about $7.5 10^{14}$ Hz to $3 10^{16}$ Hz. UV readily blackens the silver-halide emulsion of the plate.

(iii) The waves used in surgery are produced by **lasers**, which lie in the infrared, visible or ultraviolet region.

For example, a CO_2 laser at 10.6 μm has a frequency of about $2.8 10^{13}$ Hz (far infrared), and an excimer UV laser at 193 nm has a frequency of about $1.5 10^{15}$

Hz.

Final answer. (i) Microwaves (10^9–10^{11} Hz); (ii) UV (7.510^{14}–310^{16} Hz); (iii) laser (IR/UV) light as stated.

← *Back to Q146*

Q147

(i) Arrange the following electromagnetic radiation in the ascending order of their frequencies:
X-rays, microwaves, gamma rays, radio waves
(ii) Write two uses of any two of these radiation.

[2022 • Set 55-5-1]

SOLUTION

(i) Arranging the radiations in ascending order of frequency.

Radio waves have the lowest frequency, about 10^4 Hz to 10^9 Hz.

Microwaves come next, about 10^9 Hz to 10^{11} Hz.

X-rays are higher, about 10^{16} Hz to 10^{19} Hz.

Gamma rays have the highest frequency, above 10^{19} Hz.

So the ascending order is: **Radio waves < Microwaves < X-rays < Gamma rays**.

(ii) Two uses of any two of these radiations:

X-rays: (1) medical radiography to image bones and detect fractures; (2) studying crystal structure by X-ray diffraction.

Microwaves: (1) in RADAR for tracking aircraft and ships; (2) in microwave ovens to cook and heat food.

COMMON MISTAKE

Frequency and wavelength run opposite ways. Ascending frequency means descending wavelength, so do not mix the two orders up under exam pressure.

Final answer. Ascending frequency: Radio < Microwave < X-ray < Gamma; uses of X-rays and microwaves as listed.

← Back to Q147

Q148

Write the expression for the speed of light in a material medium of relative permittivity ε_r and relative magnetic permeability μ_r.

[2020 • Set 55-1-1]

SOLUTION

The speed of an electromagnetic wave in any medium of permittivity ε and permeability μ is:

$$v = \frac{1}{\sqrt{\mu\varepsilon}}$$

For a material medium, $\mu = \mu_r\mu_0$ and $\varepsilon = \varepsilon_r\varepsilon_0$, where μ_r and ε_r are the relative permeability and relative permittivity.

Substituting these,

$$v = \frac{1}{\sqrt{\mu_r\mu_0\,\varepsilon_r\varepsilon_0}}$$

$$v = \frac{1}{\sqrt{\mu_r\varepsilon_r}} \cdot \frac{1}{\sqrt{\mu_0\varepsilon_0}}$$

Since the speed of light in vacuum is $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$, this becomes:

$$v = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$$

The quantity $n = \sqrt{\mu_r\varepsilon_r}$ is the **refractive index** of the medium, so equivalently $v = \frac{c}{n}$.

Final answer. $v = \frac{c}{\sqrt{\mu_r\varepsilon_r}}$, where c is the speed of light in vacuum.

← Back to Q148

Q149

Write the wavelength range and name of the electromagnetic waves which are used in (i) radar systems for aircraft navigation, and (ii) Earth satellites to observe the growth of the crops.

[2020 • Set 55-1-1]

SOLUTION

(i) The radiation used in radar systems for aircraft navigation is **microwaves**.

Their wavelength range is about 1 mm to 30 cm (frequencies of about 1 GHz to 100 GHz). Microwaves travel in straight lines, are reflected by metal aircraft, and are not much absorbed by the atmosphere, which makes them ideal for radar.

(ii) The radiation used by Earth satellites to observe the growth of crops is **infrared (IR)** radiation (chiefly the near-infrared band).

Its wavelength range is about 700 nm to a few micrometres. Healthy green crops strongly reflect near-infrared, so satellite imaging in this band reveals the health and growth of vegetation.

Final answer. (i) Microwaves (**1** mm to **30** cm); (ii) Infrared, mainly near-IR (**700** nm to a few **μ**m).

← Back to Q149

Q150

Name the electro-magnetic waves with their frequency range, produced in
(a) some radioactive decay
(b) sparks during electric welding
(c) TV remote

[2020 • Set 55-4-1]

SOLUTION

(a) The radiation produced in some radioactive decay is **gamma rays**.

Their frequency range is about 310^{19} Hz and above (up to about 10^{22} Hz). They are emitted when an excited nucleus returns to a lower energy state.

(b) The radiation produced by the sparks (electric arc) during electric welding is **ultraviolet (UV)** radiation.

Its frequency range is about 7.510^{14} Hz to 310^{16} Hz (wavelengths from 400 nm down to about 10 nm).

(c) The radiation produced by a TV remote is **infrared (IR)** radiation.

Its frequency range is about 310^{11} Hz to 410^{14} Hz (wavelengths from about 1 mm down to 700 nm); a remote typically uses an IR LED at around 940 nm.

Final answer. (a) Gamma rays ($\mathbf{310^{19}}$ Hz); (b) UV ($\mathbf{7.510^{14}}$–$\mathbf{310^{16}}$ Hz); (c) Infrared ($\mathbf{310^{11}}$–$\mathbf{410^{14}}$ Hz).

← *Back to Q150*

Q151

(a) We feel the warmth of the sun light but not the pressure on our hands. Explain.
(b) Which out of wavelength, frequency and speed of an electro-magnetic wave does not change on passing from one medium to another?
(c) A thin ozone layer in the upper atmosphere is crucial for human survival on earth, why?

[2020 • Set 55-4-2]

SOLUTION

(a) Sunlight is an electromagnetic wave, so it carries both energy and momentum.

The intensity of sunlight reaching the Earth is large, about 1 kW/m^2, so the energy delivered to our hands is enough to be felt as warmth.

However, the radiation pressure exerted by sunlight is given by $P = \frac{I}{c}$, where I is the intensity and c is the speed of light.

Substituting, $P = \frac{1000}{3\times 10^{8}} \approx 3.3\times 10^{-6}$ Pa.

This pressure (a few micropascal) is far too small to be sensed by our skin, while the energy is large enough to warm us. That is why we feel the warmth but not the pressure.

(b) The **frequency** of an electromagnetic wave does not change when it passes from one medium to another.

Frequency is fixed by the source. Using $v = \nu\lambda$, when the wave enters a denser medium its speed v decreases, and since ν stays constant, the wavelength λ also decreases. Hence speed and wavelength both change, but frequency stays the same.

(c) The thin ozone layer in the stratosphere absorbs the harmful high-energy **ultraviolet (UV) radiation** coming from the Sun.

If this UV radiation reached the Earth's surface, it would damage living cells, cause skin cancer, cataracts and harm crops and aquatic life. By absorbing this UV, the ozone layer acts as a protective shield, making it crucial for human survival.

Final answer. Frequency stays unchanged across media; sunlight's radiation pressure is too small to feel while its energy warms us; ozone absorbs harmful UV.

← *Back to Q151*

Q152

(a) How are electro-magnetic waves produced? Depict an electro-magnetic wave propagating in Z-direction with its magnetic field $\vec{B}$ oscillating along X-direction.
(b) Write two characteristics of electro-magnetic waves.

[2020 • Set 55-4-3]

SOLUTION

(a) Electromagnetic waves are produced by **accelerating (or oscillating) electric charges**.

An oscillating charge produces an oscillating electric field, which in turn sets up an oscillating magnetic field, and these mutually regenerating fields propagate outward as an electromagnetic wave. The frequency of the wave equals the frequency of oscillation of the charge.

For a wave travelling along the $+Z$-direction with its magnetic field $\vec{B}$ oscillating along the X-direction, the electric field $\vec{E}$ must oscillate along the Y-direction.

This follows because the direction of propagation is along $\vec{E}\times\vec{B}$. With $\vec{B}$ along $+\hat{x}$, for propagation along $+\hat{z}$ we need $\hat{\boldsymbol{E}}\times\hat{\boldsymbol{x}}=\hat{\boldsymbol{z}}$, which gives $\hat{\boldsymbol{E}}=-\hat{\boldsymbol{y}}$.

Diagram description: Draw three mutually perpendicular axes with Z horizontal (direction of propagation), X and Y perpendicular to it. Show a sinusoidal curve of $\vec{B}$ oscillating in the X-Z plane (along $\pm X$) and a sinusoidal curve of $\vec{E}$ oscillating in the Y-Z plane (along $\mp Y$), both having the same wavelength and oscillating in phase (peaks and zeros line up), advancing together along $+Z$.

(b) Two characteristics of electromagnetic waves:

(1) They are **transverse waves**: $\vec{E}$, $\vec{B}$ and the direction of propagation are mutually perpendicular, with $\vec{E}$ and $\vec{B}$ in phase.

(2) They travel through vacuum with the same speed $\boldsymbol{c=\dfrac{1}{\sqrt{\mu_0\varepsilon_0}}=3\times10^8\ m/s}$, and carry both energy and momentum (they need no material medium to travel).

COMMON MISTAKE

Always fix the third direction using $\vec{E}\times\vec{B}$ pointing along propagation; a sign slip here loses the marking-scheme mark for direction.

Final answer. EM waves arise from accelerating charges; for propagation along $+\boldsymbol{Z}$ with $\vec{\boldsymbol{B}}$ along $\boldsymbol{X}$, $\vec{\boldsymbol{E}}$ oscillates along $-\boldsymbol{Y}$; they are transverse and travel at $\boldsymbol{c}$ in vacuum.

← *Back to Q152*

Q153

Identify the part of the electromagnetic spectrum used in (i) radar and (ii) eye surgery. Write their frequency range.

[2019 • Set 55-1-1]

SOLUTION

(i) The part of the electromagnetic spectrum used in **radar** is **microwaves**.

Microwaves are used in radar (RAdio Detection And Ranging) because their short wavelength allows them to be sent out in sharp narrow beams that reflect off aircraft and ships, and the echo is timed to find distance.

Frequency range of microwaves: about $\nu \sim 10^9$ Hz to 3×10^{11} Hz (i.e. roughly 1 GHz to 300 GHz), corresponding to wavelengths from about 0.3 m down to 1 mm.

(ii) The part of the electromagnetic spectrum used in **eye surgery** is the **infrared / visible region (laser light)**.

Lasers in the infrared and visible range deliver a fine, concentrated, monochromatic beam that can be focused precisely to cut or weld delicate eye tissue without damaging surrounding parts.

Frequency range of infrared: about $\nu \sim 3\times 10^{11}$ Hz to 4×10^{14} Hz, while visible light lies at about 4×10^{14} Hz to 7×10^{14} Hz.

Thus radar uses **microwaves** and eye surgery uses an **infrared/visible laser**.

Final answer. Radar uses microwaves (**~ 10^9** to **10^{11} Hz**); eye surgery uses infrared/visible laser light (**~ 10^{11}** to **10^{14} Hz**).

← *Back to Q153*

Q154

Prove that the average energy density of the oscillating electric field is equal to that of the oscillating magnetic field.

[2019 • Set 55-1-1]

SOLUTION

In an electromagnetic wave, the magnitudes of the electric and magnetic fields are related by $E_0 = c\,B_0$, where c is the speed of light and E_0, B_0 are the amplitudes.

The energy density of the electric field at any instant is $u_E = \frac{1}{2}\varepsilon_0 E^2$.

For a sinusoidal field $E = E_0 \sin(\omega t - kx)$, the time-average of E^2 over a full cycle is $E^2 = \frac{E_0^2}{2}$.

Hence the average electric energy density is $u_E = \frac{1}{2}\varepsilon_0 \cdot \frac{E_0^2}{2} = \frac{1}{4}\varepsilon_0 E_0^2$.

Similarly, the average magnetic energy density is $u_B = \frac{B^2}{2\mu_0} = \frac{1}{2\mu_0} \cdot \frac{B_0^2}{2} = \frac{B_0^2}{4\mu_0}$.

Now substitute $E_0 = cB_0$ into the electric energy density:

$$u_E = \frac{1}{4}\varepsilon_0 (cB_0)^2 = \frac{1}{4}\varepsilon_0 c^2 B_0^2.$$

Using $c^2 = \frac{1}{\mu_0 \varepsilon_0}$,

$$u_E = \frac{1}{4}\varepsilon_0 \cdot \frac{1}{\mu_0 \varepsilon_0} B_0^2 = \frac{B_0^2}{4\mu_0}.$$

This is exactly equal to u_B.

Therefore $u_E = u_B$, proving that the average energy density of the oscillating electric field equals that of the oscillating magnetic field.

COMMON MISTAKE

Remember the factor $\frac{1}{2}$ from time-averaging $\sin^2$ over a cycle; dropping it gives the instantaneous, not the average, energy density.

Final answer. It is proved that $u_E = \frac{1}{4}\varepsilon_0 E_0^2 = \frac{B_0^2}{4\mu_0} = u_B$.

← Back to Q154

Q155

Name the radiation of the electromagnetic spectrum which is used for the following:
(a) Radar
(b) To photograph internal parts of human body
(c) For taking photographs of the sky during night and foggy conditions
Give the frequency range in each case.

[2019 • Set 55-3-1]

SOLUTION

(a) The radiation used in **radar** is **microwaves**.

Microwaves form short sharp beams that reflect strongly from distant objects, making them ideal for detecting aircraft and ships.

Frequency range: $\nu \sim 10^9$ Hz to 3×10^{11} Hz.

(b) The radiation used to **photograph internal parts of the human body** is **X-rays**.

X-rays have high penetrating power. They pass through soft tissue but are absorbed by bones, producing a shadow image, so they are used for medical imaging.

Frequency range: $\nu \sim 10^{16}$ Hz to 10^{19} Hz.

(c) The radiation used for **taking photographs of the sky during night and foggy conditions** is **infrared rays**.

Infrared has longer wavelength, so it can penetrate fog, mist and haze and is emitted by warm objects even at night, allowing photography in poor visibility.

Frequency range: $\nu \sim 3\times 10^{11}$ Hz to 4×10^{14} Hz.

Final answer. Radar uses microwaves (**~ 10^9** to **10^{11} Hz**); internal body imaging uses X-rays (**~ 10^{16}** to **10^{19} Hz**); night/foggy photography uses infrared (**~ 10^{11}** to **10^{14} Hz**).

← *Back to Q155*

Q156

When a parallel plate capacitor is being charged by a dc battery, why does a galvanometer show a momentary deflection?
Obtain the expression for the displacement current and show how it is related to conduction current in a steady state.

[2019 • Set 55-3-2]

SOLUTION

When a parallel plate capacitor is connected to a DC battery, charge flows in the connecting wires only while the capacitor is being charged. As charge accumulates on the plates, the current decreases and finally becomes zero once the capacitor is fully charged. Since current flows for only this short interval, the galvanometer shows a **momentary (transient) deflection** and then returns to zero.

During charging, the electric field, and hence the electric flux Φ_E between the plates, keeps changing with time. Maxwell pointed out that this changing electric flux is equivalent to a current, called the **displacement current**.

The electric field between the plates of area A carrying charge q is $E = \frac{\sigma}{\varepsilon_0} = \frac{q}{\varepsilon_0 A}$.

The electric flux between the plates is $\Phi_E = E A = \frac{q}{\varepsilon_0}$.

The displacement current is defined as $I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$.

Substituting $\Phi_E = \frac{q}{\varepsilon_0}$,

$$I_d = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right) = \frac{dq}{dt}.$$

But $\frac{dq}{dt}$ is exactly the conduction current I_c flowing in the wires.

Therefore $I_d = I_c$ at every instant during charging. The displacement current between the plates is equal to the conduction current in the wires, ensuring continuity of total current in the circuit.

In the steady state, the capacitor is fully charged, q is constant, so $\frac{dq}{dt} = 0$ and both I_c and I_d become zero.

Final answer. The transient charging current causes a momentary deflection; the displacement current $I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \frac{dq}{dt} = I_c$ equals the conduction current.

← *Back to Q156*

Q157

Write the range of the wavelength of the following electromagnetic radiations:
(a) Infrared rays
(b) Ultraviolet rays
(c) γ-rays
Write one use of each of the above.

[2019 • Set 55-3-3]

SOLUTION

(a) **Infrared rays**: wavelength range $\lambda \sim 700$ nm (i.e. $7\times 10^{-7}\ m$) to about 1 mm ($10^{-3}\ m$).

One use: in remote sensing and thermal imaging, and in TV remote controls and physiotherapy (heat treatment).

(b) **Ultraviolet rays**: wavelength range $\lambda \sim 400$ nm ($4\times 10^{-7}\ m$) down to about 1 nm ($10^{-9}\ m$).

One use: for sterilising drinking water and surgical instruments, since UV kills bacteria and germs.

(c) **Gamma rays**: wavelength range $\lambda < 10^{-12}$ **m** (i.e. less than about 0.001 nm), the shortest wavelengths in the spectrum.

One use: in radiotherapy for the treatment of cancer, as they destroy malignant tumour cells.

Final answer. Infrared: **700 nm** to **1 mm** (thermal imaging); UV: **~ 1 nm** to **400 nm** (sterilisation); gamma rays: **$<10^{-12}$ m** (cancer radiotherapy).

← Back to Q157

Q158

Define displacement current. What role does it play while charging a capacitor by dc source. Is the value of displacement current same as that of the conduction current? Explain.

[2019 • Set 55-5-1]

SOLUTION

Displacement current is the current that exists in a region wherever the electric flux (electric field) is changing with time. It is defined as $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$, where Φ_E is the electric flux through the region and ε_0 is the permittivity of free space.

Unlike conduction current, it does not involve any actual flow of charge; it arises purely from a time-varying electric field.

Role while charging a capacitor by a DC source: While the capacitor charges, charge builds up on the plates and the electric field between the plates increases with time. There is no conduction current in the gap between the plates (no charge crosses the vacuum/dielectric), so the magnetic field there must be produced by the displacement current. This displacement current in the gap maintains the continuity of the total current across the circuit, so that the current entering one plate equals the current that effectively continues through the gap.

Is its value the same as the conduction current? Yes. The electric field between plates of area A holding charge q is $E = \frac{q}{\varepsilon_0 A}$, so the flux is $\Phi_E = E\,A = \frac{q}{\varepsilon_0}$.

Hence $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right) = \frac{dq}{dt} = I_c}$.

So the displacement current between the plates is exactly equal to the conduction current in the connecting wires at every instant during charging. Once the capacitor is fully charged, E becomes constant, $\mathbf{\frac{d\Phi_E}{dt} = 0}$, and both currents vanish.

Final answer. Displacement current $I_d = \varepsilon_0 \frac{d\Phi_E}{dt}$ maintains current continuity in the capacitor gap and equals the conduction current, $I_d = I_c$.

← Back to Q158

Q159

(a) How are electromagnetic waves produced? Explain.
(b) A plane electromagnetic wave is travelling through a medium along the +ve *z-direction. Depict the electromagnetic wave showing the directions of the oscillating electric and magnetic fields.*

[2017]

SOLUTION

(a) Electromagnetic waves are produced by **accelerating or oscillating electric charges**.

When a charge oscillates, it sets up a time-varying electric field around it. By Maxwell's theory, a changing electric field produces a changing magnetic field, and a changing magnetic field in turn produces a changing electric field. These mutually sustaining fields detach from the charge and propagate outward through space as an electromagnetic wave. The frequency of the radiated wave is equal to the frequency of oscillation of the charge. A stationary charge or a charge moving with constant velocity cannot radiate; acceleration is essential.

(b) For a plane electromagnetic wave travelling along the $+z$-direction, let the electric field oscillate along the X-axis and the magnetic field along the Y-axis.

We can write $\vec{E} = E_0 \sin(kz - \omega t)\,\hat{x}$ and $\vec{B} = B_0 \sin(kz - \omega t)\,\hat{y}$.

The direction of propagation is along $\vec{E} \times \vec{B} = \hat{x} \times \hat{y} = \hat{z}$, confirming travel along $+z$.

Diagram description: Draw the Z-axis horizontal as the direction of propagation. In the X-Z plane, draw a sinusoidal curve for $\vec{E}$ oscillating along $\pm X$. In the Y-Z plane (perpendicular to the first), draw a sinusoidal curve for $\vec{B}$ oscillating along $\pm Y$. Both have the same wavelength and are in phase (their crests, zeros and troughs occur at the same z), and they advance together along $+Z$. The fields $\vec{E}$, $\vec{B}$ and the propagation direction are mutually perpendicular.

Final answer. EM waves are produced by accelerating charges; for propagation along $+z$, $\vec{E}$ (along X) and $\vec{B}$ (along Y) oscillate in phase and mutually perpendicular, with $\vec{E}\times\vec{B}$ along $+z$.

← Back to Q159

Q160

(i) Identify the part of the electromagnetic spectrum which is:
(a) suitable for radar system used in aircraft navigation,
(b) produced by bombarding a metal target by high speed electrons.
(ii) Why does a galvanometer show a momentary deflection at the time of charging or discharging a capacitor? Write the necessary expression to explain this observation.

[2016]

SOLUTION

(i)(a) The part of the electromagnetic spectrum suitable for the **radar system used in aircraft navigation** is **microwaves**.

Microwaves can be formed into narrow directional beams that reflect off aircraft, allowing their position and distance to be tracked accurately.

(i)(b) The radiation **produced by bombarding a metal target with high-speed electrons** is **X-rays**.

When fast electrons strike a metal target and are suddenly decelerated, their kinetic energy is converted into high-frequency electromagnetic radiation, the X-rays.

(ii) When a capacitor is charged or discharged, the current in the circuit flows only for a short transient interval. While charging, charge builds up on the plates; while discharging, it flows away. In both cases the current exists only momentarily and then becomes zero, so the galvanometer shows a **momentary deflection**.

During this interval, the electric field, and hence the electric flux Φ_E between the plates, changes with time. This changing flux constitutes a displacement current given by $\mathbf{I_d = \varepsilon_0 \frac{d\Phi_E}{dt}}$.

Since the flux between plates of area A holding charge q is $\Phi_E = \frac{q}{\varepsilon_0}$, we get $I_d = \varepsilon_0 \frac{d}{dt}\left(\frac{q}{\varepsilon_0}\right) = \frac{dq}{dt} = I_c$.

Thus the displacement current between the plates equals the conduction current I_c in the wires, and it is this transient current that the galvanometer detects. When the process ends, $\frac{d\Phi_E}{dt} = 0$ and the deflection vanishes.

Final answer. Aircraft-navigation radar uses microwaves; bombarding a metal target with fast electrons gives X-rays; the momentary deflection is due to the transient current, with $I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = I_c$.

← *Back to Q160*

Q161

How are electromagnetic waves produced? What is the source of energy of these waves? Write mathematical expressions for electric and magnetic fields of an electromagnetic wave propagating along the z-axis. Write any two important properties of electromagnetic waves.

[2016]

SOLUTION

Electromagnetic waves are produced by accelerating (oscillating) electric charges. When a charge oscillates with some frequency, it sets up a time-varying electric field around it. By the Ampere-Maxwell law a changing electric field produces a time-varying magnetic field, and by Faraday's law this changing magnetic field in turn produces a time-varying electric field.

These mutually perpendicular, mutually inducing fields regenerate each other and propagate outward through space as an electromagnetic wave. The frequency of the wave equals the frequency of oscillation of the charge.

The source of energy of these waves is the kinetic energy of the accelerated (oscillating) charges, which is supplied by the external source (battery or a.c. driver) that keeps the charges oscillating.

For a plane electromagnetic wave travelling along the $+z$-axis, the electric field oscillates along the x-axis and the magnetic field along the y-axis:

$$\vec{E} = E_0 \sin(kz - \omega t)\,\hat{i}$$

$$\vec{B} = B_0 \sin(kz - \omega t)\,\hat{j}$$

Here E_0 and B_0 are the amplitudes, $k = \frac{2\pi}{\lambda}$ is the propagation constant and $\omega = 2\pi\nu$ is the angular frequency.

Two important properties of electromagnetic waves are: (1) They are **transverse** in nature, with $\vec{E}$, $\vec{B}$ and the direction of propagation mutually perpendicular. (2) They travel through vacuum with the speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8\ m/s$, requiring **no material medium**.

Final answer. EM waves are produced by accelerating charges; energy comes from the oscillating charges; $\vec{E} = E_0 \sin(kz - \omega t)\hat{i}$, $\vec{B} = B_0 \sin(kz - \omega t)\hat{j}$; they are transverse and need no medium.

← *Back to Q161*

Q162

How are electromagnetic waves produced? What is the source of energy of these waves? Draw a schematic sketch of the electromagnetic waves propagating along the +x-axis. Indicate the directions of the electric and magnetic fields. Write the relation between the velocity of propagation and the magnitudes of electric and magnetic fields.

[2015]

SOLUTION

Electromagnetic waves are produced by accelerated (oscillating) electric charges. An oscillating charge or an alternating current (as in an LC oscillator or an antenna) creates a time-varying electric field $\vec{E}$.

This changing $\vec{E}$ produces a time-varying magnetic field $\vec{B}$, which in turn produces a changing $\vec{E}$, and so on. These self-sustaining fields detach from the

source and radiate outward as an electromagnetic wave whose frequency equals the frequency of oscillation of the charge.

The source of energy of these waves is the kinetic energy of the accelerated charges, which is continuously supplied by the external driving source (battery or a.c. generator).

Schematic sketch for propagation along the $+x$-axis: Draw the x-axis horizontally as the direction of propagation. In the vertical xy-plane draw a sine curve labelled $\vec{E}$ (electric field) oscillating along the y-axis. In the horizontal xz-plane, perpendicular to the first, draw a second sine curve labelled $\vec{B}$ (magnetic field) oscillating along the z-axis. Both waves are in phase. Mark an arrow along $+x$ labelled "direction of propagation". Thus $\vec{E} \parallel \hat{j}$, $\vec{B} \parallel \hat{k}$, and $\vec{E} \times \vec{B}$ points along $+\hat{i}$.

The relation between the velocity of propagation and the magnitudes of the fields is:

$$c = \frac{E_0}{B_0} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \ m/s$$

where E_0 and B_0 are the amplitudes of the electric and magnetic fields.

COMMON MISTAKE

In the sketch, always show $\vec{E}$ and $\vec{B}$ in phase (peaks line up) and mutually perpendicular; drawing them out of phase or parallel loses marks.

Final answer. EM waves arise from accelerated charges, draw energy from those charges, and obey $\boldsymbol{c = \frac{E_0}{B_0} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3\times 10^8 \ m/s}$.

← *Back to Q162*

Q163

Name the parts of the electromagnetic spectrum which is
(a) suitable for radar systems used in aircraft navigation.
(b) used to treat muscular strain.

(c) used as a diagnostic tool in medicine.
Write in brief, how these waves can be produced.

[2015]

SOLUTION

(a) Radar systems used in aircraft navigation use **microwaves**.

Microwaves are produced by special vacuum-tube oscillators such as the klystron, magnetron, or by a Gunn diode, in which electrons oscillate at very high frequencies. Their frequencies lie in the range of about 1 GHz to 300 GHz (wavelengths from about 1 mm to 30 cm). Their short wavelength gives sharp, narrow beams suitable for radar.

(b) Treatment of muscular strain uses **infrared (IR) radiation**.

Infrared waves are produced by hot bodies and molecules, for example by an infrared lamp, a heated filament, or IR LEDs. A warm body emits radiation predominantly in the infrared region, which is why IR is also called heat radiation. The heating effect relaxes strained muscles.

(c) The diagnostic tool used in medicine is **X-rays**.

X-rays are produced by bombarding a heavy metal target (such as tungsten) with high-energy electrons in an X-ray tube. The sudden deceleration of the fast electrons in the target produces X-rays (Bremsstrahlung), along with characteristic X-rays of the target atoms. X-rays pass through soft tissue but are absorbed by bones, allowing imaging of internal structures.

Final answer. (a) Microwaves (klystron/magnetron); (b) infrared, from hot bodies/IR lamps; (c) X-rays, by striking a metal target with fast electrons.

← *Back to Q163*

Q164

Answer the following :
(a) Name the em waves which are suitable for radar systems used in aircraft navigation. Write the range of frequency of these waves.
(b) If the earth did not have atmosphere, would its average surface temperature be

higher or lower than what it is now ? Explain.
(c) An em wave exerts pressure on the surface on which it is incident. Justify.

[2014]

SOLUTION

(a) Radar systems used in aircraft navigation use **microwaves**. Their frequency range is about $\nu \sim 10^9$ Hz to 10^{11} Hz (that is, about 1 GHz to 100 GHz).

(b) If the Earth had no atmosphere, its average surface temperature would be **lower** than it is now.

The atmosphere contains greenhouse gases such as carbon dioxide and water vapour. These gases absorb the infrared radiation re-emitted by the warm Earth and trap it near the surface (the greenhouse effect), keeping the surface warm. Without an atmosphere this trapping would not occur, so the heat would escape freely into space and the average surface temperature would fall.

(c) An electromagnetic wave carries both energy and linear momentum. The momentum density of the wave is related to its energy density u by $\frac{u}{c}$.

When the wave is incident on a surface and is absorbed or reflected, this momentum is transferred to the surface. By Newton's second law, the rate of transfer of momentum to the surface is the force exerted on it, and force per unit area is pressure. Hence the electromagnetic wave exerts a **radiation pressure** on the surface, which justifies the statement.

Final answer. (a) Microwaves, $\boldsymbol{\nu \sim 10^9}$ to $\boldsymbol{10^{11}}$ Hz; (b) lower, due to loss of the greenhouse effect; (c) the wave's momentum, transferred on absorption/reflection, exerts radiation pressure.

← *Back to Q164*

Q165

Answer the following :
(a) Name the em waves which are used for the treatment of certain forms of cancer. Write their frequency range.
(b) Thin ozone layer on top of stratosphere is crucial for human survival. Why ?

(c) Why is the amount of the momentum transferred by the em waves incident on the surface so small ?

[2014]

SOLUTION

(a) The electromagnetic waves used for the treatment of certain forms of cancer are **gamma rays** (in radiotherapy). Their frequency range is about 3×10^{19} Hz to about 10^{22} Hz (and higher), corresponding to very short wavelengths below about 10^{-12} m.

(b) The thin ozone layer at the top of the stratosphere absorbs nearly all the harmful ultraviolet (UV-B and UV-C) radiation coming from the Sun.

If this UV radiation reached the Earth's surface in full strength, it would cause skin cancers, cataracts and serious damage to crops and aquatic life. By filtering out this radiation, the ozone layer protects living organisms, which is why it is crucial for human survival.

(c) The momentum carried by an electromagnetic wave is related to the energy it carries by:

$$p = \frac{U}{c}$$

Since the speed of light $\boldsymbol{c = 3 \times 10^8\ m/s}$ is extremely large, the momentum $\boldsymbol{p = \frac{U}{c}}$ obtained by dividing the energy U by this huge value is very small.

Hence, even when a fairly large amount of energy is delivered, the momentum transferred to the surface is **very small**, which is why radiation pressure is not noticeable in everyday life.

Final answer. (a) Gamma rays, $\boldsymbol{\sim 3\times 10^{19}}$ Hz to $\boldsymbol{\sim 10^{22}}$ Hz; (b) ozone absorbs harmful UV; (c) momentum $\boldsymbol{p = U/c}$ is tiny because $\boldsymbol{c}$ is very large.

← *Back to Q165*

Q166

Answer the following the questions :
(a) Name the em waves which are produced during radioactive decay of a nucleus. Write their frequency range.
(b) Welders wear special glass goggles while working. Why ? Explain.
(c) Why are infrared waves often called as heat waves ? Give their one application.

[2014]

SOLUTION

(a) The electromagnetic waves produced during the radioactive decay of a nucleus are **gamma rays**. Their frequency range is about 3×10^{19} Hz and above (up to about 10^{22} Hz), corresponding to very short wavelengths below about 10^{-12} m. They are emitted by excited nuclei.

(b) Welders wear special glass goggles because the intense welding arc emits a large amount of **ultraviolet (UV) radiation** along with bright visible and infrared light.

Direct exposure to this UV radiation causes painful inflammation of the cornea ("welder's flash") and can lead to cataracts and damage to the eyes. The special filter glass in the goggles absorbs the harmful UV (and most of the excess IR), allowing only enough visible light through for the welder to see the work safely.

(c) Infrared waves are called heat waves because most bodies at ordinary temperatures emit radiation mainly in the infrared region. When infrared radiation is absorbed by a material, its energy is taken up by the molecules of the material, increasing their vibrational motion and hence raising the temperature of the body. This direct heating effect is why infrared is called heat radiation.

One application of infrared waves: they are used in thermal imaging cameras and in night-vision devices (also in remote controls and physiotherapy).

Final answer. (a) Gamma rays, ~ 3×10^{19} Hz and above; (b) to absorb harmful UV (and excess IR) from the arc; (c) IR is absorbed and raises a body's temperature; used in thermal imaging.

← Back to Q166

Q167

Write briefly any two factors which demonstrate the need for modulating a signal. Draw a suitable diagram to show amplitude modulation using a sinusoidal signal as the modulating signal.

[2011]

SOLUTION

Two factors that demonstrate the need for modulating a signal are:

(1) **Size of the antenna or aerial.** For efficient transmission, the height of the antenna should be comparable to the wavelength of the signal, at least about $\frac{\lambda}{4}$. Audio (message) signals have low frequencies (20 Hz to 20 kHz) and hence very large wavelengths, which would need impractically long antennas (several kilometres). By modulating the message onto a high-frequency carrier of much shorter wavelength, the required antenna length becomes practical.

(2) **Effective power radiated by the antenna.** The power radiated by an antenna of length l is proportional to $(\frac{l}{\lambda})^2$. For low-frequency (large λ) signals this ratio is very small, so the radiated power is poor. Using a high-frequency carrier increases $\frac{l}{\lambda}$ and so increases the radiated power. (Modulation also allows many channels to be separated by assigning different carrier frequencies, avoiding mixing of signals.)

Amplitude modulation diagram: Draw three waveforms one below the other. (1) The modulating (message) signal $A_m \cos(\omega_m t)$, a low-frequency sine wave. (2) The carrier wave $A_c \cos(\omega_c t)$, a high-frequency sine wave of constant amplitude. (3) The amplitude-modulated wave, a high-frequency carrier whose amplitude (the dashed outline, called the envelope) rises and falls following the shape of the message signal. The modulated wave is described by:

$$c_m(t) = (A_c + A_m \omega_m t)\omega_c t$$

where the envelope traces the original message.

Final answer. Modulation is needed for a practical antenna length and for enough radiated power; AM gives a carrier whose amplitude follows the message, $c_m(t)=(A_c+A_m\omega_m t)\omega_c t$.

← Back to Q167

Q168

Draw a schematic diagram showing the (i) ground wave (ii) sky wave and (iii) space wave propagation modes for em waves.
Write the frequency range for each of the following:
(i) Standard AM broadcast
(ii) Television
(iii) Satellite communication

[2011 • Set 55-1-1]

SOLUTION

Schematic diagram of the three propagation modes (describe with labels):

(i) **Ground wave:** Draw the curved surface of the Earth. Show a transmitting antenna on the ground and a wave travelling along the curved surface of the Earth from the transmitter to the receiver, labelled "ground wave". It glides along the Earth's surface, so the antenna must be in contact with or close to the ground.

(ii) **Sky wave:** From the transmitting antenna draw a ray going up at an angle to the ionosphere (a layer drawn high above the Earth), then reflecting off the ionosphere back down to a distant receiver on the ground, labelled "sky wave". This uses reflection from the ionosphere for long-distance communication.

(iii) **Space wave:** Draw a tall transmitting tower and a tall receiving tower, with a straight (line-of-sight) ray going directly from the top of one to the top of the other, labelled "space wave". This is straight-line propagation used for very high frequencies.

Frequency ranges:

(i) Standard AM broadcast: about 540 kHz to 1600 kHz (medium frequency).

(ii) Television: about 54 MHz to 890 MHz (VHF and UHF).

(iii) Satellite communication: about 5.925 GHz to 6.425 GHz (uplink) and 3.7 GHz to 4.2 GHz (downlink), that is, in the GHz (microwave) range.

Final answer. Ground wave glides along the surface, sky wave reflects off the ionosphere, space wave is line-of-sight; AM ≈ 540-1600 kHz, TV ≈ 54-890 MHz, satellite in the GHz range.

← *Back to Q168*

Q169

Write four characteristics of electromagnetic waves. Give two uses each of (i) Radio-waves (ii) Micro-waves.

[2007]

SOLUTION

Four characteristics of electromagnetic waves:

(1) They are **transverse** waves: the electric field $\vec{E}$, the magnetic field $\vec{B}$ and the direction of propagation are mutually perpendicular ($\vec{E} \perp \vec{B}$ and both perpendicular to the direction of travel).

(2) They require **no material medium** for propagation and can travel through vacuum.

(3) In vacuum they all travel with the same speed $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8\ m/s$.

(4) The ratio of the amplitudes of the electric and magnetic fields equals the speed of light, $\frac{E_0}{B_0} = c$. (They also carry energy and momentum and are not deflected by electric or magnetic fields.)

Two uses of radio waves: (i) in radio and television broadcasting; (ii) in mobile (cellular) communication.

Two uses of microwaves: (i) in microwave ovens for cooking food; (ii) in radar and in satellite communication.

Final answer. EM waves are transverse, need no medium, travel at $c = 3 \times 10^8$ m/s, and have $E_0/B_0 = c$; radio waves are used for broadcasting and mobile communication, microwaves for ovens and radar/satellites.

← *Back to Q169*

Q170

Name the following constituent radiations of electromagnetic spectrum which
(i) produce intense heating effect.
(ii) is absorbed by the ozone layer in the atmosphere.
(iii) is used for studying crystal structure.
Write one more application for each of these radiations.

[2007]

SOLUTION

(i) The radiation that produces an intense heating effect is **infrared (IR) radiation**.

One more application: infrared waves are used in thermal imaging and night-vision devices (also in TV remote controls and in physiotherapy heat lamps).

(ii) The radiation absorbed by the ozone layer in the atmosphere is **ultraviolet (UV) radiation**.

One more application: ultraviolet rays are used for sterilisation, that is, for killing germs in water and air, and in detecting forged bank notes and signatures by fluorescence.

(iii) The radiation used for studying crystal structure is **X-rays** (through X-ray diffraction).

One more application: X-rays are used in medical radiography to examine bones and internal organs, and in industry to detect flaws and cracks inside metal castings.

Therefore: (i) Infrared, (ii) Ultraviolet, (iii) X-rays.

Final answer. (i) Infrared (heating; used in thermal imaging); (ii) ultraviolet (absorbed by ozone; used for sterilisation); (iii) X-rays (crystal structure; used in medical radiography).

← *Back to Q170*

Q171

Draw a labelled diagram of Hertz's experimental set-up to produce electromagnetic waves. Explain the generation of electromagnetic waves using this set-up.

[2006]

SOLUTION

Hertz's experimental set-up (1887) used a high-frequency LC oscillator to generate electromagnetic waves and a separate loop to detect them.

The set-up has two parts. The transmitter (oscillator) consists of two large metal plates acting as a capacitor of capacitance C, each plate joined by a wire to a small metal sphere. The two spheres are placed close together leaving a tiny air gap, which forms the spark gap. The two spheres are connected through this gap to the secondary of an induction coil, which supplies a very high alternating voltage.

The detector (receiver) is a separate copper wire bent into a ring (loop) with its two ends carrying small spheres separated by a small gap. This loop is placed a short distance away from the transmitter, with no electrical connection to it.

In a labelled diagram: on the left, draw the two horizontal metal plates (label them as the capacitor plates C), each connected by a wire to a small sphere; the two spheres face each other across the spark gap, and the induction coil is shown connected across the gap. On the right, draw the circular receiver loop with its own small gap between two spheres. Mark the space between transmitter and receiver as the region carrying the radiated electromagnetic waves (shown by outward arrows).

When the induction coil applies a large voltage across the spark gap, the air in the gap ionizes and a spark jumps across it. The circuit formed by the plates (capacitance C) and the connecting wires (inductance L) behaves as an LC

circuit, so charge oscillates back and forth at the natural frequency $\nu = \frac{1}{2\pi\sqrt{LC}}$.

These rapid oscillations of charge mean the electrons accelerate continuously. An accelerating (oscillating) charge produces a changing electric field, and by Maxwell's theory this changing electric field sets up a changing magnetic field, and so on, so that mutually perpendicular oscillating $\vec{E}$ and $\vec{B}$ fields propagate outward as an electromagnetic wave.

In Hertz's apparatus the frequency was a few hundred megahertz, giving a wavelength λ of a few metres. When these waves reached the receiver loop, the oscillating fields induced an EMF in it and a spark appeared across the receiver gap, confirming that electromagnetic waves had travelled across the empty space.

Hertz also showed that these waves could be reflected, refracted and polarized, exactly like light. **This experiment confirmed Maxwell's prediction that electromagnetic waves exist and travel through space at the speed of light.**

TIP

For full marks the diagram must be labelled (capacitor plates, spark gap, induction coil, receiver loop) and you must clearly state that an accelerating/oscillating charge is the source of EM waves.

Final answer. Hertz's spark-gap **LC** oscillator radiated electromagnetic waves that induced sparks in a nearby receiver loop, confirming Maxwell's theory.

← Back to Q171